Nicotine dependence and cessation: Reproducible Report

Contents

1 Settings

1.1 Set working environment

```r
#clear R Environment and set the seed
rm(list=ls())

# Print the current R version
R.version.string
```

```
## [1] "R version 4.5.3 (2026-03-11 ucrt)"
```

```r
# Set RNG seed
set.seed(2000)
```

1.2 Required package

dplyr and ggplot2 : Data manipulation and visualization.

gtsummary and broom : Sample characteristic table

writexl : Write results into Microsoft Excel.

lavaan and semTools : Confirmatory factor analysis and composite reliability.

psych : Corrected item-total Spearman's rank correlation.

robustbase and car : MM-estimator regression

ppcor : Partial Spearman's rank correlation.

rstatix : Kruskal-Wallis test and effect size

```r
packages <- c("dplyr", "ggplot2", "gtsummary","broom",
              "writexl",
              "lavaan","semTools","psych",
              "robustbase","car","ppcor",
              "rstatix")

for (package in packages) {

  if (!requireNamespace(package, quietly = TRUE))
    install.packages(package)

  suppressPackageStartupMessages(library(package, character.only = TRUE))

}
```

1.3 Import and inspect data attributes

```r
# load the data file
data <- as.data.frame(read.csv("nicotine_dataset.csv"))

attributes(data)[names(attributes(data)) != "row.names"]
```

```
## $names
##  [1] "ID"                        "response_time"
##  [3] "deviceid"                  "age"
##  [5] "sex"                       "smoke_stat"
##  [7] "ethnicity"                 "province"
##  [9] "marital_stat"              "residence"
## [11] "undergrad_now"             "speciality"
## [13] "university_college"        "smoke_now.cigarettes"
## [15] "smoke_now.e_cigarettes"    "smoke_now.hookah"
## [17] "smoke_past.cigarettes"     "smoke_past.e_cigarettes"
## [19] "smoke_past.hookah"         "smoke_most_ex"
## [21] "time_since_quitting"       "hc1"
## [23] "sc1"                       "ir1"
## [25] "sp1"                       "hc2"
## [27] "sc2"                       "ir2"
## [29] "sp2"                       "hc3"
## [31] "sc3"                       "ir3"
## [33] "sp3"                       "hc4"
## [35] "attention_check_a"         "sc4"
## [37] "ir4"                       "sp4"
## [39] "hc5"                       "sc5"
## [41] "ir5"                       "sp5"
## [43] "smoke_most_current"        "auto1"
## [45] "loc1"                      "tol1"
## [47] "craving1"                  "taste1"
## [49] "cog_e1"                    "w_control1"
## [51] "cue1"                      "affect1"
## [53] "auto2"                     "attach1"
## [55] "cue2"                      "cog_e2"
## [57] "auto3"                     "taste2"
## [59] "loc2"                      "craving2"
## [61] "social1"                   "w_control2"
## [63] "taste3"                    "loc3"
## [65] "attach2"                   "craving3"
## [67] "cue3"                      "auto4"
## [69] "attach3"                   "social2"
## [71] "attention_check_b"         "tol2"
## [73] "craving4"                  "social3"
## [75] "tol3"                      "cog_e3"
## [77] "affect2"                   "w_control3"
## [79] "loc4"                      "tol4"
## [81] "affect3"                   "cigg_f1"
## [83] "cigg_f2"                   "cigg_f3"
## [85] "cigg_f4"                   "cigg_f5"
## [87] "cigg_f6"                   "ecigg_f1"
## [89] "ecigg_f2"                  "ecigg_f3"
```

```
## [91] "ecigg_f4"                    "ecigg_f5"
## [93] "ecigg_f6"                    "h_f1"
## [95] "h_f2"                        "h_f3"
## [97] "h_f4"                        "h_f5"
## [99] "h_f6"
##
## $class
## [1] "data.frame"
```

2 Data preprocessing

2.1 Data cleaning

```
# Convert coded dichotomous columns into labelled factors
coded_cols <- c(14:19)
data[,coded_cols] <- lapply(data[,coded_cols],
                            function(x) factor(x, levels = c(0, 1),
                                                 labels = c("No", "Yes"))
                            )

# Data cleaning
cols_to_change <- c("ecigg_f1","ecigg_f3","ecigg_f4")
data[,cols_to_change] <- lapply(data[,cols_to_change],
                                function(x) {
  x[x == "0_1"] <- "0"
  x[x == "0_2"] <- "0"
  x[x == "1_2"] <- "1"
  x[x == "2_2"] <- "2"
  return(x)
  }
  )

# Convert cleaned columns into numeric
data[, cols_to_change] <- lapply(data[, cols_to_change], function(x) {
  as.numeric(as.character(x))
})
```

2.2 Exclude duplicate responses

```
cat("No. of responses: ",
    nrow(data),
    "\n"
    )
```

```
## No. of responses:  3310
```

```
data_unique <- data[!duplicated(data$deviceid),]
cat("No. of unique responses: ",
    nrow(data_unique),
    "\n"
    )
```

```
## No. of unique responses:  3309
```

2.3 Exclude non-undergraduate students

```
data_under <- data_unique[data_unique$undergrad_now == "yes", ]
cat("No. of undergraduate participants: ",
    nrow(data_under),
    "\n"
    )
```

```
## No. of undergraduate participants:  2215
```

2.4 Exclude non-smokers

```
sample <- data_under[data_under$smoke_stat != "none_smoker", ]
cat("No. of responses following exclusion of non-smokers: ",
    nrow(sample),
    "\n"
    )
```

```
## No. of responses following exclusion of non-smokers:  1309
```

2.5 Exclude responses with invalid age

```
sample <- sample[!is.na(sample$age), ]
cat("No. of responses following exclusion based on age (15 years - 70 years): ",
    nrow(sample),
    "\n"
    )
```

```
## No. of responses following exclusion based on age (15 years - 70 years):  1303
```

2.6 Exclude responses with invalid university

```
sample <- sample[!is.na(sample$university_college), ]
cat("No. of responses following exclusion based on valid university: ",
    nrow(sample),
    "\n"
    )
```

```
## No. of responses following exclusion based on valid university:  1251
```

2.7 Exclude students living outside of Iraq

```
sample <- sample[sample$province != "outside_Iraq", ]

cat("No. of responses following exclusion based on province (outside_Iraq removed): ",
    nrow(sample),
    "\n"
)
```

```
## No. of responses following exclusion based on province (outside_Iraq removed):  1248
```

2.8 Exclude inattentive responses

```
# Filter using attention check items
sample$attention <- ifelse(
   (!is.na(sample$attention_check_a) & sample$attention_check_a == 3) |
   (!is.na(sample$attention_check_b) & sample$attention_check_b == 6),
   "Yes", "No"
)
data_attentive <- sample[sample$attention == "Yes", ]
cat("No. of attentive participants: ",
    nrow(data_attentive),
    "\n"
    )
```

```
## No. of attentive participants:  991
```

```
# Filter using response time (12 is subtracted for ID and multiple-response variables)
data_attentive$response_threshold <- apply(data_attentive, 1,
                                           function(row) (sum(!is.na(row))*2)-12)
data_cleaned <- data_attentive[data_attentive$response_time > data_attentive$response_threshold, ]
cat("No. of responses following exclusion based on response time: ",
    nrow(data_cleaned),
    "\n"
    )
```

```
## No. of responses following exclusion based on response time:  991
```

3 Data summarization

```
# Merge smoking method columns for smokers and ex smokers into common columns
data_cleaned <- data_cleaned %>%
  mutate(cigarettes   = coalesce(smoke_now.cigarettes, smoke_past.cigarettes),
         e_cigarettes = coalesce(smoke_now.e_cigarettes, smoke_past.e_cigarettes),
         hookah       = coalesce(smoke_now.hookah, smoke_past.hookah),
         smoke_most   = coalesce(smoke_most_current, smoke_most_ex),
  ) %>% mutate(across(c(cigarettes, e_cigarettes, hookah, smoke_most), as.factor))
```

3.1 Table 1. Sample characteristics

```r
# Convert categorical variables to factors
data_cleaned <- data_cleaned %>%
  mutate(across(c(sex, ethnicity, marital_stat, residence,
                  cigarettes, e_cigarettes, hookah, smoke_most,
                  speciality),as.factor
                )
         )

# Build Table 1
tbl <- data_cleaned %>%
  tbl_summary(by        = smoke_stat,
              include   = c(age, sex, ethnicity, marital_stat, residence,
                            speciality,
                            cigarettes, e_cigarettes, hookah, smoke_most),
              type      = list(age ~ "continuous"),
              statistic = list(age ~ "{median} ({p25}, {p75})",
                               all_categorical() ~ "{n} ({p}%)"),
              missing = "ifany")

# Compare between groups (excluding field of study)
tbl <- tbl %>%
  add_p(test = list(age ~ "wilcox.test",
                    sex ~ "fisher.test",
                    ethnicity ~ "fisher.test",
                    marital_stat ~ "fisher.test",
                    residence ~ "fisher.test",
                    cigarettes ~ "fisher.test",
                    e_cigarettes ~ "fisher.test",
                    hookah ~ "fisher.test",
                    smoke_most ~ "fisher.test"
                    ),
        ) %>%

  # Ensure p-values have at least 3 decimal place
  modify_fmt_fun(p.value = function(x) style_pvalue(x, digits = 3)) %>%

  # Ensure numeric summaries have at least 1 decimal place
  modify_fmt_fun(all_continuous() ~ function(x) style_number(x, digits = 1),
                 all_categorical() ~ function(x) style_number(x, digits = 1))
```

```
## The following errors were returned during `modify_fmt_fun()`:
## x For variable `speciality` (`smoke_stat`) and "estimate", "p.value",
##   "conf.low", and "conf.high" statistics: FEXACT error 7(location). LDSTP=18480
##   is too small for this problem, (pastp=43.7097, ipn_0:=ipoin[itp=217]=2445,
##   stp[ipn_0]=39.4193). Increase workspace or consider using
##   'simulate.p.value=TRUE'
```

```r
# Display table (at end of pdf)
tbl
```

3.2 Sample geographic distribution

```r
result <- data_cleaned %>%
  count(province) %>%
  mutate(percentage = n / sum(n) * 100)

result
```

```
##        province   n percentage
## 1      Al-Anbar  28  2.8254289
## 2      Al-Basra  44  4.4399596
## 3   Al-Muthanna  18  1.8163471
## 4      Al-Najaf  59  5.9535822
## 5  Al-Qadisiyah  44  4.4399596
## 6         Babil  51  5.1463169
## 7       Baghdad 387 39.0514632
## 8       Dhi_Qar 112 11.3017154
## 9        Diyala  40  4.0363269
## 10        Duhok   1  0.1009082
## 11        Erbil   2  0.2018163
## 12      Karbala  63  6.3572149
## 13       Kirkuk  14  1.4127144
## 14       Maysan  14  1.4127144
## 15      Nineveh  36  3.6326942
## 16      Saladin  29  2.9263370
## 17  Sulaimaniya   2  0.2018163
## 18        Wasit  47  4.7426842
```

4 Item-item correlation matrix

4.1 Brief Wisconsin Inventory of Smoking Dependence Motives (WISDM-37)

```
# Store WISDM-37 item names
wisdm37_items <- c("auto1","auto2","auto3","auto4",
                   "loc1","loc2","loc3","loc4",
                   "tol1","tol2","tol3","tol4",
                   "craving1","craving2","craving3","craving4",
                   "taste1","taste2","taste3",
                   "cog_e1","cog_e2","cog_e3",
                   "w_control1","w_control2","w_control3",
                   "cue1","cue2","cue3",
                   "affect1","affect2","affect3",
                   "attach1","attach2","attach3",
                   "social1","social2","social3")

# Calculate Spearman's correlation matrix
wisdm37_cor <- as.data.frame(cor(data_cleaned[,wisdm37_items],
                                 method = "spearman",
                                 use    = "pairwise.complete.obs"))

# Write into an excel file
write_xlsx(wisdm37_cor, "WISDM-37 item-item correlation matrix.xlsx")
```

4.2 Reasons For Quitting (RFQ)

```
# Store RFQ item names
rfq_items <- c("hc1","hc2","hc3","hc4","hc5",
               "sc1","sc2","sc3","sc4","sc5",
               "ir1","ir2","ir3","ir4","ir5",
               "sp1","sp2","sp3","sp4","sp5")

# Calculate Spearman's correlation matrix
rfq_cor <- as.data.frame(cor(data_cleaned[,rfq_items],
                             method = "spearman",
                             use    = "pairwise.complete.obs"))

# Write into an excel file
write_xlsx(rfq_cor, "RFQ item-item correlation matrix.xlsx")
```

5 WISDM-37 psychometric analysis

5.1 Confirmatory factor analysis

```
df_smoker <- data_cleaned %>%
  filter(smoke_stat == "smoker")
```

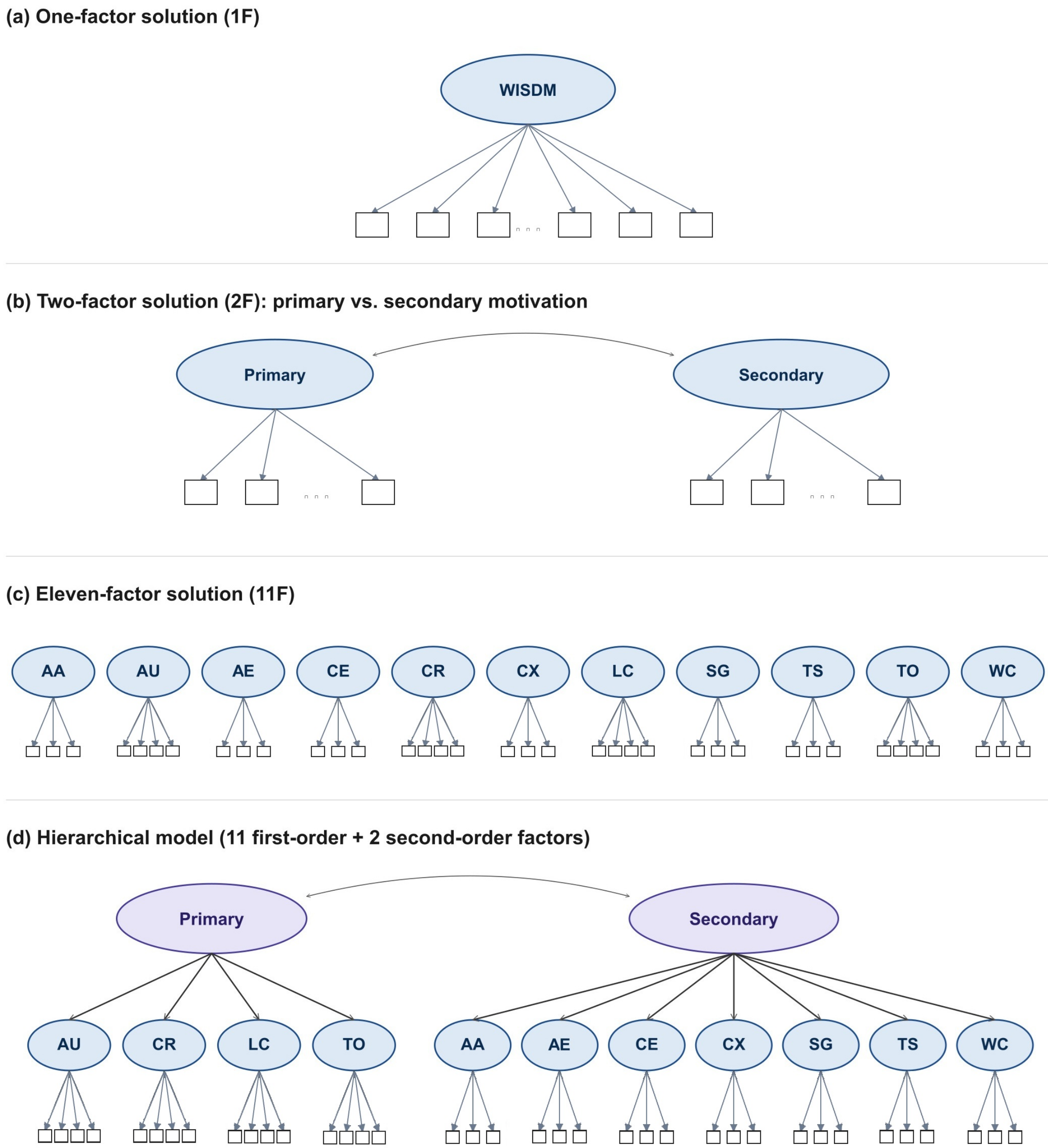


Figure 1: WISDM-37 models

5.1.1 MLR Model 1: 1-factor solution

```
cont_mod1 <-'f =~ auto1 + auto2 + auto3 + auto4 +
                  loc1 + loc2 + loc3 + loc4 +
                  tol1 + tol2 + tol3 + tol4 +
                  craving1 + craving2 + craving3 + craving4 +
                  taste1 + taste2 + taste3 +
                  cog_e1 + cog_e2 + cog_e3 +
                  w_control1 + w_control2 + w_control3 +
                  cue1 + cue2 + cue3 +
                  affect1 + affect2 + affect3 +
                  attach1 + attach2 + attach3 +
                  social1 + social2 + social3
                  '
cont_mod1_fit <- cfa(cont_mod1, data = df_smoker, estimator = "MLR")
summary(cont_mod1_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 40 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        74
##
##   Number of observations                           805
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                              6867.340    5706.204
##   Degrees of freedom                               629         629
##   P-value (Chi-square)                           0.000       0.000
##   Scaling correction factor                                  1.203
##     Yuan-Bentler correction (Mplus variant)
##
## Model Test Baseline Model:
##
##   Test statistic                             16892.465   13910.871
##   Degrees of freedom                               666         666
##   P-value                                        0.000       0.000
##   Scaling correction factor                                  1.214
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.616       0.617
##   Tucker-Lewis Index (TLI)                       0.593       0.594
##
##   Robust Comparative Fit Index (CFI)                         0.620
##   Robust Tucker-Lewis Index (TLI)                            0.598
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)             -58360.749  -58360.749
##   Scaling correction factor                                  1.039
##       for the MLR correction
```

```
##   Loglikelihood unrestricted model (H1)     -54927.079  -54927.079
##   Scaling correction factor                               1.186
##       for the MLR correction
##
##   Akaike (AIC)                              116869.497  116869.497
##   Bayesian (BIC)                            117216.620  117216.620
##   Sample-size adjusted Bayesian (SABIC)     116981.627  116981.627
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.111       0.100
##   90 Percent confidence interval - lower         0.109       0.098
##   90 Percent confidence interval - upper         0.113       0.102
##   P-value H_0: RMSEA <= 0.050                    0.000       0.000
##   P-value H_0: RMSEA >= 0.080                    1.000       1.000
##
##   Robust RMSEA                                               0.110
##   90 Percent confidence interval - lower                     0.107
##   90 Percent confidence interval - upper                     0.112
##   P-value H_0: Robust RMSEA <= 0.050                         0.000
##   P-value H_0: Robust RMSEA >= 0.080                         1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.090       0.090
##
## Parameter Estimates:
##
##   Standard errors                             Sandwich
##   Information bread                           Observed
##   Observed information based on                Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   f =~
##     auto1             1.000                               0.855    0.446
##     auto2             1.525    0.108   14.106    0.000    1.303    0.613
##     auto3             1.583    0.109   14.558    0.000    1.353    0.646
##     auto4             1.702    0.124   13.746    0.000    1.454    0.682
##     loc1              1.665    0.141   11.811    0.000    1.423    0.701
##     loc2              1.925    0.164   11.715    0.000    1.645    0.725
##     loc3              1.833    0.158   11.579    0.000    1.567    0.788
##     loc4              1.985    0.166   11.948    0.000    1.697    0.769
##     tol1              2.065    0.172   11.981    0.000    1.765    0.696
##     tol2              1.750    0.152   11.488    0.000    1.496    0.704
##     tol3              1.977    0.171   11.574    0.000    1.689    0.660
##     tol4              2.021    0.172   11.777    0.000    1.727    0.766
##     craving1          1.650    0.157   10.487    0.000    1.410    0.688
##     craving2          0.864    0.098    8.851    0.000    0.738    0.541
##     craving3          1.389    0.141    9.865    0.000    1.187    0.694
##     craving4          1.980    0.171   11.557    0.000    1.692    0.772
##     taste1            0.375    0.090    4.174    0.000    0.320    0.162
##     taste2            0.464    0.092    5.039    0.000    0.396    0.206
##     taste3            0.188    0.093    2.027    0.043    0.160    0.079
```

```
##     cog_e1            1.326    0.130   10.178    0.000    1.133    0.522
##     cog_e2            1.398    0.139   10.033    0.000    1.195    0.552
##     cog_e3            1.422    0.140   10.165    0.000    1.215    0.602
##     w_control1        0.949    0.103    9.243    0.000    0.811    0.391
##     w_control2        0.548    0.082    6.715    0.000    0.468    0.263
##     w_control3        1.087    0.121    8.956    0.000    0.929    0.427
##     cue1              1.239    0.119   10.410    0.000    1.059    0.471
##     cue2              0.749    0.103    7.246    0.000    0.640    0.285
##     cue3              0.886    0.106    8.369    0.000    0.757    0.431
##     affect1           1.146    0.127    9.011    0.000    0.980    0.511
##     affect2           1.064    0.123    8.631    0.000    0.909    0.540
##     affect3           0.896    0.104    8.600    0.000    0.765    0.457
##     attach1           1.684    0.153   11.036    0.000    1.439    0.694
##     attach2           1.678    0.153   10.999    0.000    1.434    0.646
##     attach3           1.786    0.155   11.547    0.000    1.526    0.686
##     social1           0.524    0.096    5.480    0.000    0.448    0.242
##     social2           0.523    0.101    5.165    0.000    0.447    0.216
##     social3           0.537    0.099    5.435    0.000    0.459    0.250
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .auto1             2.943    0.152   19.421    0.000    2.943    0.801
##    .auto2             2.822    0.170   16.574    0.000    2.822    0.624
##    .auto3             2.559    0.160   15.996    0.000    2.559    0.583
##    .auto4             2.430    0.159   15.278    0.000    2.430    0.535
##    .loc1              2.095    0.123   17.090    0.000    2.095    0.508
##    .loc2              2.445    0.146   16.794    0.000    2.445    0.475
##    .loc3              1.495    0.097   15.396    0.000    1.495    0.379
##    .loc4              1.988    0.116   17.133    0.000    1.988    0.409
##    .tol1              3.309    0.177   18.658    0.000    3.309    0.515
##    .tol2              2.283    0.137   16.689    0.000    2.283    0.505
##    .tol3              3.690    0.200   18.486    0.000    3.690    0.564
##    .tol4              2.094    0.125   16.744    0.000    2.094    0.413
##    .craving1          2.210    0.137   16.181    0.000    2.210    0.526
##    .craving2          1.317    0.105   12.560    0.000    1.317    0.707
##    .craving3          1.518    0.102   14.895    0.000    1.518    0.519
##    .craving4          1.944    0.123   15.837    0.000    1.944    0.404
##    .taste1            3.788    0.157   24.089    0.000    3.788    0.974
##    .taste2            3.547    0.161   22.085    0.000    3.547    0.958
##    .taste3            4.146    0.170   24.373    0.000    4.146    0.994
##    .cog_e1            3.423    0.170   20.146    0.000    3.423    0.727
##    .cog_e2            3.252    0.165   19.753    0.000    3.252    0.695
##    .cog_e3            2.600    0.144   17.992    0.000    2.600    0.638
##    .w_control1        3.644    0.160   22.796    0.000    3.644    0.847
##    .w_control2        2.949    0.171   17.222    0.000    2.949    0.931
##    .w_control3        3.877    0.154   25.171    0.000    3.877    0.818
##    .cue1              3.937    0.159   24.740    0.000    3.937    0.778
##    .cue2              4.626    0.145   31.802    0.000    4.626    0.919
##    .cue3              2.519    0.161   15.687    0.000    2.519    0.815
##    .affect1           2.721    0.156   17.475    0.000    2.721    0.739
##    .affect2           2.007    0.127   15.838    0.000    2.007    0.708
##    .affect3           2.222    0.132   16.844    0.000    2.222    0.791
##    .attach1           2.224    0.150   14.849    0.000    2.224    0.518
##    .attach2           2.877    0.165   17.465    0.000    2.877    0.583
```

```
##    .attach3           2.622    0.161   16.238    0.000    2.622    0.530
##    .social1           3.226    0.152   21.171    0.000    3.226    0.941
##    .social2           4.087    0.148   27.695    0.000    4.087    0.953
##    .social3           3.163    0.151   20.949    0.000    3.163    0.938
##     f                 0.730    0.120    6.088    0.000    1.000    1.000
```

5.1.2 MLR Model 2: 2-factor solution

```
cont_mod2 <- 'pdm =~ auto1 + auto2 + auto3 + auto4 +
                     loc1 + loc2 + loc3 + loc4 +
                     tol1 + tol2 + tol3 + tol4 +
                     craving1 + craving2 + craving3 + craving4

              sdm =~ taste1 + taste2 + taste3 +
                     cog_e1 + cog_e2 + cog_e3 +
                     w_control1 + w_control2 + w_control3 +
                     cue1 + cue2 + cue3 +
                     affect1 + affect2 + affect3 +
                     attach1 + attach2 + attach3 +
                     social1 + social2 +social3
                     '
cont_mod2_fit <- cfa(cont_mod2, data = df_smoker, estimator = "MLR")
summary(cont_mod2_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 76 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        75
##
##   Number of observations                           805
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                              6099.672    5085.137
##   Degrees of freedom                               628         628
##   P-value (Chi-square)                           0.000       0.000
##   Scaling correction factor                                  1.200
##     Yuan-Bentler correction (Mplus variant)
##
## Model Test Baseline Model:
##
##   Test statistic                             16892.465   13910.871
##   Degrees of freedom                               666         666
##   P-value                                        0.000       0.000
##   Scaling correction factor                                  1.214
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.663       0.663
##   Tucker-Lewis Index (TLI)                       0.642       0.643
##
##   Robust Comparative Fit Index (CFI)                         0.668
##   Robust Tucker-Lewis Index (TLI)                            0.647
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)             -57976.915  -57976.915
##   Scaling correction factor                                  1.075
```

```
##       for the MLR correction
##   Loglikelihood unrestricted model (H1)     -54927.079  -54927.079
##   Scaling correction factor                                  1.186
##       for the MLR correction
##
##   Akaike (AIC)                              116103.829  116103.829
##   Bayesian (BIC)                            116455.643  116455.643
##   Sample-size adjusted Bayesian (SABIC)     116217.475  116217.475
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.104       0.094
##   90 Percent confidence interval - lower         0.102       0.092
##   90 Percent confidence interval - upper         0.106       0.096
##   P-value H_0: RMSEA <= 0.050                    0.000       0.000
##   P-value H_0: RMSEA >= 0.080                    1.000       1.000
##
##   Robust RMSEA                                               0.103
##   90 Percent confidence interval - lower                     0.100
##   90 Percent confidence interval - upper                     0.105
##   P-value H_0: Robust RMSEA <= 0.050                         0.000
##   P-value H_0: Robust RMSEA >= 0.080                         1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.085       0.085
##
## Parameter Estimates:
##
##   Standard errors                             Sandwich
##   Information bread                           Observed
##   Observed information based on                Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   pdm =~
##     auto1             1.000                               0.890    0.464
##     auto2             1.502    0.104   14.456    0.000    1.337    0.629
##     auto3             1.560    0.105   14.889    0.000    1.388    0.663
##     auto4             1.682    0.118   14.217    0.000    1.497    0.702
##     loc1              1.659    0.137   12.102    0.000    1.477    0.727
##     loc2              1.869    0.157   11.927    0.000    1.663    0.733
##     loc3              1.754    0.151   11.645    0.000    1.561    0.786
##     loc4              1.994    0.163   12.203    0.000    1.775    0.804
##     tol1              2.012    0.165   12.176    0.000    1.790    0.706
##     tol2              1.724    0.148   11.619    0.000    1.534    0.722
##     tol3              1.927    0.164   11.740    0.000    1.715    0.670
##     tol4              2.022    0.169   11.963    0.000    1.800    0.799
##     craving1          1.603    0.150   10.683    0.000    1.426    0.696
##     craving2          0.795    0.089    8.882    0.000    0.707    0.518
##     craving3          1.304    0.130   10.016    0.000    1.160    0.678
##     craving4          1.930    0.164   11.744    0.000    1.718    0.783
##   sdm =~
##     taste1            1.000                               0.475    0.241
```

```
##     taste2            1.179    0.150    7.847    0.000    0.560    0.291
##     taste3            0.744    0.137    5.432    0.000    0.354    0.173
##     cog_e1            2.901    0.508    5.711    0.000    1.378    0.635
##     cog_e2            3.100    0.540    5.742    0.000    1.472    0.681
##     cog_e3            3.094    0.524    5.904    0.000    1.470    0.728
##     w_control1        1.945    0.346    5.623    0.000    0.924    0.446
##     w_control2        1.236    0.242    5.112    0.000    0.587    0.330
##     w_control3        2.079    0.379    5.486    0.000    0.987    0.454
##     cue1              2.513    0.446    5.632    0.000    1.194    0.531
##     cue2              1.569    0.301    5.212    0.000    0.745    0.332
##     cue3              1.573    0.294    5.352    0.000    0.747    0.425
##     affect1           2.533    0.413    6.140    0.000    1.203    0.627
##     affect2           2.365    0.370    6.394    0.000    1.123    0.667
##     affect3           1.881    0.289    6.508    0.000    0.894    0.533
##     attach1           2.930    0.514    5.704    0.000    1.392    0.671
##     attach2           3.136    0.522    6.013    0.000    1.490    0.671
##     attach3           3.257    0.547    5.959    0.000    1.547    0.695
##     social1           0.812    0.224    3.618    0.000    0.386    0.208
##     social2           0.951    0.240    3.957    0.000    0.452    0.218
##     social3           0.926    0.231    4.002    0.000    0.440    0.239
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   pdm ~~
##     sdm               0.332    0.061    5.477    0.000    0.786    0.786
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .auto1             2.882    0.153   18.807    0.000    2.882    0.784
##    .auto2             2.733    0.173   15.828    0.000    2.733    0.605
##    .auto3             2.463    0.162   15.183    0.000    2.463    0.561
##    .auto4             2.306    0.158   14.555    0.000    2.306    0.507
##    .loc1              1.940    0.119   16.265    0.000    1.940    0.471
##    .loc2              2.387    0.149   15.992    0.000    2.387    0.463
##    .loc3              1.512    0.103   14.650    0.000    1.512    0.383
##    .loc4              1.718    0.109   15.826    0.000    1.718    0.353
##    .tol1              3.220    0.183   17.598    0.000    3.220    0.501
##    .tol2              2.166    0.143   15.174    0.000    2.166    0.479
##    .tol3              3.602    0.205   17.565    0.000    3.602    0.550
##    .tol4              1.838    0.126   14.568    0.000    1.838    0.362
##    .craving1          2.163    0.134   16.158    0.000    2.163    0.515
##    .craving2          1.362    0.108   12.589    0.000    1.362    0.731
##    .craving3          1.582    0.105   15.119    0.000    1.582    0.540
##    .craving4          1.856    0.123   15.049    0.000    1.856    0.386
##    .taste1            3.664    0.160   22.931    0.000    3.664    0.942
##    .taste2            3.390    0.166   20.444    0.000    3.390    0.915
##    .taste3            4.047    0.173   23.407    0.000    4.047    0.970
##    .cog_e1            2.808    0.181   15.486    0.000    2.808    0.597
##    .cog_e2            2.512    0.181   13.902    0.000    2.512    0.537
##    .cog_e3            1.917    0.146   13.125    0.000    1.917    0.470
##    .w_control1        3.448    0.153   22.527    0.000    3.448    0.802
##    .w_control2        2.823    0.158   17.898    0.000    2.823    0.891
##    .w_control3        3.764    0.154   24.395    0.000    3.764    0.794
##    .cue1              3.634    0.168   21.663    0.000    3.634    0.718
```

```
##    .cue2              4.480    0.153   29.192    0.000    4.480    0.890
##    .cue3              2.534    0.161   15.704    0.000    2.534    0.819
##    .affect1           2.233    0.148   15.036    0.000    2.233    0.607
##    .affect2           1.571    0.105   15.031    0.000    1.571    0.555
##    .affect3           2.009    0.135   14.927    0.000    2.009    0.716
##    .attach1           2.359    0.157   14.998    0.000    2.359    0.549
##    .attach2           2.714    0.171   15.833    0.000    2.714    0.550
##    .attach3           2.558    0.171   15.001    0.000    2.558    0.517
##    .social1           3.278    0.158   20.769    0.000    3.278    0.957
##    .social2           4.083    0.151   27.091    0.000    4.083    0.952
##    .social3           3.180    0.156   20.432    0.000    3.180    0.943
##     pdm               0.792    0.126    6.299    0.000    1.000    1.000
##     sdm               0.226    0.074    3.063    0.002    1.000    1.000
```

5.1.3 MLR Model 3: 11-factor solution

```
cont_mod3 <-'#Primary dependence motives
             auto      =~ auto1 + auto2 + auto3 + auto4
             loc       =~ loc1 + loc2 + loc3 + loc4
             tol       =~ tol1 + tol2 + tol3 + tol4
             craving   =~ craving1 + craving2 + craving3 + craving4

             #Secondary dependence motives
             taste     =~ taste1 + taste2 + taste3
             cog_e     =~ cog_e1 + cog_e2 + cog_e3
             w_control =~ w_control1 + w_control2 + w_control3
             cue       =~ cue1 + cue2 + cue3
             affect    =~ affect1 + affect2 + affect3
             attach    =~ attach1 + attach2 + attach3
             social    =~ social1 + social2 +social3
             '
cont_mod3_fit <- cfa(cont_mod3, data = df_smoker, estimator = "MLR")
summary(cont_mod3_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 106 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                       129
##
##   Number of observations                           805
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                              2210.231    1880.598
##   Degrees of freedom                               574         574
##   P-value (Chi-square)                           0.000       0.000
##   Scaling correction factor                                  1.175
##     Yuan-Bentler correction (Mplus variant)
##
## Model Test Baseline Model:
##
##   Test statistic                             16892.465   13910.871
##   Degrees of freedom                               666         666
##   P-value                                        0.000       0.000
##   Scaling correction factor                                  1.214
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.899       0.901
##   Tucker-Lewis Index (TLI)                       0.883       0.886
##
##   Robust Comparative Fit Index (CFI)                         0.905
##   Robust Tucker-Lewis Index (TLI)                            0.889
##
## Loglikelihood and Information Criteria:
##
```

```
##   Loglikelihood user model (H0)            -56032.194  -56032.194
##   Scaling correction factor                                 1.235
##       for the MLR correction
##   Loglikelihood unrestricted model (H1)     -54927.079  -54927.079
##   Scaling correction factor                                 1.186
##       for the MLR correction
##
##   Akaike (AIC)                              112322.388  112322.388
##   Bayesian (BIC)                            112927.507  112927.507
##   Sample-size adjusted Bayesian (SABIC)     112517.858  112517.858
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.060       0.053
##   90 Percent confidence interval - lower         0.057       0.051
##   90 Percent confidence interval - upper         0.062       0.056
##   P-value H_0: RMSEA <= 0.050                    0.000       0.017
##   P-value H_0: RMSEA >= 0.080                    0.000       0.000
##
##   Robust RMSEA                                               0.058
##   90 Percent confidence interval - lower                     0.055
##   90 Percent confidence interval - upper                     0.061
##   P-value H_0: Robust RMSEA <= 0.050                         0.000
##   P-value H_0: Robust RMSEA >= 0.080                         0.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.044       0.044
##
## Parameter Estimates:
##
##   Standard errors                             Sandwich
##   Information bread                           Observed
##   Observed information based on                Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   auto =~
##     auto1             1.000                               1.174    0.612
##     auto2             1.449    0.084   17.342    0.000    1.701    0.800
##     auto3             1.550    0.083   18.623    0.000    1.819    0.868
##     auto4             1.558    0.087   17.922    0.000    1.828    0.858
##   loc =~
##     loc1              1.000                               1.497    0.737
##     loc2              1.131    0.041   27.832    0.000    1.692    0.745
##     loc3              1.067    0.042   25.103    0.000    1.596    0.803
##     loc4              1.185    0.049   24.345    0.000    1.774    0.804
##   tol =~
##     tol1              1.000                               1.835    0.724
##     tol2              0.928    0.058   16.041    0.000    1.703    0.801
##     tol3              0.977    0.028   35.404    0.000    1.793    0.701
##     tol4              1.073    0.058   18.395    0.000    1.970    0.874
##   craving =~
##     craving1          1.000                               1.509    0.737
```

```
##     craving2          0.542    0.038   14.378    0.000    0.818    0.600
##     craving3          0.849    0.042   20.084    0.000    1.281    0.749
##     craving4          1.155    0.049   23.441    0.000    1.743    0.795
##   taste =~
##     taste1            1.000                               1.383    0.701
##     taste2            1.185    0.070   16.976    0.000    1.639    0.851
##     taste3            1.155    0.069   16.739    0.000    1.598    0.782
##   cog_e =~
##     cog_e1            1.000                               1.674    0.771
##     cog_e2            1.111    0.036   30.511    0.000    1.860    0.860
##     cog_e3            1.022    0.048   21.110    0.000    1.711    0.847
##   w_control =~
##     w_control1        1.000                               1.419    0.684
##     w_control2        0.721    0.061   11.826    0.000    1.023    0.575
##     w_control3        1.034    0.088   11.704    0.000    1.467    0.674
##   cue =~
##     cue1              1.000                               1.285    0.571
##     cue2              0.695    0.079    8.826    0.000    0.893    0.398
##     cue3              0.725    0.090    8.055    0.000    0.932    0.530
##   affect =~
##     affect1           1.000                               1.432    0.746
##     affect2           0.955    0.048   19.932    0.000    1.366    0.812
##     affect3           0.720    0.051   14.091    0.000    1.031    0.615
##   attach =~
##     attach1           1.000                               1.525    0.736
##     attach2           1.076    0.048   22.289    0.000    1.641    0.739
##     attach3           1.089    0.056   19.532    0.000    1.661    0.746
##   social =~
##     social1           1.000                               1.331    0.719
##     social2           1.134    0.064   17.758    0.000    1.510    0.729
##     social3           1.283    0.068   18.784    0.000    1.708    0.930
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   auto ~~
##     loc               1.303    0.106   12.332    0.000    0.742    0.742
##     tol               1.443    0.134   10.775    0.000    0.670    0.670
##     craving           1.133    0.099   11.403    0.000    0.640    0.640
##     taste             0.250    0.073    3.454    0.001    0.154    0.154
##     cog_e             0.847    0.105    8.051    0.000    0.431    0.431
##     w_control         0.638    0.092    6.915    0.000    0.383    0.383
##     cue               0.922    0.107    8.601    0.000    0.611    0.611
##     affect            0.612    0.089    6.838    0.000    0.364    0.364
##     attach            1.172    0.114   10.269    0.000    0.655    0.655
##     social            0.328    0.075    4.401    0.000    0.210    0.210
##   loc ~~
##     tol               2.478    0.163   15.236    0.000    0.902    0.902
##     craving           2.159    0.140   15.390    0.000    0.956    0.956
##     taste             0.237    0.090    2.623    0.009    0.115    0.115
##     cog_e             1.415    0.127   11.171    0.000    0.565    0.565
##     w_control         1.049    0.107    9.798    0.000    0.494    0.494
##     cue               1.348    0.131   10.258    0.000    0.701    0.701
##     affect            1.238    0.117   10.571    0.000    0.578    0.578
##     attach            1.908    0.145   13.170    0.000    0.836    0.836
```

```
##     social            0.509    0.094    5.432    0.000    0.256    0.256
##   tol ~~
##     craving           2.267    0.172   13.182    0.000    0.819    0.819
##     taste             0.243    0.110    2.212    0.027    0.096    0.096
##     cog_e             1.676    0.171    9.814    0.000    0.546    0.546
##     w_control         1.297    0.149    8.682    0.000    0.498    0.498
##     cue               1.367    0.163    8.411    0.000    0.580    0.580
##     affect            1.352    0.143    9.422    0.000    0.515    0.515
##     attach            2.157    0.178   12.122    0.000    0.771    0.771
##     social            0.527    0.104    5.053    0.000    0.216    0.216
##   craving ~~
##     taste             0.283    0.095    2.975    0.003    0.135    0.135
##     cog_e             1.432    0.129   11.095    0.000    0.567    0.567
##     w_control         0.941    0.108    8.736    0.000    0.439    0.439
##     cue               1.465    0.130   11.268    0.000    0.755    0.755
##     affect            1.400    0.125   11.156    0.000    0.648    0.648
##     attach            1.994    0.153   13.031    0.000    0.867    0.867
##     social            0.558    0.096    5.818    0.000    0.278    0.278
##   taste ~~
##     cog_e             0.364    0.104    3.490    0.000    0.157    0.157
##     w_control         0.418    0.097    4.306    0.000    0.213    0.213
##     cue               0.468    0.103    4.536    0.000    0.263    0.263
##     affect            0.672    0.098    6.866    0.000    0.339    0.339
##     attach            0.542    0.099    5.459    0.000    0.257    0.257
##     social            0.207    0.076    2.719    0.007    0.112    0.112
##   cog_e ~~
##     w_control         1.264    0.133    9.480    0.000    0.532    0.532
##     cue               1.248    0.159    7.856    0.000    0.581    0.581
##     affect            1.651    0.137   12.070    0.000    0.689    0.689
##     attach            1.636    0.150   10.884    0.000    0.641    0.641
##     social            0.260    0.099    2.620    0.009    0.117    0.117
##   w_control ~~
##     cue               1.145    0.163    7.034    0.000    0.628    0.628
##     affect            0.865    0.106    8.122    0.000    0.426    0.426
##     attach            1.091    0.117    9.327    0.000    0.504    0.504
##     social            0.383    0.089    4.310    0.000    0.203    0.203
##   cue ~~
##     affect            1.282    0.147    8.738    0.000    0.697    0.697
##     attach            1.623    0.158   10.265    0.000    0.828    0.828
##     social            0.759    0.100    7.559    0.000    0.444    0.444
##   affect ~~
##     attach            1.572    0.137   11.474    0.000    0.720    0.720
##     social            0.203    0.086    2.362    0.018    0.107    0.107
##   attach ~~
##     social            0.481    0.101    4.769    0.000    0.237    0.237
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .auto1             2.296    0.152   15.106    0.000    2.296    0.625
##    .auto2             1.626    0.143   11.350    0.000    1.626    0.360
##    .auto3             1.080    0.112    9.680    0.000    1.080    0.246
##    .auto4             1.202    0.132    9.095    0.000    1.202    0.264
##    .loc1              1.881    0.130   14.480    0.000    1.881    0.457
##    .loc2              2.290    0.159   14.379    0.000    2.290    0.444
```

```
##    .loc3              1.401    0.107   13.055    0.000    1.401    0.355
##    .loc4              1.721    0.125   13.801    0.000    1.721    0.354
##    .tol1              3.057    0.252   12.107    0.000    3.057    0.476
##    .tol2              1.621    0.166    9.746    0.000    1.621    0.359
##    .tol3              3.327    0.274   12.130    0.000    3.327    0.509
##    .tol4              1.197    0.148    8.071    0.000    1.197    0.236
##    .craving1          1.920    0.140   13.760    0.000    1.920    0.457
##    .craving2          1.192    0.100   11.870    0.000    1.192    0.640
##    .craving3          1.286    0.107   11.962    0.000    1.286    0.439
##    .craving4          1.769    0.144   12.246    0.000    1.769    0.368
##    .taste1            1.977    0.151   13.113    0.000    1.977    0.508
##    .taste2            1.019    0.161    6.309    0.000    1.019    0.275
##    .taste3            1.620    0.182    8.919    0.000    1.620    0.388
##    .cog_e1            1.906    0.164   11.600    0.000    1.906    0.405
##    .cog_e2            1.222    0.148    8.281    0.000    1.222    0.261
##    .cog_e3            1.151    0.129    8.952    0.000    1.151    0.282
##    .w_control1        2.287    0.217   10.530    0.000    2.287    0.532
##    .w_control2        2.121    0.146   14.514    0.000    2.121    0.669
##    .w_control3        2.587    0.215   12.054    0.000    2.587    0.546
##    .cue1              3.408    0.220   15.486    0.000    3.408    0.674
##    .cue2              4.238    0.196   21.662    0.000    4.238    0.841
##    .cue3              2.224    0.173   12.857    0.000    2.224    0.719
##    .affect1           1.631    0.147   11.120    0.000    1.631    0.443
##    .affect2           0.966    0.099    9.802    0.000    0.966    0.341
##    .affect3           1.744    0.145   12.049    0.000    1.744    0.621
##    .attach1           1.970    0.165   11.950    0.000    1.970    0.459
##    .attach2           2.241    0.167   13.446    0.000    2.241    0.454
##    .attach3           2.194    0.168   13.055    0.000    2.194    0.443
##    .social1           1.655    0.129   12.859    0.000    1.655    0.483
##    .social2           2.008    0.176   11.429    0.000    2.008    0.468
##    .social3           0.455    0.130    3.497    0.000    0.455    0.135
##     auto              1.378    0.152    9.067    0.000    1.000    1.000
##     loc               2.240    0.168   13.325    0.000    1.000    1.000
##     tol               3.367    0.277   12.151    0.000    1.000    1.000
##     craving           2.278    0.181   12.613    0.000    1.000    1.000
##     taste             1.913    0.184   10.377    0.000    1.000    1.000
##     cog_e             2.801    0.208   13.471    0.000    1.000    1.000
##     w_control         2.015    0.235    8.564    0.000    1.000    1.000
##     cue               1.651    0.228    7.251    0.000    1.000    1.000
##     affect            2.049    0.180   11.358    0.000    1.000    1.000
##     attach            2.325    0.211   11.044    0.000    1.000    1.000
##     social            1.772    0.172   10.307    0.000    1.000    1.000
```

5.1.4 MLR Model 4: Hierarchial factor solution

```
cont_mod4 <- '#1st order
             auto      =~ auto1 + auto2 + auto3 + auto4
             loc       =~ loc1 + loc2 + loc3 + loc4
             tol       =~ tol1 + tol2 + tol3 + tol4
             craving   =~ craving1 + craving2 + craving3 + craving4
             taste     =~ taste1 + taste2 + taste3
             cog_e     =~ cog_e1 + cog_e2 + cog_e3
             w_control =~ w_control1 + w_control2 + w_control3
             cue       =~ cue1 + cue2 + cue3
             affect    =~ affect1 + affect2 + affect3
             attach    =~ attach1 + attach2 + attach3
             social    =~ social1 + social2 +social3

             #2nd order
             pdm =~ auto + loc + tol + craving
             sdm =~ taste + cog_e + w_control + cue + affect + attach + social
             '
cont_mod4_fit <- cfa(cont_mod4, data = df_smoker, estimator = "MLR")
```

```
## Warning: lavaan->lav_object_post_check():
##    some estimated lv variances are negative
```

```
summary(cont_mod4_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 90 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        86
##
##   Number of observations                           805
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                              2520.879    2139.813
##   Degrees of freedom                               617         617
##   P-value (Chi-square)                           0.000       0.000
##   Scaling correction factor                                  1.178
##     Yuan-Bentler correction (Mplus variant)
##
## Model Test Baseline Model:
##
##   Test statistic                             16892.465   13910.871
##   Degrees of freedom                               666         666
##   P-value                                        0.000       0.000
##   Scaling correction factor                                  1.214
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.883       0.885
```

```
##   Tucker-Lewis Index (TLI)                       0.873       0.876
##
##   Robust Comparative Fit Index (CFI)                         0.888
##   Robust Tucker-Lewis Index (TLI)                            0.880
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)             -56187.518  -56187.518
##   Scaling correction factor                                  1.245
##       for the MLR correction
##   Loglikelihood unrestricted model (H1)     -54927.079  -54927.079
##   Scaling correction factor                                  1.186
##       for the MLR correction
##
##   Akaike (AIC)                              112547.037  112547.037
##   Bayesian (BIC)                            112950.449  112950.449
##   Sample-size adjusted Bayesian (SABIC)     112677.350  112677.350
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.062       0.055
##   90 Percent confidence interval - lower         0.059       0.053
##   90 Percent confidence interval - upper         0.064       0.058
##   P-value H_0: RMSEA <= 0.050                    0.000       0.000
##   P-value H_0: RMSEA >= 0.080                    0.000       0.000
##
##   Robust RMSEA                                               0.060
##   90 Percent confidence interval - lower                     0.057
##   90 Percent confidence interval - upper                     0.063
##   P-value H_0: Robust RMSEA <= 0.050                         0.000
##   P-value H_0: Robust RMSEA >= 0.080                         0.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.055       0.055
##
## Parameter Estimates:
##
##   Standard errors                             Sandwich
##   Information bread                           Observed
##   Observed information based on                Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   auto =~
##     auto1             1.000                               1.168    0.609
##     auto2             1.459    0.084   17.319    0.000    1.704    0.801
##     auto3             1.559    0.084   18.530    0.000    1.821    0.869
##     auto4             1.564    0.088   17.852    0.000    1.827    0.857
##   loc =~
##     loc1              1.000                               1.492    0.735
##     loc2              1.137    0.040   28.166    0.000    1.696    0.747
##     loc3              1.078    0.042   25.551    0.000    1.608    0.809
##     loc4              1.180    0.047   25.125    0.000    1.761    0.798
```

```
##   tol =~
##     tol1              1.000                               1.835    0.724
##     tol2              0.930    0.056   16.570    0.000    1.708    0.803
##     tol3              0.973    0.027   36.005    0.000    1.785    0.698
##     tol4              1.073    0.056   19.242    0.000    1.970    0.874
##   craving =~
##     craving1          1.000                               1.511    0.737
##     craving2          0.523    0.036   14.571    0.000    0.790    0.579
##     craving3          0.835    0.042   19.914    0.000    1.261    0.737
##     craving4          1.177    0.050   23.744    0.000    1.779    0.811
##   taste =~
##     taste1            1.000                               1.382    0.701
##     taste2            1.194    0.073   16.240    0.000    1.650    0.857
##     taste3            1.146    0.068   16.859    0.000    1.585    0.776
##   cog_e =~
##     cog_e1            1.000                               1.688    0.778
##     cog_e2            1.117    0.038   29.650    0.000    1.885    0.871
##     cog_e3            0.996    0.045   22.091    0.000    1.681    0.833
##   w_control =~
##     w_control1        1.000                               1.387    0.669
##     w_control2        0.747    0.061   12.220    0.000    1.036    0.582
##     w_control3        1.074    0.092   11.676    0.000    1.489    0.684
##   cue =~
##     cue1              1.000                               1.331    0.592
##     cue2              0.692    0.074    9.390    0.000    0.921    0.411
##     cue3              0.662    0.074    8.983    0.000    0.881    0.501
##   affect =~
##     affect1           1.000                               1.431    0.746
##     affect2           0.943    0.049   19.068    0.000    1.349    0.801
##     affect3           0.738    0.053   14.036    0.000    1.056    0.630
##   attach =~
##     attach1           1.000                               1.506    0.727
##     attach2           1.097    0.050   21.900    0.000    1.652    0.744
##     attach3           1.110    0.058   19.172    0.000    1.672    0.751
##   social =~
##     social1           1.000                               1.311    0.708
##     social2           1.140    0.063   18.043    0.000    1.495    0.722
##     social3           1.322    0.071   18.520    0.000    1.734    0.944
##   pdm =~
##     auto              1.000                               0.728    0.728
##     loc               1.761    0.124   14.256    0.000    1.003    1.003
##     tol               1.921    0.144   13.375    0.000    0.890    0.890
##     craving           1.691    0.133   12.741    0.000    0.951    0.951
##   sdm =~
##     taste             1.000                               0.260    0.260
##     cog_e             3.315    0.574    5.773    0.000    0.706    0.706
##     w_control         2.299    0.416    5.532    0.000    0.596    0.596
##     cue               3.247    0.592    5.481    0.000    0.877    0.877
##     affect            3.053    0.490    6.226    0.000    0.767    0.767
##     attach            4.020    0.704    5.713    0.000    0.959    0.959
##     social            0.972    0.228    4.265    0.000    0.266    0.266
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
```

```
##   pdm ~~
##     sdm               0.260    0.046    5.592    0.000    0.851    0.851
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .auto1             2.310    0.153   15.128    0.000    2.310    0.629
##    .auto2             1.617    0.144   11.242    0.000    1.617    0.358
##    .auto3             1.074    0.111    9.661    0.000    1.074    0.245
##    .auto4             1.206    0.133    9.065    0.000    1.206    0.265
##    .loc1              1.895    0.130   14.576    0.000    1.895    0.460
##    .loc2              2.276    0.157   14.519    0.000    2.276    0.442
##    .loc3              1.363    0.102   13.380    0.000    1.363    0.345
##    .loc4              1.767    0.119   14.791    0.000    1.767    0.363
##    .tol1              3.056    0.245   12.474    0.000    3.056    0.476
##    .tol2              1.604    0.162    9.914    0.000    1.604    0.355
##    .tol3              3.357    0.266   12.631    0.000    3.357    0.513
##    .tol4              1.196    0.140    8.525    0.000    1.196    0.236
##    .craving1          1.915    0.139   13.817    0.000    1.915    0.456
##    .craving2          1.237    0.103   12.027    0.000    1.237    0.664
##    .craving3          1.337    0.108   12.388    0.000    1.337    0.457
##    .craving4          1.642    0.130   12.581    0.000    1.642    0.342
##    .taste1            1.979    0.154   12.888    0.000    1.979    0.509
##    .taste2            0.982    0.166    5.929    0.000    0.982    0.265
##    .taste3            1.660    0.184    9.032    0.000    1.660    0.398
##    .cog_e1            1.859    0.159   11.678    0.000    1.859    0.395
##    .cog_e2            1.129    0.139    8.103    0.000    1.129    0.241
##    .cog_e3            1.250    0.128    9.802    0.000    1.250    0.307
##    .w_control1        2.379    0.215   11.062    0.000    2.379    0.553
##    .w_control2        2.095    0.140   14.927    0.000    2.095    0.661
##    .w_control3        2.521    0.219   11.533    0.000    2.521    0.532
##    .cue1              3.287    0.208   15.791    0.000    3.287    0.650
##    .cue2              4.187    0.194   21.592    0.000    4.187    0.831
##    .cue3              2.316    0.168   13.820    0.000    2.316    0.749
##    .affect1           1.634    0.150   10.888    0.000    1.634    0.444
##    .affect2           1.014    0.108    9.404    0.000    1.014    0.358
##    .affect3           1.693    0.144   11.718    0.000    1.693    0.603
##    .attach1           2.026    0.163   12.428    0.000    2.026    0.472
##    .attach2           2.203    0.168   13.088    0.000    2.203    0.447
##    .attach3           2.157    0.170   12.658    0.000    2.157    0.436
##    .social1           1.707    0.128   13.289    0.000    1.707    0.498
##    .social2           2.053    0.181   11.363    0.000    2.053    0.479
##    .social3           0.367    0.137    2.673    0.008    0.367    0.109
##    .auto              0.642    0.085    7.583    0.000    0.471    0.471
##    .loc              -0.015    0.042   -0.346    0.730   -0.007   -0.007
##    .tol               0.702    0.095    7.415    0.000    0.208    0.208
##    .craving           0.218    0.054    4.034    0.000    0.096    0.096
##    .taste             1.781    0.182    9.808    0.000    0.932    0.932
##    .cog_e             1.428    0.153    9.330    0.000    0.502    0.502
##    .w_control         1.240    0.172    7.214    0.000    0.645    0.645
##    .cue               0.409    0.150    2.728    0.006    0.231    0.231
##    .affect            0.843    0.119    7.068    0.000    0.412    0.412
##    .attach            0.181    0.083    2.169    0.030    0.080    0.080
##    .social            1.597    0.158   10.120    0.000    0.929    0.929
##     pdm               0.722    0.097    7.451    0.000    1.000    1.000
```

```
## sdm              0.129  0.043  2.998  0.003  1.000  1.000
```

5.1.5 Relative fitness between MLR models

```r
# Model 1 vs Model 2
lavTestLRT(cont_mod1_fit, cont_mod2_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##                Df    AIC    BIC  Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
## cont_mod2_fit 628 116104 116456 6099.7
## cont_mod1_fit 629 116869 117217 6867.3     207.43 0.97418       1  < 2.2e-16
##
## cont_mod2_fit
## cont_mod1_fit ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Model 1 vs Model 3
lavTestLRT(cont_mod1_fit, cont_mod3_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##                Df    AIC    BIC  Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
## cont_mod3_fit 574 112322 112928 2210.2
## cont_mod1_fit 629 116869 117217 6867.3     3109.2 0.32144      55  < 2.2e-16
##
## cont_mod3_fit
## cont_mod1_fit ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Model 1 vs Model 4
lavTestLRT(cont_mod1_fit, cont_mod4_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##                Df    AIC    BIC  Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
## cont_mod4_fit 617 112547 112950 2520.9
## cont_mod1_fit 629 116869 117217 6867.3     1731.9 0.66845      12  < 2.2e-16
##
## cont_mod4_fit
## cont_mod1_fit ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Model 3 vs Model 4
lavTestLRT(cont_mod3_fit, cont_mod4_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##                Df    AIC    BIC  Chisq Chisq diff    RMSEA Df diff Pr(>Chisq)
## cont_mod3_fit 574 112322 112928 2210.2
## cont_mod4_fit 617 112547 112950 2520.9     255.57 0.086397      43  < 2.2e-16
##
## cont_mod3_fit
## cont_mod4_fit ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5.1.6 WLSMV Model 1: 1-factor solution

```
ord_mod1 <- 'f =~ auto1 + auto2 + auto3 + auto4 +
                  loc1 + loc2 + loc3 + loc4 +
                  tol1 + tol2 + tol3 + tol4 +
                  craving1 + craving2 + craving3 + craving4 +
                  taste1 + taste2 + taste3 +
                  cog_e1 + cog_e2 + cog_e3 +
                  w_control1 + w_control2 + w_control3 +
                  cue1 + cue2 + cue3 +
                  affect1 + affect2 + affect3 +
                  attach1 + attach2 + attach3 +
                  social1 + social2 +social3
                  '
ord_mod1_fit <- cfa(ord_mod1, data = df_smoker, estimator = "WLSMV",
                    ordered = wisdm37_items)
summary(ord_mod1_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 53 iterations
##
##   Estimator                                         DWLS
##   Optimization method                             NLMINB
##   Number of model parameters                         259
##
##   Number of observations                             805
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                             14396.513   12552.377
##   Degrees of freedom                               629         629
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  1.177
##   Shift parameter                                          324.564
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                            158009.714   38794.468
##   Degrees of freedom                               666         666
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.127
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.913       0.687
##   Tucker-Lewis Index (TLI)                       0.907       0.669
##
##   Robust Comparative Fit Index (CFI)                         0.580
##   Robust Tucker-Lewis Index (TLI)                            0.556
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.165       0.154
```

```
##   90 Percent confidence interval - lower        0.163       0.151
##   90 Percent confidence interval - upper        0.167       0.156
##   P-value H_0: RMSEA <= 0.050                   0.000       0.000
##   P-value H_0: RMSEA >= 0.080                   1.000       1.000
##
##   Robust RMSEA                                              0.135
##   90 Percent confidence interval - lower                    0.132
##   90 Percent confidence interval - upper                    0.138
##   P-value H_0: Robust RMSEA <= 0.050                        0.000
##   P-value H_0: Robust RMSEA >= 0.080                        1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                          0.108       0.108
##
## Parameter Estimates:
##
##   Parameterization                              Delta
##   Standard errors                          Robust.sem
##   Information                                Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   f =~
##     auto1             1.000                                0.493    0.493
##     auto2             1.432    0.068   21.154    0.000    0.706    0.706
##     auto3             1.537    0.072   21.226    0.000    0.757    0.757
##     auto4             1.558    0.077   20.309    0.000    0.768    0.768
##     loc1              1.463    0.077   18.987    0.000    0.721    0.721
##     loc2              1.526    0.082   18.685    0.000    0.752    0.752
##     loc3              1.654    0.085   19.474    0.000    0.815    0.815
##     loc4              1.696    0.088   19.301    0.000    0.835    0.835
##     tol1              1.590    0.082   19.354    0.000    0.783    0.783
##     tol2              1.540    0.081   18.931    0.000    0.759    0.759
##     tol3              1.533    0.083   18.450    0.000    0.755    0.755
##     tol4              1.745    0.091   19.126    0.000    0.860    0.860
##     craving1          1.454    0.083   17.510    0.000    0.717    0.717
##     craving2          1.309    0.079   16.491    0.000    0.645    0.645
##     craving3          1.546    0.088   17.482    0.000    0.762    0.762
##     craving4          1.606    0.085   18.825    0.000    0.791    0.791
##     taste1            0.575    0.068    8.487    0.000    0.283    0.283
##     taste2            0.724    0.067   10.747    0.000    0.357    0.357
##     taste3            0.518    0.067    7.736    0.000    0.255    0.255
##     cog_e1            1.272    0.070   18.070    0.000    0.627    0.627
##     cog_e2            1.396    0.077   18.139    0.000    0.688    0.688
##     cog_e3            1.452    0.079   18.337    0.000    0.716    0.716
##     w_control1        0.962    0.069   13.911    0.000    0.474    0.474
##     w_control2        0.674    0.075    8.966    0.000    0.332    0.332
##     w_control3        0.939    0.072   13.020    0.000    0.463    0.463
##     cue1              1.058    0.065   16.261    0.000    0.521    0.521
##     cue2              0.677    0.065   10.488    0.000    0.333    0.333
##     cue3              1.030    0.074   13.966    0.000    0.508    0.508
##     affect1           1.212    0.075   16.248    0.000    0.597    0.597
```

```
##     affect2           1.339    0.079   16.848    0.000    0.660    0.660
##     affect3           1.106    0.073   15.182    0.000    0.545    0.545
##     attach1           1.512    0.085   17.691    0.000    0.745    0.745
##     attach2           1.426    0.079   18.046    0.000    0.703    0.703
##     attach3           1.465    0.079   18.640    0.000    0.722    0.722
##     social1           0.698    0.071    9.834    0.000    0.344    0.344
##     social2           0.699    0.068   10.243    0.000    0.344    0.344
##     social3           0.808    0.074   10.862    0.000    0.398    0.398
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     auto1|t1         -1.244    0.059  -21.018    0.000   -1.244   -1.244
##     auto1|t2         -0.924    0.052  -17.846    0.000   -0.924   -0.924
##     auto1|t3         -0.668    0.048  -13.917    0.000   -0.668   -0.668
##     auto1|t4         -0.177    0.044   -3.978    0.000   -0.177   -0.177
##     auto1|t5          0.295    0.045    6.577    0.000    0.295    0.295
##     auto1|t6          0.828    0.050   16.508    0.000    0.828    0.828
##     auto2|t1         -1.019    0.054  -18.999    0.000   -1.019   -1.019
##     auto2|t2         -0.675    0.048  -14.052    0.000   -0.675   -0.675
##     auto2|t3         -0.498    0.046  -10.763    0.000   -0.498   -0.498
##     auto2|t4         -0.123    0.044   -2.782    0.005   -0.123   -0.123
##     auto2|t5          0.253    0.045    5.665    0.000    0.253    0.253
##     auto2|t6          0.731    0.049   14.993    0.000    0.731    0.731
##     auto3|t1         -1.035    0.054  -19.175    0.000   -1.035   -1.035
##     auto3|t2         -0.842    0.050  -16.703    0.000   -0.842   -0.842
##     auto3|t3         -0.622    0.047  -13.102    0.000   -0.622   -0.622
##     auto3|t4         -0.318    0.045   -7.068    0.000   -0.318   -0.318
##     auto3|t5          0.155    0.044    3.486    0.000    0.155    0.155
##     auto3|t6          0.618    0.047   13.033    0.000    0.618    0.618
##     auto4|t1         -0.988    0.053  -18.642    0.000   -0.988   -0.988
##     auto4|t2         -0.764    0.049  -15.525    0.000   -0.764   -0.764
##     auto4|t3         -0.562    0.047  -12.006    0.000   -0.562   -0.562
##     auto4|t4         -0.231    0.045   -5.173    0.000   -0.231   -0.231
##     auto4|t5          0.190    0.044    4.260    0.000    0.190    0.190
##     auto4|t6          0.656    0.048   13.714    0.000    0.656    0.656
##     loc1|t1          -1.084    0.055  -19.688    0.000   -1.084   -1.084
##     loc1|t2          -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     loc1|t3          -0.484    0.046  -10.486    0.000   -0.484   -0.484
##     loc1|t4          -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     loc1|t5           0.312    0.045    6.928    0.000    0.312    0.312
##     loc1|t6           0.882    0.051   17.279    0.000    0.882    0.882
##     loc2|t1          -0.672    0.048  -13.985    0.000   -0.672   -0.672
##     loc2|t2          -0.459    0.046  -10.000    0.000   -0.459   -0.459
##     loc2|t3          -0.231    0.045   -5.173    0.000   -0.231   -0.231
##     loc2|t4           0.083    0.044    1.867    0.062    0.083    0.083
##     loc2|t5           0.439    0.046    9.583    0.000    0.439    0.439
##     loc2|t6           0.798    0.050   16.052    0.000    0.798    0.798
##     loc3|t1          -1.237    0.059  -20.972    0.000   -1.237   -1.237
##     loc3|t2          -0.983    0.053  -18.582    0.000   -0.983   -0.983
##     loc3|t3          -0.752    0.049  -15.326    0.000   -0.752   -0.752
##     loc3|t4          -0.391    0.045   -8.607    0.000   -0.391   -0.391
##     loc3|t5           0.045    0.044    1.021    0.307    0.045    0.045
##     loc3|t6           0.548    0.047   11.730    0.000    0.548    0.548
##     loc4|t1          -0.618    0.047  -13.033    0.000   -0.618   -0.618
```

```
##     loc4|t2          -0.315    0.045   -6.998    0.000   -0.315   -0.315
##     loc4|t3          -0.058    0.044   -1.303    0.193   -0.058   -0.058
##     loc4|t4           0.228    0.045    5.103    0.000    0.228    0.228
##     loc4|t5           0.614    0.047   12.965    0.000    0.614    0.614
##     loc4|t6           1.009    0.053   18.881    0.000    1.009    1.009
##     tol1|t1          -0.505    0.046  -10.901    0.000   -0.505   -0.505
##     tol1|t2          -0.312    0.045   -6.928    0.000   -0.312   -0.312
##     tol1|t3          -0.202    0.045   -4.541    0.000   -0.202   -0.202
##     tol1|t4          -0.045    0.044   -1.021    0.307   -0.045   -0.045
##     tol1|t5           0.171    0.044    3.838    0.000    0.171    0.171
##     tol1|t6           0.498    0.046   10.763    0.000    0.498    0.498
##     tol2|t1          -0.973    0.053  -18.461    0.000   -0.973   -0.973
##     tol2|t2          -0.683    0.048  -14.187    0.000   -0.683   -0.683
##     tol2|t3          -0.335    0.045   -7.418    0.000   -0.335   -0.335
##     tol2|t4          -0.098    0.044   -2.219    0.027   -0.098   -0.098
##     tol2|t5           0.315    0.045    6.998    0.000    0.315    0.315
##     tol2|t6           0.820    0.050   16.378    0.000    0.820    0.820
##     tol3|t1          -0.446    0.046   -9.722    0.000   -0.446   -0.446
##     tol3|t2          -0.273    0.045   -6.086    0.000   -0.273   -0.273
##     tol3|t3          -0.161    0.044   -3.627    0.000   -0.161   -0.161
##     tol3|t4          -0.030    0.044   -0.669    0.503   -0.030   -0.030
##     tol3|t5           0.183    0.044    4.119    0.000    0.183    0.183
##     tol3|t6           0.523    0.046   11.247    0.000    0.523    0.523
##     tol4|t1          -0.707    0.048  -14.591    0.000   -0.707   -0.707
##     tol4|t2          -0.449    0.046   -9.791    0.000   -0.449   -0.449
##     tol4|t3          -0.221    0.045   -4.963    0.000   -0.221   -0.221
##     tol4|t4           0.079    0.044    1.796    0.072    0.079    0.079
##     tol4|t5           0.432    0.046    9.443    0.000    0.432    0.432
##     tol4|t6           0.828    0.050   16.508    0.000    0.828    0.828
##     craving1|t1      -1.237    0.059  -20.972    0.000   -1.237   -1.237
##     craving1|t2      -0.949    0.052  -18.156    0.000   -0.949   -0.949
##     craving1|t3      -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     craving1|t4      -0.418    0.046   -9.165    0.000   -0.418   -0.418
##     craving1|t5      -0.064    0.044   -1.444    0.149   -0.064   -0.064
##     craving1|t6       0.388    0.045    8.537    0.000    0.388    0.388
##     craving2|t1      -1.984    0.096  -20.642    0.000   -1.984   -1.984
##     craving2|t2      -1.754    0.080  -21.820    0.000   -1.754   -1.754
##     craving2|t3      -1.613    0.073  -22.104    0.000   -1.613   -1.613
##     craving2|t4      -1.217    0.058  -20.832    0.000   -1.217   -1.217
##     craving2|t5      -0.614    0.047  -12.965    0.000   -0.614   -0.614
##     craving2|t6       0.002    0.044    0.035    0.972    0.002    0.002
##     craving3|t1      -1.636    0.074  -22.082    0.000   -1.636   -1.636
##     craving3|t2      -1.392    0.064  -21.786    0.000   -1.392   -1.392
##     craving3|t3      -1.179    0.057  -20.539    0.000   -1.179   -1.179
##     craving3|t4      -0.851    0.051  -16.832    0.000   -0.851   -0.851
##     craving3|t5      -0.435    0.046   -9.513    0.000   -0.435   -0.435
##     craving3|t6       0.073    0.044    1.655    0.098    0.073    0.073
##     craving4|t1      -0.973    0.053  -18.461    0.000   -0.973   -0.973
##     craving4|t2      -0.683    0.048  -14.187    0.000   -0.683   -0.683
##     craving4|t3      -0.512    0.046  -11.040    0.000   -0.512   -0.512
##     craving4|t4      -0.269    0.045   -6.016    0.000   -0.269   -0.269
##     craving4|t5       0.117    0.044    2.641    0.008    0.117    0.117
##     craving4|t6       0.618    0.047   13.033    0.000    0.618    0.618
##     taste1|t1        -1.329    0.062  -21.516    0.000   -1.329   -1.329
```

```
##     taste1|t2        -0.993    0.053  -18.703    0.000   -0.993   -0.993
##     taste1|t3        -0.777    0.049  -15.723    0.000   -0.777   -0.777
##     taste1|t4        -0.331    0.045   -7.348    0.000   -0.331   -0.331
##     taste1|t5         0.036    0.044    0.810    0.418    0.036    0.036
##     taste1|t6         0.491    0.046   10.624    0.000    0.491    0.491
##     taste2|t1        -1.292    0.061  -21.320    0.000   -1.292   -1.292
##     taste2|t2        -1.051    0.054  -19.349    0.000   -1.051   -1.051
##     taste2|t3        -0.768    0.049  -15.591    0.000   -0.768   -0.768
##     taste2|t4        -0.354    0.045   -7.838    0.000   -0.354   -0.354
##     taste2|t5         0.067    0.044    1.515    0.130    0.067    0.067
##     taste2|t6         0.629    0.048   13.238    0.000    0.629    0.629
##     taste3|t1        -1.154    0.057  -20.335    0.000   -1.154   -1.154
##     taste3|t2        -0.958    0.052  -18.278    0.000   -0.958   -0.958
##     taste3|t3        -0.748    0.049  -15.259    0.000   -0.748   -0.748
##     taste3|t4        -0.422    0.046   -9.234    0.000   -0.422   -0.422
##     taste3|t5         0.014    0.044    0.317    0.751    0.014    0.014
##     taste3|t6         0.480    0.046   10.416    0.000    0.480    0.480
##     cog_e1|t1        -0.873    0.051  -17.152    0.000   -0.873   -0.873
##     cog_e1|t2        -0.707    0.048  -14.591    0.000   -0.707   -0.707
##     cog_e1|t3        -0.541    0.047  -11.592    0.000   -0.541   -0.541
##     cog_e1|t4        -0.196    0.045   -4.400    0.000   -0.196   -0.196
##     cog_e1|t5         0.180    0.044    4.049    0.000    0.180    0.180
##     cog_e1|t6         0.752    0.049   15.326    0.000    0.752    0.752
##     cog_e2|t1        -0.824    0.050  -16.443    0.000   -0.824   -0.824
##     cog_e2|t2        -0.668    0.048  -13.917    0.000   -0.668   -0.668
##     cog_e2|t3        -0.487    0.046  -10.555    0.000   -0.487   -0.487
##     cog_e2|t4        -0.158    0.044   -3.556    0.000   -0.158   -0.158
##     cog_e2|t5         0.292    0.045    6.507    0.000    0.292    0.292
##     cog_e2|t6         0.833    0.050   16.573    0.000    0.833    0.833
##     cog_e3|t1        -1.079    0.055  -19.632    0.000   -1.079   -1.079
##     cog_e3|t2        -0.924    0.052  -17.846    0.000   -0.924   -0.924
##     cog_e3|t3        -0.719    0.049  -14.792    0.000   -0.719   -0.719
##     cog_e3|t4        -0.345    0.045   -7.628    0.000   -0.345   -0.345
##     cog_e3|t5         0.120    0.044    2.712    0.007    0.120    0.120
##     cog_e3|t6         0.695    0.048   14.389    0.000    0.695    0.695
##     w_control1|t1     0.002    0.044    0.035    0.972    0.002    0.002
##     w_control1|t2     0.241    0.045    5.384    0.000    0.241    0.241
##     w_control1|t3     0.439    0.046    9.583    0.000    0.439    0.439
##     w_control1|t4     0.752    0.049   15.326    0.000    0.752    0.752
##     w_control1|t5     1.062    0.055   19.463    0.000    1.062    1.062
##     w_control1|t6     1.392    0.064   21.786    0.000    1.392    1.392
##     w_control2|t1     0.266    0.045    5.946    0.000    0.266    0.266
##     w_control2|t2     0.466    0.046   10.139    0.000    0.466    0.466
##     w_control2|t3     0.711    0.049   14.658    0.000    0.711    0.711
##     w_control2|t4     1.230    0.059   20.926    0.000    1.230    1.230
##     w_control2|t5     1.384    0.064   21.755    0.000    1.384    1.384
##     w_control2|t6     1.660    0.075   22.048    0.000    1.660    1.660
##     w_control3|t1    -0.401    0.046   -8.816    0.000   -0.401   -0.401
##     w_control3|t2    -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     w_control3|t3     0.045    0.044    1.021    0.307    0.045    0.045
##     w_control3|t4     0.439    0.046    9.583    0.000    0.439    0.439
##     w_control3|t5     0.781    0.049   15.789    0.000    0.781    0.781
##     w_control3|t6     1.173    0.057   20.489    0.000    1.173    1.173
##     cue1|t1          -0.807    0.050  -16.183    0.000   -0.807   -0.807
```

```
##     cue1|t2          -0.629    0.048  -13.238    0.000   -0.629   -0.629
##     cue1|t3          -0.480    0.046  -10.416    0.000   -0.480   -0.480
##     cue1|t4          -0.139    0.044   -3.134    0.002   -0.139   -0.139
##     cue1|t5           0.221    0.045    4.963    0.000    0.221    0.221
##     cue1|t6           0.644    0.048   13.510    0.000    0.644    0.644
##     cue2|t1          -0.785    0.050  -15.855    0.000   -0.785   -0.785
##     cue2|t2          -0.530    0.047  -11.385    0.000   -0.530   -0.530
##     cue2|t3          -0.331    0.045   -7.348    0.000   -0.331   -0.331
##     cue2|t4           0.008    0.044    0.176    0.860    0.008    0.008
##     cue2|t5           0.341    0.045    7.558    0.000    0.341    0.341
##     cue2|t6           0.748    0.049   15.259    0.000    0.748    0.748
##     cue3|t1          -1.498    0.068  -22.059    0.000   -1.498   -1.498
##     cue3|t2          -1.329    0.062  -21.516    0.000   -1.329   -1.329
##     cue3|t3          -1.142    0.056  -20.231    0.000   -1.142   -1.142
##     cue3|t4          -0.864    0.051  -17.024    0.000   -0.864   -0.864
##     cue3|t5          -0.401    0.046   -8.816    0.000   -0.401   -0.401
##     cue3|t6           0.161    0.044    3.627    0.000    0.161    0.161
##     affect1|t1       -1.299    0.061  -21.360    0.000   -1.299   -1.299
##     affect1|t2       -1.096    0.055  -19.799    0.000   -1.096   -1.096
##     affect1|t3       -0.901    0.051  -17.533    0.000   -0.901   -0.901
##     affect1|t4       -0.562    0.047  -12.006    0.000   -0.562   -0.562
##     affect1|t5       -0.089    0.044   -2.007    0.045   -0.089   -0.089
##     affect1|t6        0.459    0.046   10.000    0.000    0.459    0.459
##     affect2|t1       -1.590    0.072  -22.115    0.000   -1.590   -1.590
##     affect2|t2       -1.392    0.064  -21.786    0.000   -1.392   -1.392
##     affect2|t3       -1.160    0.057  -20.387    0.000   -1.160   -1.160
##     affect2|t4       -0.855    0.051  -16.896    0.000   -0.855   -0.855
##     affect2|t5       -0.279    0.045   -6.227    0.000   -0.279   -0.279
##     affect2|t6        0.305    0.045    6.788    0.000    0.305    0.305
##     affect3|t1       -1.590    0.072  -22.115    0.000   -1.590   -1.590
##     affect3|t2       -1.400    0.064  -21.815    0.000   -1.400   -1.400
##     affect3|t3       -1.185    0.058  -20.589    0.000   -1.185   -1.185
##     affect3|t4       -0.695    0.048  -14.389    0.000   -0.695   -0.695
##     affect3|t5       -0.130    0.044   -2.923    0.003   -0.130   -0.130
##     affect3|t6        0.415    0.046    9.095    0.000    0.415    0.415
##     attach1|t1       -1.166    0.057  -20.438    0.000   -1.166   -1.166
##     attach1|t2       -1.035    0.054  -19.175    0.000   -1.035   -1.035
##     attach1|t3       -0.873    0.051  -17.152    0.000   -0.873   -0.873
##     attach1|t4       -0.588    0.047  -12.486    0.000   -0.588   -0.588
##     attach1|t5       -0.234    0.045   -5.244    0.000   -0.234   -0.234
##     attach1|t6        0.139    0.044    3.134    0.002    0.139    0.139
##     attach2|t1       -0.939    0.052  -18.032    0.000   -0.939   -0.939
##     attach2|t2       -0.735    0.049  -15.060    0.000   -0.735   -0.735
##     attach2|t3       -0.551    0.047  -11.799    0.000   -0.551   -0.551
##     attach2|t4       -0.247    0.045   -5.525    0.000   -0.247   -0.247
##     attach2|t5        0.092    0.044    2.078    0.038    0.092    0.092
##     attach2|t6        0.498    0.046   10.763    0.000    0.498    0.498
##     attach3|t1       -0.864    0.051  -17.024    0.000   -0.864   -0.864
##     attach3|t2       -0.660    0.048  -13.782    0.000   -0.660   -0.660
##     attach3|t3       -0.466    0.046  -10.139    0.000   -0.466   -0.466
##     attach3|t4       -0.145    0.044   -3.275    0.001   -0.145   -0.145
##     attach3|t5        0.244    0.045    5.454    0.000    0.244    0.244
##     attach3|t6        0.618    0.047   13.033    0.000    0.618    0.618
##     social1|t1       -1.508    0.068  -22.073    0.000   -1.508   -1.508
```

```
##     social1|t2       -1.185    0.058  -20.589    0.000   -1.185   -1.185
##     social1|t3       -0.949    0.052  -18.156    0.000   -0.949   -0.949
##     social1|t4       -0.508    0.046  -10.971    0.000   -0.508   -0.508
##     social1|t5       -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     social1|t6        0.378    0.045    8.327    0.000    0.378    0.378
##     social2|t1       -1.030    0.054  -19.117    0.000   -1.030   -1.030
##     social2|t2       -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     social2|t3       -0.555    0.047  -11.868    0.000   -0.555   -0.555
##     social2|t4       -0.177    0.044   -3.978    0.000   -0.177   -0.177
##     social2|t5        0.253    0.045    5.665    0.000    0.253    0.253
##     social2|t6        0.798    0.050   16.052    0.000    0.798    0.798
##     social3|t1       -1.470    0.067  -22.009    0.000   -1.470   -1.470
##     social3|t2       -1.124    0.056  -20.072    0.000   -1.124   -1.124
##     social3|t3       -0.878    0.051  -17.216    0.000   -0.878   -0.878
##     social3|t4       -0.494    0.046  -10.694    0.000   -0.494   -0.494
##     social3|t5       -0.092    0.044   -2.078    0.038   -0.092   -0.092
##     social3|t6        0.603    0.047   12.760    0.000    0.603    0.603
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .auto1             0.757                               0.757    0.757
##    .auto2             0.502                               0.502    0.502
##    .auto3             0.427                               0.427    0.427
##    .auto4             0.411                               0.411    0.411
##    .loc1              0.480                               0.480    0.480
##    .loc2              0.435                               0.435    0.435
##    .loc3              0.336                               0.336    0.336
##    .loc4              0.302                               0.302    0.302
##    .tol1              0.386                               0.386    0.386
##    .tol2              0.424                               0.424    0.424
##    .tol3              0.430                               0.430    0.430
##    .tol4              0.261                               0.261    0.261
##    .craving1          0.486                               0.486    0.486
##    .craving2          0.584                               0.584    0.584
##    .craving3          0.420                               0.420    0.420
##    .craving4          0.374                               0.374    0.374
##    .taste1            0.920                               0.920    0.920
##    .taste2            0.873                               0.873    0.873
##    .taste3            0.935                               0.935    0.935
##    .cog_e1            0.607                               0.607    0.607
##    .cog_e2            0.527                               0.527    0.527
##    .cog_e3            0.488                               0.488    0.488
##    .w_control1        0.775                               0.775    0.775
##    .w_control2        0.890                               0.890    0.890
##    .w_control3        0.786                               0.786    0.786
##    .cue1              0.728                               0.728    0.728
##    .cue2              0.889                               0.889    0.889
##    .cue3              0.742                               0.742    0.742
##    .affect1           0.643                               0.643    0.643
##    .affect2           0.565                               0.565    0.565
##    .affect3           0.703                               0.703    0.703
##    .attach1           0.445                               0.445    0.445
##    .attach2           0.506                               0.506    0.506
##    .attach3           0.479                               0.479    0.479
```

```
##    .social1           0.882                               0.882    0.882
##    .social2           0.881                               0.881    0.881
##    .social3           0.842                               0.842    0.842
##     f                 0.243    0.025    9.663    0.000    1.000    1.000
```

5.1.7 WLSMV Model 2: 2-factor solution

```
ord_mod2 <- 'pdm =~ auto1 + auto2 + auto3 + auto4 +
                    loc1 + loc2 + loc3 + loc4 +
                    tol1 + tol2 + tol3 + tol4 +
                    craving1 + craving2 + craving3 + craving4

              sdm =~ taste1 + taste2 + taste3 +
                    cog_e1 + cog_e2 + cog_e3 +
                    w_control1 + w_control2 + w_control3 +
                    cue1 + cue2 + cue3 +
                    affect1 + affect2 + affect3 +
                    attach1 + attach2 + attach3 +
                    social1 + social2 +social3
                    '
ord_mod2_fit <- cfa(ord_mod2, data = df_smoker, estimator = "WLSMV",
                   ordered = wisdm37_items)
summary(ord_mod2_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 70 iterations
##
##   Estimator                                         DWLS
##   Optimization method                             NLMINB
##   Number of model parameters                         260
##
##   Number of observations                             805
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                             12014.160   10366.561
##   Degrees of freedom                               628         628
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  1.196
##   Shift parameter                                          323.677
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                            158009.714   38794.468
##   Degrees of freedom                               666         666
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.127
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.928       0.745
##   Tucker-Lewis Index (TLI)                       0.923       0.729
##
##   Robust Comparative Fit Index (CFI)                         0.625
##   Robust Tucker-Lewis Index (TLI)                            0.602
##
## Root Mean Square Error of Approximation:
##
```

```
##   RMSEA                                         0.150       0.139
##   90 Percent confidence interval - lower        0.148       0.137
##   90 Percent confidence interval - upper        0.153       0.141
##   P-value H_0: RMSEA <= 0.050                   0.000       0.000
##   P-value H_0: RMSEA >= 0.080                   1.000       1.000
##
##   Robust RMSEA                                              0.128
##   90 Percent confidence interval - lower                    0.125
##   90 Percent confidence interval - upper                    0.131
##   P-value H_0: Robust RMSEA <= 0.050                        0.000
##   P-value H_0: Robust RMSEA >= 0.080                        1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                          0.102       0.102
##
## Parameter Estimates:
##
##   Parameterization                               Delta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   pdm =~
##     auto1             1.000                               0.515    0.515
##     auto2             1.417    0.066   21.567    0.000    0.730    0.730
##     auto3             1.517    0.070   21.629    0.000    0.781    0.781
##     auto4             1.537    0.074   20.727    0.000    0.792    0.792
##     loc1              1.445    0.074   19.420    0.000    0.745    0.745
##     loc2              1.506    0.079   19.113    0.000    0.776    0.776
##     loc3              1.630    0.082   19.928    0.000    0.840    0.840
##     loc4              1.657    0.084   19.822    0.000    0.854    0.854
##     tol1              1.565    0.079   19.804    0.000    0.806    0.806
##     tol2              1.512    0.078   19.403    0.000    0.779    0.779
##     tol3              1.509    0.080   18.872    0.000    0.778    0.778
##     tol4              1.703    0.087   19.645    0.000    0.877    0.877
##     craving1          1.437    0.080   17.919    0.000    0.740    0.740
##     craving2          1.298    0.077   16.752    0.000    0.668    0.668
##     craving3          1.526    0.085   17.853    0.000    0.786    0.786
##     craving4          1.580    0.082   19.274    0.000    0.814    0.814
##   sdm =~
##     taste1            1.000                               0.335    0.335
##     taste2            1.263    0.107   11.779    0.000    0.424    0.424
##     taste3            0.963    0.096   10.034    0.000    0.323    0.323
##     cog_e1            2.037    0.196   10.380    0.000    0.683    0.683
##     cog_e2            2.235    0.212   10.559    0.000    0.749    0.749
##     cog_e3            2.320    0.219   10.606    0.000    0.778    0.778
##     w_control1        1.560    0.169    9.207    0.000    0.523    0.523
##     w_control2        1.121    0.151    7.418    0.000    0.376    0.376
##     w_control3        1.519    0.168    9.020    0.000    0.510    0.510
##     cue1              1.704    0.177    9.610    0.000    0.572    0.572
##     cue2              1.101    0.137    8.036    0.000    0.369    0.369
```

```
##     cue3              1.658    0.176    9.403    0.000    0.556    0.556
##     affect1           1.956    0.184   10.617    0.000    0.656    0.656
##     affect2           2.155    0.195   11.066    0.000    0.723    0.723
##     affect3           1.785    0.166   10.769    0.000    0.599    0.599
##     attach1           2.460    0.233   10.540    0.000    0.825    0.825
##     attach2           2.299    0.215   10.684    0.000    0.771    0.771
##     attach3           2.375    0.224   10.597    0.000    0.796    0.796
##     social1           1.170    0.148    7.928    0.000    0.392    0.392
##     social2           1.193    0.142    8.400    0.000    0.400    0.400
##     social3           1.376    0.153    8.972    0.000    0.462    0.462
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   pdm ~~
##     sdm               0.129    0.014    9.210    0.000    0.747    0.747
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     auto1|t1         -1.244    0.059  -21.018    0.000   -1.244   -1.244
##     auto1|t2         -0.924    0.052  -17.846    0.000   -0.924   -0.924
##     auto1|t3         -0.668    0.048  -13.917    0.000   -0.668   -0.668
##     auto1|t4         -0.177    0.044   -3.978    0.000   -0.177   -0.177
##     auto1|t5          0.295    0.045    6.577    0.000    0.295    0.295
##     auto1|t6          0.828    0.050   16.508    0.000    0.828    0.828
##     auto2|t1         -1.019    0.054  -18.999    0.000   -1.019   -1.019
##     auto2|t2         -0.675    0.048  -14.052    0.000   -0.675   -0.675
##     auto2|t3         -0.498    0.046  -10.763    0.000   -0.498   -0.498
##     auto2|t4         -0.123    0.044   -2.782    0.005   -0.123   -0.123
##     auto2|t5          0.253    0.045    5.665    0.000    0.253    0.253
##     auto2|t6          0.731    0.049   14.993    0.000    0.731    0.731
##     auto3|t1         -1.035    0.054  -19.175    0.000   -1.035   -1.035
##     auto3|t2         -0.842    0.050  -16.703    0.000   -0.842   -0.842
##     auto3|t3         -0.622    0.047  -13.102    0.000   -0.622   -0.622
##     auto3|t4         -0.318    0.045   -7.068    0.000   -0.318   -0.318
##     auto3|t5          0.155    0.044    3.486    0.000    0.155    0.155
##     auto3|t6          0.618    0.047   13.033    0.000    0.618    0.618
##     auto4|t1         -0.988    0.053  -18.642    0.000   -0.988   -0.988
##     auto4|t2         -0.764    0.049  -15.525    0.000   -0.764   -0.764
##     auto4|t3         -0.562    0.047  -12.006    0.000   -0.562   -0.562
##     auto4|t4         -0.231    0.045   -5.173    0.000   -0.231   -0.231
##     auto4|t5          0.190    0.044    4.260    0.000    0.190    0.190
##     auto4|t6          0.656    0.048   13.714    0.000    0.656    0.656
##     loc1|t1          -1.084    0.055  -19.688    0.000   -1.084   -1.084
##     loc1|t2          -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     loc1|t3          -0.484    0.046  -10.486    0.000   -0.484   -0.484
##     loc1|t4          -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     loc1|t5           0.312    0.045    6.928    0.000    0.312    0.312
##     loc1|t6           0.882    0.051   17.279    0.000    0.882    0.882
##     loc2|t1          -0.672    0.048  -13.985    0.000   -0.672   -0.672
##     loc2|t2          -0.459    0.046  -10.000    0.000   -0.459   -0.459
##     loc2|t3          -0.231    0.045   -5.173    0.000   -0.231   -0.231
##     loc2|t4           0.083    0.044    1.867    0.062    0.083    0.083
##     loc2|t5           0.439    0.046    9.583    0.000    0.439    0.439
##     loc2|t6           0.798    0.050   16.052    0.000    0.798    0.798
```

```
##     loc3|t1         -1.237    0.059  -20.972    0.000   -1.237   -1.237
##     loc3|t2         -0.983    0.053  -18.582    0.000   -0.983   -0.983
##     loc3|t3         -0.752    0.049  -15.326    0.000   -0.752   -0.752
##     loc3|t4         -0.391    0.045   -8.607    0.000   -0.391   -0.391
##     loc3|t5          0.045    0.044    1.021    0.307    0.045    0.045
##     loc3|t6          0.548    0.047   11.730    0.000    0.548    0.548
##     loc4|t1         -0.618    0.047  -13.033    0.000   -0.618   -0.618
##     loc4|t2         -0.315    0.045   -6.998    0.000   -0.315   -0.315
##     loc4|t3         -0.058    0.044   -1.303    0.193   -0.058   -0.058
##     loc4|t4          0.228    0.045    5.103    0.000    0.228    0.228
##     loc4|t5          0.614    0.047   12.965    0.000    0.614    0.614
##     loc4|t6          1.009    0.053   18.881    0.000    1.009    1.009
##     tol1|t1         -0.505    0.046  -10.901    0.000   -0.505   -0.505
##     tol1|t2         -0.312    0.045   -6.928    0.000   -0.312   -0.312
##     tol1|t3         -0.202    0.045   -4.541    0.000   -0.202   -0.202
##     tol1|t4         -0.045    0.044   -1.021    0.307   -0.045   -0.045
##     tol1|t5          0.171    0.044    3.838    0.000    0.171    0.171
##     tol1|t6          0.498    0.046   10.763    0.000    0.498    0.498
##     tol2|t1         -0.973    0.053  -18.461    0.000   -0.973   -0.973
##     tol2|t2         -0.683    0.048  -14.187    0.000   -0.683   -0.683
##     tol2|t3         -0.335    0.045   -7.418    0.000   -0.335   -0.335
##     tol2|t4         -0.098    0.044   -2.219    0.027   -0.098   -0.098
##     tol2|t5          0.315    0.045    6.998    0.000    0.315    0.315
##     tol2|t6          0.820    0.050   16.378    0.000    0.820    0.820
##     tol3|t1         -0.446    0.046   -9.722    0.000   -0.446   -0.446
##     tol3|t2         -0.273    0.045   -6.086    0.000   -0.273   -0.273
##     tol3|t3         -0.161    0.044   -3.627    0.000   -0.161   -0.161
##     tol3|t4         -0.030    0.044   -0.669    0.503   -0.030   -0.030
##     tol3|t5          0.183    0.044    4.119    0.000    0.183    0.183
##     tol3|t6          0.523    0.046   11.247    0.000    0.523    0.523
##     tol4|t1         -0.707    0.048  -14.591    0.000   -0.707   -0.707
##     tol4|t2         -0.449    0.046   -9.791    0.000   -0.449   -0.449
##     tol4|t3         -0.221    0.045   -4.963    0.000   -0.221   -0.221
##     tol4|t4          0.079    0.044    1.796    0.072    0.079    0.079
##     tol4|t5          0.432    0.046    9.443    0.000    0.432    0.432
##     tol4|t6          0.828    0.050   16.508    0.000    0.828    0.828
##     craving1|t1     -1.237    0.059  -20.972    0.000   -1.237   -1.237
##     craving1|t2     -0.949    0.052  -18.156    0.000   -0.949   -0.949
##     craving1|t3     -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     craving1|t4     -0.418    0.046   -9.165    0.000   -0.418   -0.418
##     craving1|t5     -0.064    0.044   -1.444    0.149   -0.064   -0.064
##     craving1|t6      0.388    0.045    8.537    0.000    0.388    0.388
##     craving2|t1     -1.984    0.096  -20.642    0.000   -1.984   -1.984
##     craving2|t2     -1.754    0.080  -21.820    0.000   -1.754   -1.754
##     craving2|t3     -1.613    0.073  -22.104    0.000   -1.613   -1.613
##     craving2|t4     -1.217    0.058  -20.832    0.000   -1.217   -1.217
##     craving2|t5     -0.614    0.047  -12.965    0.000   -0.614   -0.614
##     craving2|t6      0.002    0.044    0.035    0.972    0.002    0.002
##     craving3|t1     -1.636    0.074  -22.082    0.000   -1.636   -1.636
##     craving3|t2     -1.392    0.064  -21.786    0.000   -1.392   -1.392
##     craving3|t3     -1.179    0.057  -20.539    0.000   -1.179   -1.179
##     craving3|t4     -0.851    0.051  -16.832    0.000   -0.851   -0.851
##     craving3|t5     -0.435    0.046   -9.513    0.000   -0.435   -0.435
##     craving3|t6      0.073    0.044    1.655    0.098    0.073    0.073
```

```
##     craving4|t1      -0.973     0.053  -18.461     0.000   -0.973   -0.973
##     craving4|t2      -0.683     0.048  -14.187     0.000   -0.683   -0.683
##     craving4|t3      -0.512     0.046  -11.040     0.000   -0.512   -0.512
##     craving4|t4      -0.269     0.045   -6.016     0.000   -0.269   -0.269
##     craving4|t5       0.117     0.044    2.641     0.008    0.117    0.117
##     craving4|t6       0.618     0.047   13.033     0.000    0.618    0.618
##     taste1|t1        -1.329     0.062  -21.516     0.000   -1.329   -1.329
##     taste1|t2        -0.993     0.053  -18.703     0.000   -0.993   -0.993
##     taste1|t3        -0.777     0.049  -15.723     0.000   -0.777   -0.777
##     taste1|t4        -0.331     0.045   -7.348     0.000   -0.331   -0.331
##     taste1|t5         0.036     0.044    0.810     0.418    0.036    0.036
##     taste1|t6         0.491     0.046   10.624     0.000    0.491    0.491
##     taste2|t1        -1.292     0.061  -21.320     0.000   -1.292   -1.292
##     taste2|t2        -1.051     0.054  -19.349     0.000   -1.051   -1.051
##     taste2|t3        -0.768     0.049  -15.591     0.000   -0.768   -0.768
##     taste2|t4        -0.354     0.045   -7.838     0.000   -0.354   -0.354
##     taste2|t5         0.067     0.044    1.515     0.130    0.067    0.067
##     taste2|t6         0.629     0.048   13.238     0.000    0.629    0.629
##     taste3|t1        -1.154     0.057  -20.335     0.000   -1.154   -1.154
##     taste3|t2        -0.958     0.052  -18.278     0.000   -0.958   -0.958
##     taste3|t3        -0.748     0.049  -15.259     0.000   -0.748   -0.748
##     taste3|t4        -0.422     0.046   -9.234     0.000   -0.422   -0.422
##     taste3|t5         0.014     0.044    0.317     0.751    0.014    0.014
##     taste3|t6         0.480     0.046   10.416     0.000    0.480    0.480
##     cog_e1|t1        -0.873     0.051  -17.152     0.000   -0.873   -0.873
##     cog_e1|t2        -0.707     0.048  -14.591     0.000   -0.707   -0.707
##     cog_e1|t3        -0.541     0.047  -11.592     0.000   -0.541   -0.541
##     cog_e1|t4        -0.196     0.045   -4.400     0.000   -0.196   -0.196
##     cog_e1|t5         0.180     0.044    4.049     0.000    0.180    0.180
##     cog_e1|t6         0.752     0.049   15.326     0.000    0.752    0.752
##     cog_e2|t1        -0.824     0.050  -16.443     0.000   -0.824   -0.824
##     cog_e2|t2        -0.668     0.048  -13.917     0.000   -0.668   -0.668
##     cog_e2|t3        -0.487     0.046  -10.555     0.000   -0.487   -0.487
##     cog_e2|t4        -0.158     0.044   -3.556     0.000   -0.158   -0.158
##     cog_e2|t5         0.292     0.045    6.507     0.000    0.292    0.292
##     cog_e2|t6         0.833     0.050   16.573     0.000    0.833    0.833
##     cog_e3|t1        -1.079     0.055  -19.632     0.000   -1.079   -1.079
##     cog_e3|t2        -0.924     0.052  -17.846     0.000   -0.924   -0.924
##     cog_e3|t3        -0.719     0.049  -14.792     0.000   -0.719   -0.719
##     cog_e3|t4        -0.345     0.045   -7.628     0.000   -0.345   -0.345
##     cog_e3|t5         0.120     0.044    2.712     0.007    0.120    0.120
##     cog_e3|t6         0.695     0.048   14.389     0.000    0.695    0.695
##     w_control1|t1     0.002     0.044    0.035     0.972    0.002    0.002
##     w_control1|t2     0.241     0.045    5.384     0.000    0.241    0.241
##     w_control1|t3     0.439     0.046    9.583     0.000    0.439    0.439
##     w_control1|t4     0.752     0.049   15.326     0.000    0.752    0.752
##     w_control1|t5     1.062     0.055   19.463     0.000    1.062    1.062
##     w_control1|t6     1.392     0.064   21.786     0.000    1.392    1.392
##     w_control2|t1     0.266     0.045    5.946     0.000    0.266    0.266
##     w_control2|t2     0.466     0.046   10.139     0.000    0.466    0.466
##     w_control2|t3     0.711     0.049   14.658     0.000    0.711    0.711
##     w_control2|t4     1.230     0.059   20.926     0.000    1.230    1.230
##     w_control2|t5     1.384     0.064   21.755     0.000    1.384    1.384
##     w_control2|t6     1.660     0.075   22.048     0.000    1.660    1.660
```

```
##     w_control3|t1    -0.401    0.046   -8.816    0.000   -0.401   -0.401
##     w_control3|t2    -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     w_control3|t3     0.045    0.044    1.021    0.307    0.045    0.045
##     w_control3|t4     0.439    0.046    9.583    0.000    0.439    0.439
##     w_control3|t5     0.781    0.049   15.789    0.000    0.781    0.781
##     w_control3|t6     1.173    0.057   20.489    0.000    1.173    1.173
##     cue1|t1          -0.807    0.050  -16.183    0.000   -0.807   -0.807
##     cue1|t2          -0.629    0.048  -13.238    0.000   -0.629   -0.629
##     cue1|t3          -0.480    0.046  -10.416    0.000   -0.480   -0.480
##     cue1|t4          -0.139    0.044   -3.134    0.002   -0.139   -0.139
##     cue1|t5           0.221    0.045    4.963    0.000    0.221    0.221
##     cue1|t6           0.644    0.048   13.510    0.000    0.644    0.644
##     cue2|t1          -0.785    0.050  -15.855    0.000   -0.785   -0.785
##     cue2|t2          -0.530    0.047  -11.385    0.000   -0.530   -0.530
##     cue2|t3          -0.331    0.045   -7.348    0.000   -0.331   -0.331
##     cue2|t4           0.008    0.044    0.176    0.860    0.008    0.008
##     cue2|t5           0.341    0.045    7.558    0.000    0.341    0.341
##     cue2|t6           0.748    0.049   15.259    0.000    0.748    0.748
##     cue3|t1          -1.498    0.068  -22.059    0.000   -1.498   -1.498
##     cue3|t2          -1.329    0.062  -21.516    0.000   -1.329   -1.329
##     cue3|t3          -1.142    0.056  -20.231    0.000   -1.142   -1.142
##     cue3|t4          -0.864    0.051  -17.024    0.000   -0.864   -0.864
##     cue3|t5          -0.401    0.046   -8.816    0.000   -0.401   -0.401
##     cue3|t6           0.161    0.044    3.627    0.000    0.161    0.161
##     affect1|t1       -1.299    0.061  -21.360    0.000   -1.299   -1.299
##     affect1|t2       -1.096    0.055  -19.799    0.000   -1.096   -1.096
##     affect1|t3       -0.901    0.051  -17.533    0.000   -0.901   -0.901
##     affect1|t4       -0.562    0.047  -12.006    0.000   -0.562   -0.562
##     affect1|t5       -0.089    0.044   -2.007    0.045   -0.089   -0.089
##     affect1|t6        0.459    0.046   10.000    0.000    0.459    0.459
##     affect2|t1       -1.590    0.072  -22.115    0.000   -1.590   -1.590
##     affect2|t2       -1.392    0.064  -21.786    0.000   -1.392   -1.392
##     affect2|t3       -1.160    0.057  -20.387    0.000   -1.160   -1.160
##     affect2|t4       -0.855    0.051  -16.896    0.000   -0.855   -0.855
##     affect2|t5       -0.279    0.045   -6.227    0.000   -0.279   -0.279
##     affect2|t6        0.305    0.045    6.788    0.000    0.305    0.305
##     affect3|t1       -1.590    0.072  -22.115    0.000   -1.590   -1.590
##     affect3|t2       -1.400    0.064  -21.815    0.000   -1.400   -1.400
##     affect3|t3       -1.185    0.058  -20.589    0.000   -1.185   -1.185
##     affect3|t4       -0.695    0.048  -14.389    0.000   -0.695   -0.695
##     affect3|t5       -0.130    0.044   -2.923    0.003   -0.130   -0.130
##     affect3|t6        0.415    0.046    9.095    0.000    0.415    0.415
##     attach1|t1       -1.166    0.057  -20.438    0.000   -1.166   -1.166
##     attach1|t2       -1.035    0.054  -19.175    0.000   -1.035   -1.035
##     attach1|t3       -0.873    0.051  -17.152    0.000   -0.873   -0.873
##     attach1|t4       -0.588    0.047  -12.486    0.000   -0.588   -0.588
##     attach1|t5       -0.234    0.045   -5.244    0.000   -0.234   -0.234
##     attach1|t6        0.139    0.044    3.134    0.002    0.139    0.139
##     attach2|t1       -0.939    0.052  -18.032    0.000   -0.939   -0.939
##     attach2|t2       -0.735    0.049  -15.060    0.000   -0.735   -0.735
##     attach2|t3       -0.551    0.047  -11.799    0.000   -0.551   -0.551
##     attach2|t4       -0.247    0.045   -5.525    0.000   -0.247   -0.247
##     attach2|t5        0.092    0.044    2.078    0.038    0.092    0.092
##     attach2|t6        0.498    0.046   10.763    0.000    0.498    0.498
```

```
##     attach3|t1       -0.864    0.051  -17.024    0.000   -0.864   -0.864
##     attach3|t2       -0.660    0.048  -13.782    0.000   -0.660   -0.660
##     attach3|t3       -0.466    0.046  -10.139    0.000   -0.466   -0.466
##     attach3|t4       -0.145    0.044   -3.275    0.001   -0.145   -0.145
##     attach3|t5        0.244    0.045    5.454    0.000    0.244    0.244
##     attach3|t6        0.618    0.047   13.033    0.000    0.618    0.618
##     social1|t1       -1.508    0.068  -22.073    0.000   -1.508   -1.508
##     social1|t2       -1.185    0.058  -20.589    0.000   -1.185   -1.185
##     social1|t3       -0.949    0.052  -18.156    0.000   -0.949   -0.949
##     social1|t4       -0.508    0.046  -10.971    0.000   -0.508   -0.508
##     social1|t5       -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     social1|t6        0.378    0.045    8.327    0.000    0.378    0.378
##     social2|t1       -1.030    0.054  -19.117    0.000   -1.030   -1.030
##     social2|t2       -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     social2|t3       -0.555    0.047  -11.868    0.000   -0.555   -0.555
##     social2|t4       -0.177    0.044   -3.978    0.000   -0.177   -0.177
##     social2|t5        0.253    0.045    5.665    0.000    0.253    0.253
##     social2|t6        0.798    0.050   16.052    0.000    0.798    0.798
##     social3|t1       -1.470    0.067  -22.009    0.000   -1.470   -1.470
##     social3|t2       -1.124    0.056  -20.072    0.000   -1.124   -1.124
##     social3|t3       -0.878    0.051  -17.216    0.000   -0.878   -0.878
##     social3|t4       -0.494    0.046  -10.694    0.000   -0.494   -0.494
##     social3|t5       -0.092    0.044   -2.078    0.038   -0.092   -0.092
##     social3|t6        0.603    0.047   12.760    0.000    0.603    0.603
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .auto1             0.735                               0.735    0.735
##    .auto2             0.467                               0.467    0.467
##    .auto3             0.390                               0.390    0.390
##    .auto4             0.373                               0.373    0.373
##    .loc1              0.446                               0.446    0.446
##    .loc2              0.399                               0.399    0.399
##    .loc3              0.295                               0.295    0.295
##    .loc4              0.271                               0.271    0.271
##    .tol1              0.350                               0.350    0.350
##    .tol2              0.394                               0.394    0.394
##    .tol3              0.395                               0.395    0.395
##    .tol4              0.231                               0.231    0.231
##    .craving1          0.452                               0.452    0.452
##    .craving2          0.553                               0.553    0.553
##    .craving3          0.382                               0.382    0.382
##    .craving4          0.337                               0.337    0.337
##    .taste1            0.888                               0.888    0.888
##    .taste2            0.821                               0.821    0.821
##    .taste3            0.896                               0.896    0.896
##    .cog_e1            0.533                               0.533    0.533
##    .cog_e2            0.438                               0.438    0.438
##    .cog_e3            0.394                               0.394    0.394
##    .w_control1        0.726                               0.726    0.726
##    .w_control2        0.859                               0.859    0.859
##    .w_control3        0.740                               0.740    0.740
##    .cue1              0.673                               0.673    0.673
##    .cue2              0.864                               0.864    0.864
```

```
##    .cue3              0.691                                 0.691    0.691
##    .affect1           0.569                                 0.569    0.569
##    .affect2           0.477                                 0.477    0.477
##    .affect3           0.642                                 0.642    0.642
##    .attach1           0.319                                 0.319    0.319
##    .attach2           0.406                                 0.406    0.406
##    .attach3           0.366                                 0.366    0.366
##    .social1           0.846                                 0.846    0.846
##    .social2           0.840                                 0.840    0.840
##    .social3           0.787                                 0.787    0.787
##     pdm               0.265    0.027    9.973    0.000    1.000    1.000
##     sdm               0.112    0.021    5.444    0.000    1.000    1.000
```

5.1.8 WLSMV Model 3: 11-factor solution

```
ord_mod3 <- '#Primary dependence motives
             auto      =~ auto1 + auto2 + auto3 + auto4
             loc       =~ loc1 + loc2 + loc3 + loc4
             tol       =~ tol1 + tol2 + tol3 + tol4
             craving   =~ craving1 + craving2 + craving3 + craving4

             #Secondary dependence motives
             taste     =~ taste1 + taste2 + taste3
             cog_e     =~ cog_e1 + cog_e2 + cog_e3
             w_control =~ w_control1 + w_control2 + w_control3
             cue       =~ cue1 + cue2 + cue3
             affect    =~ affect1 + affect2 + affect3
             attach    =~ attach1 + attach2 + attach3
             social    =~ social1 + social2 +social3
             '
ord_mod3_fit <- cfa(ord_mod3, data = df_smoker, estimator = "WLSMV",
                    ordered = wisdm37_items)
```

```
## Warning: lavaan->lav_object_post_check():
##    covariance matrix of latent variables is not positive definite ; use
##    lavInspect(fit, "cov.lv") to investigate.
```

```
summary(ord_mod3_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## Warning: lavaan->lav_object_post_check():
##    covariance matrix of latent variables is not positive definite ; use
##    lavInspect(fit, "cov.lv") to investigate.
## Warning: lavaan->lav_object_post_check():
##    covariance matrix of latent variables is not positive definite ; use
##    lavInspect(fit, "cov.lv") to investigate.
```

```
## lavaan 0.6-21 ended normally after 74 iterations
##
##   Estimator                                         DWLS
##   Optimization method                             NLMINB
##   Number of model parameters                         314
##
##   Number of observations                             805
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                              1799.114    2329.920
##   Degrees of freedom                               574         574
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  0.870
##   Shift parameter                                          263.156
##     simple second-order correction
##
## Model Test Baseline Model:
```

```
##
##   Test statistic                             158009.714   38794.468
##   Degrees of freedom                                666         666
##   P-value                                            NA       0.000
##   Scaling correction factor                                   4.127
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                     0.992       0.954
##   Tucker-Lewis Index (TLI)                        0.991       0.947
##
##   Robust Comparative Fit Index (CFI)                          0.860
##   Robust Tucker-Lewis Index (TLI)                             0.837
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                           0.052       0.062
##   90 Percent confidence interval - lower          0.049       0.059
##   90 Percent confidence interval - upper          0.054       0.064
##   P-value H_0: RMSEA <= 0.050                     0.173       0.000
##   P-value H_0: RMSEA >= 0.080                     0.000       0.000
##
##   Robust RMSEA                                                0.082
##   90 Percent confidence interval - lower                      0.079
##   90 Percent confidence interval - upper                      0.085
##   P-value H_0: Robust RMSEA <= 0.050                          0.000
##   P-value H_0: Robust RMSEA >= 0.080                          0.827
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                            0.048       0.048
##
## Parameter Estimates:
##
##   Parameterization                               Delta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   auto =~
##     auto1             1.000                               0.618    0.618
##     auto2             1.361    0.058   23.329    0.000    0.841    0.841
##     auto3             1.436    0.061   23.579    0.000    0.888    0.888
##     auto4             1.479    0.066   22.396    0.000    0.915    0.915
##   loc =~
##     loc1              1.000                               0.760    0.760
##     loc2              1.041    0.025   42.259    0.000    0.791    0.791
##     loc3              1.121    0.025   44.085    0.000    0.851    0.851
##     loc4              1.149    0.025   46.398    0.000    0.872    0.872
##   tol =~
##     tol1              1.000                               0.868    0.868
##     tol2              0.955    0.021   44.480    0.000    0.829    0.829
```

```
##     tol3              0.963    0.023   42.494    0.000    0.836    0.836
##     tol4              1.089    0.020   54.309    0.000    0.946    0.946
##   craving =~
##     craving1          1.000                               0.772    0.772
##     craving2          0.902    0.032   28.273    0.000    0.696    0.696
##     craving3          1.058    0.030   34.939    0.000    0.817    0.817
##     craving4          1.105    0.028   38.791    0.000    0.853    0.853
##   taste =~
##     taste1            1.000                               0.746    0.746
##     taste2            1.274    0.062   20.454    0.000    0.951    0.951
##     taste3            1.024    0.042   24.617    0.000    0.764    0.764
##   cog_e =~
##     cog_e1            1.000                               0.802    0.802
##     cog_e2            1.077    0.029   37.433    0.000    0.863    0.863
##     cog_e3            1.144    0.031   36.475    0.000    0.917    0.917
##   w_control =~
##     w_control1        1.000                               0.786    0.786
##     w_control2        0.726    0.064   11.431    0.000    0.570    0.570
##     w_control3        0.979    0.064   15.347    0.000    0.769    0.769
##   cue =~
##     cue1              1.000                               0.633    0.633
##     cue2              0.645    0.057   11.256    0.000    0.408    0.408
##     cue3              0.977    0.065   15.124    0.000    0.619    0.619
##   affect =~
##     affect1           1.000                               0.773    0.773
##     affect2           1.097    0.037   29.692    0.000    0.848    0.848
##     affect3           0.910    0.040   22.644    0.000    0.704    0.704
##   attach =~
##     attach1           1.000                               0.827    0.827
##     attach2           0.940    0.029   32.777    0.000    0.777    0.777
##     attach3           0.972    0.031   31.617    0.000    0.803    0.803
##   social =~
##     social1           1.000                               0.786    0.786
##     social2           1.000    0.035   28.598    0.000    0.786    0.786
##     social3           1.183    0.051   23.124    0.000    0.930    0.930
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   auto ~~
##     loc               0.360    0.021   16.828    0.000    0.767    0.767
##     tol               0.356    0.023   15.707    0.000    0.662    0.662
##     craving           0.314    0.021   14.961    0.000    0.658    0.658
##     taste             0.087    0.018    4.929    0.000    0.188    0.188
##     cog_e             0.227    0.021   10.727    0.000    0.459    0.459
##     w_control         0.198    0.023    8.751    0.000    0.408    0.408
##     cue               0.249    0.021   11.643    0.000    0.637    0.637
##     affect            0.198    0.021    9.557    0.000    0.414    0.414
##     attach            0.347    0.022   15.476    0.000    0.679    0.679
##     social            0.122    0.019    6.424    0.000    0.251    0.251
##   loc ~~
##     tol               0.596    0.021   28.888    0.000    0.904    0.904
##     craving           0.568    0.021   26.876    0.000    0.968    0.968
##     taste             0.080    0.021    3.756    0.000    0.142    0.142
##     cog_e             0.354    0.022   15.949    0.000    0.582    0.582
```

```
##     w_control         0.303    0.026   11.880    0.000    0.508    0.508
##     cue               0.341    0.023   14.943    0.000    0.709    0.709
##     affect            0.364    0.023   15.619    0.000    0.619    0.619
##     attach            0.532    0.022   23.964    0.000    0.847    0.847
##     social            0.166    0.023    7.214    0.000    0.278    0.278
##   tol ~~
##     craving           0.542    0.022   24.382    0.000    0.808    0.808
##     taste             0.066    0.024    2.704    0.007    0.102    0.102
##     cog_e             0.393    0.024   16.292    0.000    0.565    0.565
##     w_control         0.345    0.028   12.323    0.000    0.505    0.505
##     cue               0.313    0.025   12.581    0.000    0.570    0.570
##     affect            0.368    0.025   14.599    0.000    0.549    0.549
##     attach            0.542    0.023   23.370    0.000    0.755    0.755
##     social            0.142    0.025    5.778    0.000    0.209    0.209
##   craving ~~
##     taste             0.128    0.023    5.651    0.000    0.222    0.222
##     cog_e             0.363    0.023   15.632    0.000    0.586    0.586
##     w_control         0.264    0.027    9.873    0.000    0.436    0.436
##     cue               0.395    0.025   16.098    0.000    0.808    0.808
##     affect            0.423    0.023   17.992    0.000    0.708    0.708
##     attach            0.556    0.023   24.425    0.000    0.871    0.871
##     social            0.195    0.024    8.071    0.000    0.322    0.322
##   taste ~~
##     cog_e             0.121    0.023    5.341    0.000    0.202    0.202
##     w_control         0.118    0.025    4.751    0.000    0.202    0.202
##     cue               0.153    0.023    6.559    0.000    0.324    0.324
##     affect            0.244    0.023   10.581    0.000    0.424    0.424
##     attach            0.196    0.025    7.964    0.000    0.318    0.318
##     social            0.095    0.022    4.243    0.000    0.161    0.161
##   cog_e ~~
##     w_control         0.352    0.028   12.617    0.000    0.559    0.559
##     cue               0.302    0.024   12.380    0.000    0.596    0.596
##     affect            0.465    0.024   19.664    0.000    0.751    0.751
##     attach            0.439    0.024   17.979    0.000    0.663    0.663
##     social            0.071    0.025    2.876    0.004    0.113    0.113
##   w_control ~~
##     cue               0.318    0.027   11.614    0.000    0.639    0.639
##     affect            0.276    0.028    9.721    0.000    0.454    0.454
##     attach            0.334    0.029   11.493    0.000    0.514    0.514
##     social            0.136    0.027    5.072    0.000    0.220    0.220
##   cue ~~
##     affect            0.359    0.025   14.325    0.000    0.733    0.733
##     attach            0.439    0.025   17.344    0.000    0.839    0.839
##     social            0.249    0.023   10.587    0.000    0.500    0.500
##   affect ~~
##     attach            0.484    0.025   19.153    0.000    0.757    0.757
##     social            0.079    0.025    3.194    0.001    0.130    0.130
##   attach ~~
##     social            0.170    0.026    6.562    0.000    0.262    0.262
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     auto1|t1         -1.244    0.059  -21.018    0.000   -1.244   -1.244
##     auto1|t2         -0.924    0.052  -17.846    0.000   -0.924   -0.924
```

```
##     auto1|t3          -0.668    0.048  -13.917    0.000   -0.668   -0.668
##     auto1|t4          -0.177    0.044   -3.978    0.000   -0.177   -0.177
##     auto1|t5           0.295    0.045    6.577    0.000    0.295    0.295
##     auto1|t6           0.828    0.050   16.508    0.000    0.828    0.828
##     auto2|t1          -1.019    0.054  -18.999    0.000   -1.019   -1.019
##     auto2|t2          -0.675    0.048  -14.052    0.000   -0.675   -0.675
##     auto2|t3          -0.498    0.046  -10.763    0.000   -0.498   -0.498
##     auto2|t4          -0.123    0.044   -2.782    0.005   -0.123   -0.123
##     auto2|t5           0.253    0.045    5.665    0.000    0.253    0.253
##     auto2|t6           0.731    0.049   14.993    0.000    0.731    0.731
##     auto3|t1          -1.035    0.054  -19.175    0.000   -1.035   -1.035
##     auto3|t2          -0.842    0.050  -16.703    0.000   -0.842   -0.842
##     auto3|t3          -0.622    0.047  -13.102    0.000   -0.622   -0.622
##     auto3|t4          -0.318    0.045   -7.068    0.000   -0.318   -0.318
##     auto3|t5           0.155    0.044    3.486    0.000    0.155    0.155
##     auto3|t6           0.618    0.047   13.033    0.000    0.618    0.618
##     auto4|t1          -0.988    0.053  -18.642    0.000   -0.988   -0.988
##     auto4|t2          -0.764    0.049  -15.525    0.000   -0.764   -0.764
##     auto4|t3          -0.562    0.047  -12.006    0.000   -0.562   -0.562
##     auto4|t4          -0.231    0.045   -5.173    0.000   -0.231   -0.231
##     auto4|t5           0.190    0.044    4.260    0.000    0.190    0.190
##     auto4|t6           0.656    0.048   13.714    0.000    0.656    0.656
##     loc1|t1           -1.084    0.055  -19.688    0.000   -1.084   -1.084
##     loc1|t2           -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     loc1|t3           -0.484    0.046  -10.486    0.000   -0.484   -0.484
##     loc1|t4           -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     loc1|t5            0.312    0.045    6.928    0.000    0.312    0.312
##     loc1|t6            0.882    0.051   17.279    0.000    0.882    0.882
##     loc2|t1           -0.672    0.048  -13.985    0.000   -0.672   -0.672
##     loc2|t2           -0.459    0.046  -10.000    0.000   -0.459   -0.459
##     loc2|t3           -0.231    0.045   -5.173    0.000   -0.231   -0.231
##     loc2|t4            0.083    0.044    1.867    0.062    0.083    0.083
##     loc2|t5            0.439    0.046    9.583    0.000    0.439    0.439
##     loc2|t6            0.798    0.050   16.052    0.000    0.798    0.798
##     loc3|t1           -1.237    0.059  -20.972    0.000   -1.237   -1.237
##     loc3|t2           -0.983    0.053  -18.582    0.000   -0.983   -0.983
##     loc3|t3           -0.752    0.049  -15.326    0.000   -0.752   -0.752
##     loc3|t4           -0.391    0.045   -8.607    0.000   -0.391   -0.391
##     loc3|t5            0.045    0.044    1.021    0.307    0.045    0.045
##     loc3|t6            0.548    0.047   11.730    0.000    0.548    0.548
##     loc4|t1           -0.618    0.047  -13.033    0.000   -0.618   -0.618
##     loc4|t2           -0.315    0.045   -6.998    0.000   -0.315   -0.315
##     loc4|t3           -0.058    0.044   -1.303    0.193   -0.058   -0.058
##     loc4|t4            0.228    0.045    5.103    0.000    0.228    0.228
##     loc4|t5            0.614    0.047   12.965    0.000    0.614    0.614
##     loc4|t6            1.009    0.053   18.881    0.000    1.009    1.009
##     tol1|t1           -0.505    0.046  -10.901    0.000   -0.505   -0.505
##     tol1|t2           -0.312    0.045   -6.928    0.000   -0.312   -0.312
##     tol1|t3           -0.202    0.045   -4.541    0.000   -0.202   -0.202
##     tol1|t4           -0.045    0.044   -1.021    0.307   -0.045   -0.045
##     tol1|t5            0.171    0.044    3.838    0.000    0.171    0.171
##     tol1|t6            0.498    0.046   10.763    0.000    0.498    0.498
##     tol2|t1           -0.973    0.053  -18.461    0.000   -0.973   -0.973
##     tol2|t2           -0.683    0.048  -14.187    0.000   -0.683   -0.683
```

```
##     tol2|t3          -0.335    0.045   -7.418    0.000   -0.335   -0.335
##     tol2|t4          -0.098    0.044   -2.219    0.027   -0.098   -0.098
##     tol2|t5           0.315    0.045    6.998    0.000    0.315    0.315
##     tol2|t6           0.820    0.050   16.378    0.000    0.820    0.820
##     tol3|t1          -0.446    0.046   -9.722    0.000   -0.446   -0.446
##     tol3|t2          -0.273    0.045   -6.086    0.000   -0.273   -0.273
##     tol3|t3          -0.161    0.044   -3.627    0.000   -0.161   -0.161
##     tol3|t4          -0.030    0.044   -0.669    0.503   -0.030   -0.030
##     tol3|t5           0.183    0.044    4.119    0.000    0.183    0.183
##     tol3|t6           0.523    0.046   11.247    0.000    0.523    0.523
##     tol4|t1          -0.707    0.048  -14.591    0.000   -0.707   -0.707
##     tol4|t2          -0.449    0.046   -9.791    0.000   -0.449   -0.449
##     tol4|t3          -0.221    0.045   -4.963    0.000   -0.221   -0.221
##     tol4|t4           0.079    0.044    1.796    0.072    0.079    0.079
##     tol4|t5           0.432    0.046    9.443    0.000    0.432    0.432
##     tol4|t6           0.828    0.050   16.508    0.000    0.828    0.828
##     craving1|t1      -1.237    0.059  -20.972    0.000   -1.237   -1.237
##     craving1|t2      -0.949    0.052  -18.156    0.000   -0.949   -0.949
##     craving1|t3      -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     craving1|t4      -0.418    0.046   -9.165    0.000   -0.418   -0.418
##     craving1|t5      -0.064    0.044   -1.444    0.149   -0.064   -0.064
##     craving1|t6       0.388    0.045    8.537    0.000    0.388    0.388
##     craving2|t1      -1.984    0.096  -20.642    0.000   -1.984   -1.984
##     craving2|t2      -1.754    0.080  -21.820    0.000   -1.754   -1.754
##     craving2|t3      -1.613    0.073  -22.104    0.000   -1.613   -1.613
##     craving2|t4      -1.217    0.058  -20.832    0.000   -1.217   -1.217
##     craving2|t5      -0.614    0.047  -12.965    0.000   -0.614   -0.614
##     craving2|t6       0.002    0.044    0.035    0.972    0.002    0.002
##     craving3|t1      -1.636    0.074  -22.082    0.000   -1.636   -1.636
##     craving3|t2      -1.392    0.064  -21.786    0.000   -1.392   -1.392
##     craving3|t3      -1.179    0.057  -20.539    0.000   -1.179   -1.179
##     craving3|t4      -0.851    0.051  -16.832    0.000   -0.851   -0.851
##     craving3|t5      -0.435    0.046   -9.513    0.000   -0.435   -0.435
##     craving3|t6       0.073    0.044    1.655    0.098    0.073    0.073
##     craving4|t1      -0.973    0.053  -18.461    0.000   -0.973   -0.973
##     craving4|t2      -0.683    0.048  -14.187    0.000   -0.683   -0.683
##     craving4|t3      -0.512    0.046  -11.040    0.000   -0.512   -0.512
##     craving4|t4      -0.269    0.045   -6.016    0.000   -0.269   -0.269
##     craving4|t5       0.117    0.044    2.641    0.008    0.117    0.117
##     craving4|t6       0.618    0.047   13.033    0.000    0.618    0.618
##     taste1|t1        -1.329    0.062  -21.516    0.000   -1.329   -1.329
##     taste1|t2        -0.993    0.053  -18.703    0.000   -0.993   -0.993
##     taste1|t3        -0.777    0.049  -15.723    0.000   -0.777   -0.777
##     taste1|t4        -0.331    0.045   -7.348    0.000   -0.331   -0.331
##     taste1|t5         0.036    0.044    0.810    0.418    0.036    0.036
##     taste1|t6         0.491    0.046   10.624    0.000    0.491    0.491
##     taste2|t1        -1.292    0.061  -21.320    0.000   -1.292   -1.292
##     taste2|t2        -1.051    0.054  -19.349    0.000   -1.051   -1.051
##     taste2|t3        -0.768    0.049  -15.591    0.000   -0.768   -0.768
##     taste2|t4        -0.354    0.045   -7.838    0.000   -0.354   -0.354
##     taste2|t5         0.067    0.044    1.515    0.130    0.067    0.067
##     taste2|t6         0.629    0.048   13.238    0.000    0.629    0.629
##     taste3|t1        -1.154    0.057  -20.335    0.000   -1.154   -1.154
##     taste3|t2        -0.958    0.052  -18.278    0.000   -0.958   -0.958
```

```
##     taste3|t3       -0.748    0.049  -15.259    0.000   -0.748   -0.748
##     taste3|t4       -0.422    0.046   -9.234    0.000   -0.422   -0.422
##     taste3|t5        0.014    0.044    0.317    0.751    0.014    0.014
##     taste3|t6        0.480    0.046   10.416    0.000    0.480    0.480
##     cog_e1|t1       -0.873    0.051  -17.152    0.000   -0.873   -0.873
##     cog_e1|t2       -0.707    0.048  -14.591    0.000   -0.707   -0.707
##     cog_e1|t3       -0.541    0.047  -11.592    0.000   -0.541   -0.541
##     cog_e1|t4       -0.196    0.045   -4.400    0.000   -0.196   -0.196
##     cog_e1|t5        0.180    0.044    4.049    0.000    0.180    0.180
##     cog_e1|t6        0.752    0.049   15.326    0.000    0.752    0.752
##     cog_e2|t1       -0.824    0.050  -16.443    0.000   -0.824   -0.824
##     cog_e2|t2       -0.668    0.048  -13.917    0.000   -0.668   -0.668
##     cog_e2|t3       -0.487    0.046  -10.555    0.000   -0.487   -0.487
##     cog_e2|t4       -0.158    0.044   -3.556    0.000   -0.158   -0.158
##     cog_e2|t5        0.292    0.045    6.507    0.000    0.292    0.292
##     cog_e2|t6        0.833    0.050   16.573    0.000    0.833    0.833
##     cog_e3|t1       -1.079    0.055  -19.632    0.000   -1.079   -1.079
##     cog_e3|t2       -0.924    0.052  -17.846    0.000   -0.924   -0.924
##     cog_e3|t3       -0.719    0.049  -14.792    0.000   -0.719   -0.719
##     cog_e3|t4       -0.345    0.045   -7.628    0.000   -0.345   -0.345
##     cog_e3|t5        0.120    0.044    2.712    0.007    0.120    0.120
##     cog_e3|t6        0.695    0.048   14.389    0.000    0.695    0.695
##     w_control1|t1    0.002    0.044    0.035    0.972    0.002    0.002
##     w_control1|t2    0.241    0.045    5.384    0.000    0.241    0.241
##     w_control1|t3    0.439    0.046    9.583    0.000    0.439    0.439
##     w_control1|t4    0.752    0.049   15.326    0.000    0.752    0.752
##     w_control1|t5    1.062    0.055   19.463    0.000    1.062    1.062
##     w_control1|t6    1.392    0.064   21.786    0.000    1.392    1.392
##     w_control2|t1    0.266    0.045    5.946    0.000    0.266    0.266
##     w_control2|t2    0.466    0.046   10.139    0.000    0.466    0.466
##     w_control2|t3    0.711    0.049   14.658    0.000    0.711    0.711
##     w_control2|t4    1.230    0.059   20.926    0.000    1.230    1.230
##     w_control2|t5    1.384    0.064   21.755    0.000    1.384    1.384
##     w_control2|t6    1.660    0.075   22.048    0.000    1.660    1.660
##     w_control3|t1   -0.401    0.046   -8.816    0.000   -0.401   -0.401
##     w_control3|t2   -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     w_control3|t3    0.045    0.044    1.021    0.307    0.045    0.045
##     w_control3|t4    0.439    0.046    9.583    0.000    0.439    0.439
##     w_control3|t5    0.781    0.049   15.789    0.000    0.781    0.781
##     w_control3|t6    1.173    0.057   20.489    0.000    1.173    1.173
##     cue1|t1         -0.807    0.050  -16.183    0.000   -0.807   -0.807
##     cue1|t2         -0.629    0.048  -13.238    0.000   -0.629   -0.629
##     cue1|t3         -0.480    0.046  -10.416    0.000   -0.480   -0.480
##     cue1|t4         -0.139    0.044   -3.134    0.002   -0.139   -0.139
##     cue1|t5          0.221    0.045    4.963    0.000    0.221    0.221
##     cue1|t6          0.644    0.048   13.510    0.000    0.644    0.644
##     cue2|t1         -0.785    0.050  -15.855    0.000   -0.785   -0.785
##     cue2|t2         -0.530    0.047  -11.385    0.000   -0.530   -0.530
##     cue2|t3         -0.331    0.045   -7.348    0.000   -0.331   -0.331
##     cue2|t4          0.008    0.044    0.176    0.860    0.008    0.008
##     cue2|t5          0.341    0.045    7.558    0.000    0.341    0.341
##     cue2|t6          0.748    0.049   15.259    0.000    0.748    0.748
##     cue3|t1         -1.498    0.068  -22.059    0.000   -1.498   -1.498
##     cue3|t2         -1.329    0.062  -21.516    0.000   -1.329   -1.329
```

```
##     cue3|t3          -1.142    0.056  -20.231    0.000   -1.142   -1.142
##     cue3|t4          -0.864    0.051  -17.024    0.000   -0.864   -0.864
##     cue3|t5          -0.401    0.046   -8.816    0.000   -0.401   -0.401
##     cue3|t6           0.161    0.044    3.627    0.000    0.161    0.161
##     affect1|t1       -1.299    0.061  -21.360    0.000   -1.299   -1.299
##     affect1|t2       -1.096    0.055  -19.799    0.000   -1.096   -1.096
##     affect1|t3       -0.901    0.051  -17.533    0.000   -0.901   -0.901
##     affect1|t4       -0.562    0.047  -12.006    0.000   -0.562   -0.562
##     affect1|t5       -0.089    0.044   -2.007    0.045   -0.089   -0.089
##     affect1|t6        0.459    0.046   10.000    0.000    0.459    0.459
##     affect2|t1       -1.590    0.072  -22.115    0.000   -1.590   -1.590
##     affect2|t2       -1.392    0.064  -21.786    0.000   -1.392   -1.392
##     affect2|t3       -1.160    0.057  -20.387    0.000   -1.160   -1.160
##     affect2|t4       -0.855    0.051  -16.896    0.000   -0.855   -0.855
##     affect2|t5       -0.279    0.045   -6.227    0.000   -0.279   -0.279
##     affect2|t6        0.305    0.045    6.788    0.000    0.305    0.305
##     affect3|t1       -1.590    0.072  -22.115    0.000   -1.590   -1.590
##     affect3|t2       -1.400    0.064  -21.815    0.000   -1.400   -1.400
##     affect3|t3       -1.185    0.058  -20.589    0.000   -1.185   -1.185
##     affect3|t4       -0.695    0.048  -14.389    0.000   -0.695   -0.695
##     affect3|t5       -0.130    0.044   -2.923    0.003   -0.130   -0.130
##     affect3|t6        0.415    0.046    9.095    0.000    0.415    0.415
##     attach1|t1       -1.166    0.057  -20.438    0.000   -1.166   -1.166
##     attach1|t2       -1.035    0.054  -19.175    0.000   -1.035   -1.035
##     attach1|t3       -0.873    0.051  -17.152    0.000   -0.873   -0.873
##     attach1|t4       -0.588    0.047  -12.486    0.000   -0.588   -0.588
##     attach1|t5       -0.234    0.045   -5.244    0.000   -0.234   -0.234
##     attach1|t6        0.139    0.044    3.134    0.002    0.139    0.139
##     attach2|t1       -0.939    0.052  -18.032    0.000   -0.939   -0.939
##     attach2|t2       -0.735    0.049  -15.060    0.000   -0.735   -0.735
##     attach2|t3       -0.551    0.047  -11.799    0.000   -0.551   -0.551
##     attach2|t4       -0.247    0.045   -5.525    0.000   -0.247   -0.247
##     attach2|t5        0.092    0.044    2.078    0.038    0.092    0.092
##     attach2|t6        0.498    0.046   10.763    0.000    0.498    0.498
##     attach3|t1       -0.864    0.051  -17.024    0.000   -0.864   -0.864
##     attach3|t2       -0.660    0.048  -13.782    0.000   -0.660   -0.660
##     attach3|t3       -0.466    0.046  -10.139    0.000   -0.466   -0.466
##     attach3|t4       -0.145    0.044   -3.275    0.001   -0.145   -0.145
##     attach3|t5        0.244    0.045    5.454    0.000    0.244    0.244
##     attach3|t6        0.618    0.047   13.033    0.000    0.618    0.618
##     social1|t1       -1.508    0.068  -22.073    0.000   -1.508   -1.508
##     social1|t2       -1.185    0.058  -20.589    0.000   -1.185   -1.185
##     social1|t3       -0.949    0.052  -18.156    0.000   -0.949   -0.949
##     social1|t4       -0.508    0.046  -10.971    0.000   -0.508   -0.508
##     social1|t5       -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     social1|t6        0.378    0.045    8.327    0.000    0.378    0.378
##     social2|t1       -1.030    0.054  -19.117    0.000   -1.030   -1.030
##     social2|t2       -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     social2|t3       -0.555    0.047  -11.868    0.000   -0.555   -0.555
##     social2|t4       -0.177    0.044   -3.978    0.000   -0.177   -0.177
##     social2|t5        0.253    0.045    5.665    0.000    0.253    0.253
##     social2|t6        0.798    0.050   16.052    0.000    0.798    0.798
##     social3|t1       -1.470    0.067  -22.009    0.000   -1.470   -1.470
##     social3|t2       -1.124    0.056  -20.072    0.000   -1.124   -1.124
```

```
##     social3|t3       -0.878    0.051  -17.216    0.000   -0.878   -0.878
##     social3|t4       -0.494    0.046  -10.694    0.000   -0.494   -0.494
##     social3|t5       -0.092    0.044   -2.078    0.038   -0.092   -0.092
##     social3|t6        0.603    0.047   12.760    0.000    0.603    0.603
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .auto1             0.617                               0.617    0.617
##    .auto2             0.292                               0.292    0.292
##    .auto3             0.212                               0.212    0.212
##    .auto4             0.163                               0.163    0.163
##    .loc1              0.423                               0.423    0.423
##    .loc2              0.375                               0.375    0.375
##    .loc3              0.275                               0.275    0.275
##    .loc4              0.239                               0.239    0.239
##    .tol1              0.246                               0.246    0.246
##    .tol2              0.313                               0.313    0.313
##    .tol3              0.301                               0.301    0.301
##    .tol4              0.106                               0.106    0.106
##    .craving1          0.404                               0.404    0.404
##    .craving2          0.515                               0.515    0.515
##    .craving3          0.333                               0.333    0.333
##    .craving4          0.273                               0.273    0.273
##    .taste1            0.443                               0.443    0.443
##    .taste2            0.096                               0.096    0.096
##    .taste3            0.416                               0.416    0.416
##    .cog_e1            0.357                               0.357    0.357
##    .cog_e2            0.255                               0.255    0.255
##    .cog_e3            0.159                               0.159    0.159
##    .w_control1        0.383                               0.383    0.383
##    .w_control2        0.675                               0.675    0.675
##    .w_control3        0.409                               0.409    0.409
##    .cue1              0.599                               0.599    0.599
##    .cue2              0.833                               0.833    0.833
##    .cue3              0.617                               0.617    0.617
##    .affect1           0.402                               0.402    0.402
##    .affect2           0.280                               0.280    0.280
##    .affect3           0.505                               0.505    0.505
##    .attach1           0.317                               0.317    0.317
##    .attach2           0.396                               0.396    0.396
##    .attach3           0.355                               0.355    0.355
##    .social1           0.382                               0.382    0.382
##    .social2           0.382                               0.382    0.382
##    .social3           0.136                               0.136    0.136
##     auto              0.383    0.032   11.962    0.000    1.000    1.000
##     loc               0.577    0.024   23.719    0.000    1.000    1.000
##     tol               0.754    0.024   30.795    0.000    1.000    1.000
##     craving           0.596    0.028   21.297    0.000    1.000    1.000
##     taste             0.557    0.036   15.488    0.000    1.000    1.000
##     cog_e             0.643    0.028   23.191    0.000    1.000    1.000
##     w_control         0.617    0.053   11.694    0.000    1.000    1.000
##     cue               0.401    0.040   10.015    0.000    1.000    1.000
##     affect            0.598    0.032   18.549    0.000    1.000    1.000
##     attach            0.683    0.032   21.336    0.000    1.000    1.000
```

```
##     social              0.618    0.034   18.306    0.000    1.000    1.000
```

5.1.9 WLSMV Model 4: Hierarchial factor solution

```
ord_mod4 <- '#1st order
             auto      =~ auto1 + auto2 + auto3 + auto4
             loc       =~ loc1 + loc2 + loc3 + loc4
             tol       =~ tol1 + tol2 + tol3 + tol4
             craving   =~ craving1 + craving2 + craving3 + craving4
             taste     =~ taste1 + taste2 + taste3
             cog_e     =~ cog_e1 + cog_e2 + cog_e3
             w_control =~ w_control1 + w_control2 + w_control3
             cue       =~ cue1 + cue2 + cue3
             affect    =~ affect1 + affect2 + affect3
             attach    =~ attach1 + attach2 + attach3
             social    =~ social1 + social2 +social3

             #2nd order
             pdm =~ auto + loc + tol + craving
             sdm =~ taste + cog_e + w_control + cue + affect + attach + social
             '
ord_mod4_fit <- cfa(ord_mod4, data = df_smoker, estimator = "WLSMV",
                    ordered = wisdm37_items)
```

```
## Warning: lavaan->lav_object_post_check():
##    some estimated lv variances are negative
```

```
summary(ord_mod4_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## Warning: lavaan->lav_object_post_check():
##    some estimated lv variances are negative
## Warning: lavaan->lav_object_post_check():
##    some estimated lv variances are negative
```

```
## lavaan 0.6-21 ended normally after 100 iterations
##
##   Estimator                                       DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                       271
##
##   Number of observations                           805
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                              2789.298    2813.601
##   Degrees of freedom                               617         617
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  1.112
##   Shift parameter                                          304.716
##     simple second-order correction
##
## Model Test Baseline Model:
##
```

```
##   Test statistic                             158009.714   38794.468
##   Degrees of freedom                               666         666
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.127
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.986       0.942
##   Tucker-Lewis Index (TLI)                       0.985       0.938
##
##   Robust Comparative Fit Index (CFI)                         0.836
##   Robust Tucker-Lewis Index (TLI)                            0.823
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.066       0.067
##   90 Percent confidence interval - lower         0.064       0.064
##   90 Percent confidence interval - upper         0.069       0.069
##   P-value H_0: RMSEA <= 0.050                    0.000       0.000
##   P-value H_0: RMSEA >= 0.080                    0.000       0.000
##
##   Robust RMSEA                                               0.085
##   90 Percent confidence interval - lower                     0.082
##   90 Percent confidence interval - upper                     0.089
##   P-value H_0: Robust RMSEA <= 0.050                         0.000
##   P-value H_0: Robust RMSEA >= 0.080                         0.997
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.061       0.061
##
## Parameter Estimates:
##
##   Parameterization                               Delta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model         Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   auto =~
##     auto1             1.000                                0.618    0.618
##     auto2             1.362    0.059   23.241    0.000    0.842    0.842
##     auto3             1.437    0.061   23.474    0.000    0.888    0.888
##     auto4             1.482    0.067   22.276    0.000    0.915    0.915
##   loc =~
##     loc1              1.000                                0.758    0.758
##     loc2              1.042    0.025   42.044    0.000    0.789    0.789
##     loc3              1.124    0.026   43.982    0.000    0.851    0.851
##     loc4              1.156    0.025   46.359    0.000    0.876    0.876
##   tol =~
##     tol1              1.000                                0.868    0.868
##     tol2              0.954    0.022   44.356    0.000    0.828    0.828
##     tol3              0.961    0.023   42.296    0.000    0.834    0.834
```

```
##     tol4              1.093    0.020   53.984    0.000    0.948    0.948
##   craving =~
##     craving1          1.000                               0.773    0.773
##     craving2          0.900    0.032   28.127    0.000    0.696    0.696
##     craving3          1.058    0.030   34.760    0.000    0.817    0.817
##     craving4          1.103    0.028   38.852    0.000    0.852    0.852
##   taste =~
##     taste1            1.000                               0.746    0.746
##     taste2            1.295    0.070   18.560    0.000    0.966    0.966
##     taste3            1.003    0.042   23.707    0.000    0.749    0.749
##   cog_e =~
##     cog_e1            1.000                               0.800    0.800
##     cog_e2            1.078    0.029   36.879    0.000    0.862    0.862
##     cog_e3            1.151    0.032   35.821    0.000    0.920    0.920
##   w_control =~
##     w_control1        1.000                               0.784    0.784
##     w_control2        0.726    0.064   11.265    0.000    0.570    0.570
##     w_control3        0.983    0.065   15.034    0.000    0.771    0.771
##   cue =~
##     cue1              1.000                               0.637    0.637
##     cue2              0.641    0.058   11.087    0.000    0.408    0.408
##     cue3              0.966    0.065   14.781    0.000    0.615    0.615
##   affect =~
##     affect1           1.000                               0.772    0.772
##     affect2           1.103    0.039   28.316    0.000    0.852    0.852
##     affect3           0.908    0.041   21.978    0.000    0.701    0.701
##   attach =~
##     attach1           1.000                               0.828    0.828
##     attach2           0.937    0.029   32.724    0.000    0.776    0.776
##     attach3           0.969    0.031   31.548    0.000    0.803    0.803
##   social =~
##     social1           1.000                               0.780    0.780
##     social2           1.006    0.035   28.387    0.000    0.785    0.785
##     social3           1.197    0.054   22.288    0.000    0.934    0.934
##   pdm =~
##     auto              1.000                               0.748    0.748
##     loc               1.645    0.091   18.061    0.000    1.003    1.003
##     tol               1.645    0.089   18.386    0.000    0.876    0.876
##     craving           1.607    0.095   16.888    0.000    0.961    0.961
##   sdm =~
##     taste             1.000                               0.290    0.290
##     cog_e             2.743    0.358    7.672    0.000    0.742    0.742
##     w_control         2.253    0.318    7.077    0.000    0.621    0.621
##     cue               2.647    0.361    7.335    0.000    0.899    0.899
##     affect            2.887    0.359    8.044    0.000    0.808    0.808
##     attach            3.820    0.487    7.840    0.000    0.997    0.997
##     social            1.137    0.195    5.817    0.000    0.315    0.315
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   pdm ~~
##     sdm               0.084    0.012    7.049    0.000    0.841    0.841
##
## Thresholds:
```

```
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     auto1|t1         -1.244    0.059  -21.018    0.000   -1.244   -1.244
##     auto1|t2         -0.924    0.052  -17.846    0.000   -0.924   -0.924
##     auto1|t3         -0.668    0.048  -13.917    0.000   -0.668   -0.668
##     auto1|t4         -0.177    0.044   -3.978    0.000   -0.177   -0.177
##     auto1|t5          0.295    0.045    6.577    0.000    0.295    0.295
##     auto1|t6          0.828    0.050   16.508    0.000    0.828    0.828
##     auto2|t1         -1.019    0.054  -18.999    0.000   -1.019   -1.019
##     auto2|t2         -0.675    0.048  -14.052    0.000   -0.675   -0.675
##     auto2|t3         -0.498    0.046  -10.763    0.000   -0.498   -0.498
##     auto2|t4         -0.123    0.044   -2.782    0.005   -0.123   -0.123
##     auto2|t5          0.253    0.045    5.665    0.000    0.253    0.253
##     auto2|t6          0.731    0.049   14.993    0.000    0.731    0.731
##     auto3|t1         -1.035    0.054  -19.175    0.000   -1.035   -1.035
##     auto3|t2         -0.842    0.050  -16.703    0.000   -0.842   -0.842
##     auto3|t3         -0.622    0.047  -13.102    0.000   -0.622   -0.622
##     auto3|t4         -0.318    0.045   -7.068    0.000   -0.318   -0.318
##     auto3|t5          0.155    0.044    3.486    0.000    0.155    0.155
##     auto3|t6          0.618    0.047   13.033    0.000    0.618    0.618
##     auto4|t1         -0.988    0.053  -18.642    0.000   -0.988   -0.988
##     auto4|t2         -0.764    0.049  -15.525    0.000   -0.764   -0.764
##     auto4|t3         -0.562    0.047  -12.006    0.000   -0.562   -0.562
##     auto4|t4         -0.231    0.045   -5.173    0.000   -0.231   -0.231
##     auto4|t5          0.190    0.044    4.260    0.000    0.190    0.190
##     auto4|t6          0.656    0.048   13.714    0.000    0.656    0.656
##     loc1|t1          -1.084    0.055  -19.688    0.000   -1.084   -1.084
##     loc1|t2          -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     loc1|t3          -0.484    0.046  -10.486    0.000   -0.484   -0.484
##     loc1|t4          -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     loc1|t5           0.312    0.045    6.928    0.000    0.312    0.312
##     loc1|t6           0.882    0.051   17.279    0.000    0.882    0.882
##     loc2|t1          -0.672    0.048  -13.985    0.000   -0.672   -0.672
##     loc2|t2          -0.459    0.046  -10.000    0.000   -0.459   -0.459
##     loc2|t3          -0.231    0.045   -5.173    0.000   -0.231   -0.231
##     loc2|t4           0.083    0.044    1.867    0.062    0.083    0.083
##     loc2|t5           0.439    0.046    9.583    0.000    0.439    0.439
##     loc2|t6           0.798    0.050   16.052    0.000    0.798    0.798
##     loc3|t1          -1.237    0.059  -20.972    0.000   -1.237   -1.237
##     loc3|t2          -0.983    0.053  -18.582    0.000   -0.983   -0.983
##     loc3|t3          -0.752    0.049  -15.326    0.000   -0.752   -0.752
##     loc3|t4          -0.391    0.045   -8.607    0.000   -0.391   -0.391
##     loc3|t5           0.045    0.044    1.021    0.307    0.045    0.045
##     loc3|t6           0.548    0.047   11.730    0.000    0.548    0.548
##     loc4|t1          -0.618    0.047  -13.033    0.000   -0.618   -0.618
##     loc4|t2          -0.315    0.045   -6.998    0.000   -0.315   -0.315
##     loc4|t3          -0.058    0.044   -1.303    0.193   -0.058   -0.058
##     loc4|t4           0.228    0.045    5.103    0.000    0.228    0.228
##     loc4|t5           0.614    0.047   12.965    0.000    0.614    0.614
##     loc4|t6           1.009    0.053   18.881    0.000    1.009    1.009
##     tol1|t1          -0.505    0.046  -10.901    0.000   -0.505   -0.505
##     tol1|t2          -0.312    0.045   -6.928    0.000   -0.312   -0.312
##     tol1|t3          -0.202    0.045   -4.541    0.000   -0.202   -0.202
##     tol1|t4          -0.045    0.044   -1.021    0.307   -0.045   -0.045
##     tol1|t5           0.171    0.044    3.838    0.000    0.171    0.171
```

```
##     tol1|t6          0.498    0.046   10.763    0.000    0.498    0.498
##     tol2|t1         -0.973    0.053  -18.461    0.000   -0.973   -0.973
##     tol2|t2         -0.683    0.048  -14.187    0.000   -0.683   -0.683
##     tol2|t3         -0.335    0.045   -7.418    0.000   -0.335   -0.335
##     tol2|t4         -0.098    0.044   -2.219    0.027   -0.098   -0.098
##     tol2|t5          0.315    0.045    6.998    0.000    0.315    0.315
##     tol2|t6          0.820    0.050   16.378    0.000    0.820    0.820
##     tol3|t1         -0.446    0.046   -9.722    0.000   -0.446   -0.446
##     tol3|t2         -0.273    0.045   -6.086    0.000   -0.273   -0.273
##     tol3|t3         -0.161    0.044   -3.627    0.000   -0.161   -0.161
##     tol3|t4         -0.030    0.044   -0.669    0.503   -0.030   -0.030
##     tol3|t5          0.183    0.044    4.119    0.000    0.183    0.183
##     tol3|t6          0.523    0.046   11.247    0.000    0.523    0.523
##     tol4|t1         -0.707    0.048  -14.591    0.000   -0.707   -0.707
##     tol4|t2         -0.449    0.046   -9.791    0.000   -0.449   -0.449
##     tol4|t3         -0.221    0.045   -4.963    0.000   -0.221   -0.221
##     tol4|t4          0.079    0.044    1.796    0.072    0.079    0.079
##     tol4|t5          0.432    0.046    9.443    0.000    0.432    0.432
##     tol4|t6          0.828    0.050   16.508    0.000    0.828    0.828
##     craving1|t1     -1.237    0.059  -20.972    0.000   -1.237   -1.237
##     craving1|t2     -0.949    0.052  -18.156    0.000   -0.949   -0.949
##     craving1|t3     -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     craving1|t4     -0.418    0.046   -9.165    0.000   -0.418   -0.418
##     craving1|t5     -0.064    0.044   -1.444    0.149   -0.064   -0.064
##     craving1|t6      0.388    0.045    8.537    0.000    0.388    0.388
##     craving2|t1     -1.984    0.096  -20.642    0.000   -1.984   -1.984
##     craving2|t2     -1.754    0.080  -21.820    0.000   -1.754   -1.754
##     craving2|t3     -1.613    0.073  -22.104    0.000   -1.613   -1.613
##     craving2|t4     -1.217    0.058  -20.832    0.000   -1.217   -1.217
##     craving2|t5     -0.614    0.047  -12.965    0.000   -0.614   -0.614
##     craving2|t6      0.002    0.044    0.035    0.972    0.002    0.002
##     craving3|t1     -1.636    0.074  -22.082    0.000   -1.636   -1.636
##     craving3|t2     -1.392    0.064  -21.786    0.000   -1.392   -1.392
##     craving3|t3     -1.179    0.057  -20.539    0.000   -1.179   -1.179
##     craving3|t4     -0.851    0.051  -16.832    0.000   -0.851   -0.851
##     craving3|t5     -0.435    0.046   -9.513    0.000   -0.435   -0.435
##     craving3|t6      0.073    0.044    1.655    0.098    0.073    0.073
##     craving4|t1     -0.973    0.053  -18.461    0.000   -0.973   -0.973
##     craving4|t2     -0.683    0.048  -14.187    0.000   -0.683   -0.683
##     craving4|t3     -0.512    0.046  -11.040    0.000   -0.512   -0.512
##     craving4|t4     -0.269    0.045   -6.016    0.000   -0.269   -0.269
##     craving4|t5      0.117    0.044    2.641    0.008    0.117    0.117
##     craving4|t6      0.618    0.047   13.033    0.000    0.618    0.618
##     taste1|t1       -1.329    0.062  -21.516    0.000   -1.329   -1.329
##     taste1|t2       -0.993    0.053  -18.703    0.000   -0.993   -0.993
##     taste1|t3       -0.777    0.049  -15.723    0.000   -0.777   -0.777
##     taste1|t4       -0.331    0.045   -7.348    0.000   -0.331   -0.331
##     taste1|t5        0.036    0.044    0.810    0.418    0.036    0.036
##     taste1|t6        0.491    0.046   10.624    0.000    0.491    0.491
##     taste2|t1       -1.292    0.061  -21.320    0.000   -1.292   -1.292
##     taste2|t2       -1.051    0.054  -19.349    0.000   -1.051   -1.051
##     taste2|t3       -0.768    0.049  -15.591    0.000   -0.768   -0.768
##     taste2|t4       -0.354    0.045   -7.838    0.000   -0.354   -0.354
##     taste2|t5        0.067    0.044    1.515    0.130    0.067    0.067
```

```
##     taste2|t6         0.629    0.048   13.238    0.000    0.629    0.629
##     taste3|t1        -1.154    0.057  -20.335    0.000   -1.154   -1.154
##     taste3|t2        -0.958    0.052  -18.278    0.000   -0.958   -0.958
##     taste3|t3        -0.748    0.049  -15.259    0.000   -0.748   -0.748
##     taste3|t4        -0.422    0.046   -9.234    0.000   -0.422   -0.422
##     taste3|t5         0.014    0.044    0.317    0.751    0.014    0.014
##     taste3|t6         0.480    0.046   10.416    0.000    0.480    0.480
##     cog_e1|t1        -0.873    0.051  -17.152    0.000   -0.873   -0.873
##     cog_e1|t2        -0.707    0.048  -14.591    0.000   -0.707   -0.707
##     cog_e1|t3        -0.541    0.047  -11.592    0.000   -0.541   -0.541
##     cog_e1|t4        -0.196    0.045   -4.400    0.000   -0.196   -0.196
##     cog_e1|t5         0.180    0.044    4.049    0.000    0.180    0.180
##     cog_e1|t6         0.752    0.049   15.326    0.000    0.752    0.752
##     cog_e2|t1        -0.824    0.050  -16.443    0.000   -0.824   -0.824
##     cog_e2|t2        -0.668    0.048  -13.917    0.000   -0.668   -0.668
##     cog_e2|t3        -0.487    0.046  -10.555    0.000   -0.487   -0.487
##     cog_e2|t4        -0.158    0.044   -3.556    0.000   -0.158   -0.158
##     cog_e2|t5         0.292    0.045    6.507    0.000    0.292    0.292
##     cog_e2|t6         0.833    0.050   16.573    0.000    0.833    0.833
##     cog_e3|t1        -1.079    0.055  -19.632    0.000   -1.079   -1.079
##     cog_e3|t2        -0.924    0.052  -17.846    0.000   -0.924   -0.924
##     cog_e3|t3        -0.719    0.049  -14.792    0.000   -0.719   -0.719
##     cog_e3|t4        -0.345    0.045   -7.628    0.000   -0.345   -0.345
##     cog_e3|t5         0.120    0.044    2.712    0.007    0.120    0.120
##     cog_e3|t6         0.695    0.048   14.389    0.000    0.695    0.695
##     w_control1|t1     0.002    0.044    0.035    0.972    0.002    0.002
##     w_control1|t2     0.241    0.045    5.384    0.000    0.241    0.241
##     w_control1|t3     0.439    0.046    9.583    0.000    0.439    0.439
##     w_control1|t4     0.752    0.049   15.326    0.000    0.752    0.752
##     w_control1|t5     1.062    0.055   19.463    0.000    1.062    1.062
##     w_control1|t6     1.392    0.064   21.786    0.000    1.392    1.392
##     w_control2|t1     0.266    0.045    5.946    0.000    0.266    0.266
##     w_control2|t2     0.466    0.046   10.139    0.000    0.466    0.466
##     w_control2|t3     0.711    0.049   14.658    0.000    0.711    0.711
##     w_control2|t4     1.230    0.059   20.926    0.000    1.230    1.230
##     w_control2|t5     1.384    0.064   21.755    0.000    1.384    1.384
##     w_control2|t6     1.660    0.075   22.048    0.000    1.660    1.660
##     w_control3|t1    -0.401    0.046   -8.816    0.000   -0.401   -0.401
##     w_control3|t2    -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     w_control3|t3     0.045    0.044    1.021    0.307    0.045    0.045
##     w_control3|t4     0.439    0.046    9.583    0.000    0.439    0.439
##     w_control3|t5     0.781    0.049   15.789    0.000    0.781    0.781
##     w_control3|t6     1.173    0.057   20.489    0.000    1.173    1.173
##     cue1|t1          -0.807    0.050  -16.183    0.000   -0.807   -0.807
##     cue1|t2          -0.629    0.048  -13.238    0.000   -0.629   -0.629
##     cue1|t3          -0.480    0.046  -10.416    0.000   -0.480   -0.480
##     cue1|t4          -0.139    0.044   -3.134    0.002   -0.139   -0.139
##     cue1|t5           0.221    0.045    4.963    0.000    0.221    0.221
##     cue1|t6           0.644    0.048   13.510    0.000    0.644    0.644
##     cue2|t1          -0.785    0.050  -15.855    0.000   -0.785   -0.785
##     cue2|t2          -0.530    0.047  -11.385    0.000   -0.530   -0.530
##     cue2|t3          -0.331    0.045   -7.348    0.000   -0.331   -0.331
##     cue2|t4           0.008    0.044    0.176    0.860    0.008    0.008
##     cue2|t5           0.341    0.045    7.558    0.000    0.341    0.341
```

```
##     cue2|t6          0.748    0.049   15.259    0.000    0.748    0.748
##     cue3|t1         -1.498    0.068  -22.059    0.000   -1.498   -1.498
##     cue3|t2         -1.329    0.062  -21.516    0.000   -1.329   -1.329
##     cue3|t3         -1.142    0.056  -20.231    0.000   -1.142   -1.142
##     cue3|t4         -0.864    0.051  -17.024    0.000   -0.864   -0.864
##     cue3|t5         -0.401    0.046   -8.816    0.000   -0.401   -0.401
##     cue3|t6          0.161    0.044    3.627    0.000    0.161    0.161
##     affect1|t1      -1.299    0.061  -21.360    0.000   -1.299   -1.299
##     affect1|t2      -1.096    0.055  -19.799    0.000   -1.096   -1.096
##     affect1|t3      -0.901    0.051  -17.533    0.000   -0.901   -0.901
##     affect1|t4      -0.562    0.047  -12.006    0.000   -0.562   -0.562
##     affect1|t5      -0.089    0.044   -2.007    0.045   -0.089   -0.089
##     affect1|t6       0.459    0.046   10.000    0.000    0.459    0.459
##     affect2|t1      -1.590    0.072  -22.115    0.000   -1.590   -1.590
##     affect2|t2      -1.392    0.064  -21.786    0.000   -1.392   -1.392
##     affect2|t3      -1.160    0.057  -20.387    0.000   -1.160   -1.160
##     affect2|t4      -0.855    0.051  -16.896    0.000   -0.855   -0.855
##     affect2|t5      -0.279    0.045   -6.227    0.000   -0.279   -0.279
##     affect2|t6       0.305    0.045    6.788    0.000    0.305    0.305
##     affect3|t1      -1.590    0.072  -22.115    0.000   -1.590   -1.590
##     affect3|t2      -1.400    0.064  -21.815    0.000   -1.400   -1.400
##     affect3|t3      -1.185    0.058  -20.589    0.000   -1.185   -1.185
##     affect3|t4      -0.695    0.048  -14.389    0.000   -0.695   -0.695
##     affect3|t5      -0.130    0.044   -2.923    0.003   -0.130   -0.130
##     affect3|t6       0.415    0.046    9.095    0.000    0.415    0.415
##     attach1|t1      -1.166    0.057  -20.438    0.000   -1.166   -1.166
##     attach1|t2      -1.035    0.054  -19.175    0.000   -1.035   -1.035
##     attach1|t3      -0.873    0.051  -17.152    0.000   -0.873   -0.873
##     attach1|t4      -0.588    0.047  -12.486    0.000   -0.588   -0.588
##     attach1|t5      -0.234    0.045   -5.244    0.000   -0.234   -0.234
##     attach1|t6       0.139    0.044    3.134    0.002    0.139    0.139
##     attach2|t1      -0.939    0.052  -18.032    0.000   -0.939   -0.939
##     attach2|t2      -0.735    0.049  -15.060    0.000   -0.735   -0.735
##     attach2|t3      -0.551    0.047  -11.799    0.000   -0.551   -0.551
##     attach2|t4      -0.247    0.045   -5.525    0.000   -0.247   -0.247
##     attach2|t5       0.092    0.044    2.078    0.038    0.092    0.092
##     attach2|t6       0.498    0.046   10.763    0.000    0.498    0.498
##     attach3|t1      -0.864    0.051  -17.024    0.000   -0.864   -0.864
##     attach3|t2      -0.660    0.048  -13.782    0.000   -0.660   -0.660
##     attach3|t3      -0.466    0.046  -10.139    0.000   -0.466   -0.466
##     attach3|t4      -0.145    0.044   -3.275    0.001   -0.145   -0.145
##     attach3|t5       0.244    0.045    5.454    0.000    0.244    0.244
##     attach3|t6       0.618    0.047   13.033    0.000    0.618    0.618
##     social1|t1      -1.508    0.068  -22.073    0.000   -1.508   -1.508
##     social1|t2      -1.185    0.058  -20.589    0.000   -1.185   -1.185
##     social1|t3      -0.949    0.052  -18.156    0.000   -0.949   -0.949
##     social1|t4      -0.508    0.046  -10.971    0.000   -0.508   -0.508
##     social1|t5      -0.155    0.044   -3.486    0.000   -0.155   -0.155
##     social1|t6       0.378    0.045    8.327    0.000    0.378    0.378
##     social2|t1      -1.030    0.054  -19.117    0.000   -1.030   -1.030
##     social2|t2      -0.756    0.049  -15.392    0.000   -0.756   -0.756
##     social2|t3      -0.555    0.047  -11.868    0.000   -0.555   -0.555
##     social2|t4      -0.177    0.044   -3.978    0.000   -0.177   -0.177
##     social2|t5       0.253    0.045    5.665    0.000    0.253    0.253
```

```
##     social2|t6        0.798    0.050   16.052    0.000    0.798    0.798
##     social3|t1       -1.470    0.067  -22.009    0.000   -1.470   -1.470
##     social3|t2       -1.124    0.056  -20.072    0.000   -1.124   -1.124
##     social3|t3       -0.878    0.051  -17.216    0.000   -0.878   -0.878
##     social3|t4       -0.494    0.046  -10.694    0.000   -0.494   -0.494
##     social3|t5       -0.092    0.044   -2.078    0.038   -0.092   -0.092
##     social3|t6        0.603    0.047   12.760    0.000    0.603    0.603
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .auto1             0.619                               0.619    0.619
##    .auto2             0.292                               0.292    0.292
##    .auto3             0.212                               0.212    0.212
##    .auto4             0.162                               0.162    0.162
##    .loc1              0.426                               0.426    0.426
##    .loc2              0.377                               0.377    0.377
##    .loc3              0.275                               0.275    0.275
##    .loc4              0.233                               0.233    0.233
##    .tol1              0.247                               0.247    0.247
##    .tol2              0.314                               0.314    0.314
##    .tol3              0.304                               0.304    0.304
##    .tol4              0.101                               0.101    0.101
##    .craving1          0.403                               0.403    0.403
##    .craving2          0.516                               0.516    0.516
##    .craving3          0.332                               0.332    0.332
##    .craving4          0.274                               0.274    0.274
##    .taste1            0.443                               0.443    0.443
##    .taste2            0.066                               0.066    0.066
##    .taste3            0.440                               0.440    0.440
##    .cog_e1            0.361                               0.361    0.361
##    .cog_e2            0.257                               0.257    0.257
##    .cog_e3            0.153                               0.153    0.153
##    .w_control1        0.385                               0.385    0.385
##    .w_control2        0.676                               0.676    0.676
##    .w_control3        0.406                               0.406    0.406
##    .cue1              0.595                               0.595    0.595
##    .cue2              0.834                               0.834    0.834
##    .cue3              0.622                               0.622    0.622
##    .affect1           0.404                               0.404    0.404
##    .affect2           0.274                               0.274    0.274
##    .affect3           0.509                               0.509    0.509
##    .attach1           0.314                               0.314    0.314
##    .attach2           0.398                               0.398    0.398
##    .attach3           0.356                               0.356    0.356
##    .social1           0.391                               0.391    0.391
##    .social2           0.383                               0.383    0.383
##    .social3           0.127                               0.127    0.127
##    .auto              0.168    0.016   10.799    0.000    0.441    0.441
##    .loc              -0.004    0.008   -0.456    0.648   -0.006   -0.006
##    .tol               0.176    0.014   12.881    0.000    0.233    0.233
##    .craving           0.046    0.010    4.579    0.000    0.077    0.077
##    .taste             0.510    0.035   14.612    0.000    0.916    0.916
##    .cog_e             0.288    0.019   14.808    0.000    0.450    0.450
##    .w_control         0.378    0.043    8.713    0.000    0.614    0.614
```

```
##    .cue              0.078    0.027    2.870    0.004    0.193    0.193
##    .affect           0.207    0.022    9.370    0.000    0.347    0.347
##    .attach           0.004    0.018    0.233    0.816    0.006    0.006
##    .social           0.548    0.032   16.934    0.000    0.901    0.901
##     pdm              0.213    0.023    9.220    0.000    1.000    1.000
##     sdm              0.047    0.012    3.959    0.000    1.000    1.000
```

5.1.10 Relative fitness between WLSMV models

```r
# Model 1 vs Model 2
lavTestLRT(ord_mod1_fit, ord_mod2_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.2000")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##               Df AIC BIC Chisq Chisq diff  RMSEA Df diff Pr(>Chisq)
## ord_mod2_fit 628         12014
## ord_mod1_fit 629         14397     389.36 1.7199       1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Model 1 vs Model 3
lavTestLRT(ord_mod1_fit, ord_mod3_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.2000")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##               Df AIC BIC   Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
## ord_mod3_fit 574          1799.1
## ord_mod1_fit 629         14396.5     4930.1 0.53224      55  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Model 1 vs Model 4
lavTestLRT(ord_mod1_fit, ord_mod4_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.2000")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##               Df AIC BIC   Chisq Chisq diff  RMSEA Df diff Pr(>Chisq)
## ord_mod4_fit 617          2789.3
## ord_mod1_fit 629         14396.5     3886.3 1.0956      12  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5.2 Corrected item-total correlation

```r
# Define vectors to store item names
wisdm_items <- list(AUTO = c("auto1","auto2","auto3","auto4"),
                    LOC = c("loc1","loc2","loc3","loc4"),
                    TOL = c("tol1","tol2","tol3","tol4"),
                    CRAVING = c("craving1","craving2","craving3","craving4"),
                    TASTE = c("taste1","taste2","taste3"),
                    COG_E = c("cog_e1","cog_e2","cog_e3"),
                    W_CONTROL = c("w_control1","w_control2","w_control3"),
                    CUE = c("cue1","cue2","cue3"),
                    AFFECT = c("affect1","affect2","affect3"),
                    ATTACH = c("attach1","attach2","attach3"),
                    SOCIAL = c("social1","social2","social3"))

# Define factors
wisdm_scales <- lapply(wisdm_items, function(items) {
  data_cleaned %>% dplyr::select(dplyr::all_of(items))
})

# Perform reliability analysis and store item statistics
wisdm_rel <- lapply(wisdm_scales, function(df) {
  alpha(cor(df, method = "spearman", use = "pairwise.complete.obs"))
})
wisdm_item_stats <- lapply(wisdm_rel, function(x) {
  x$item.stats
})

# Store results into a single data frame
wisdm_citc <- bind_rows(lapply(names(wisdm_item_stats), function(scale) {
  df <- wisdm_item_stats[[scale]]
  df$Item <- rownames(df)
  df$Scale <- scale
  df}))
```

```r
# Display corrected item-total correlations
wisdm_citc %>% dplyr::select(Scale, Item, r.drop)
```

```
##                 Scale       Item    r.drop
## auto1            AUTO      auto1 0.5827469
## auto2            AUTO      auto2 0.7316918
## auto3            AUTO      auto3 0.7829146
## auto4            AUTO      auto4 0.7635958
## loc1              LOC       loc1 0.7111690
## loc2              LOC       loc2 0.7011613
## loc3              LOC       loc3 0.6975813
## loc4              LOC       loc4 0.7004674
## tol1              TOL       tol1 0.7076954
## tol2              TOL       tol2 0.6785563
## tol3              TOL       tol3 0.6888439
## tol4              TOL       tol4 0.7592075
## craving1      CRAVING   craving1 0.6329662
## craving2      CRAVING   craving2 0.5980353
## craving3      CRAVING   craving3 0.6890441
## craving4      CRAVING   craving4 0.6510394
## taste1          TASTE     taste1 0.6318629
## taste2          TASTE     taste2 0.7312832
## taste3          TASTE     taste3 0.6886521
## cog_e1          COG_E     cog_e1 0.6981742
## cog_e2          COG_E     cog_e2 0.7831804
## cog_e3          COG_E     cog_e3 0.7136303
## w_control1 W_CONTROL w_control1 0.5111423
## w_control2 W_CONTROL w_control2 0.4786560
## w_control3 W_CONTROL w_control3 0.5080936
## cue1              CUE       cue1 0.3082981
## cue2              CUE       cue2 0.3598690
## cue3              CUE       cue3 0.3366233
## affect1        AFFECT    affect1 0.6012276
## affect2        AFFECT    affect2 0.6491456
## affect3        AFFECT    affect3 0.5679739
## attach1        ATTACH    attach1 0.5918500
## attach2        ATTACH    attach2 0.6652499
## attach3        ATTACH    attach3 0.6002822
## social1        SOCIAL    social1 0.6388196
## social2        SOCIAL    social2 0.6512057
## social3        SOCIAL    social3 0.7707376
```

5.3 Item-factor correlations

```r
# Calculate the mean score for each WISDM-37 factor
data_cleaned$auto <- rowMeans(data_cleaned[, wisdm_items$AUTO], na.rm = TRUE)
data_cleaned$loc <- rowMeans(data_cleaned[, wisdm_items$LOC], na.rm = TRUE)
data_cleaned$tol <- rowMeans(data_cleaned[, wisdm_items$TOL], na.rm = TRUE)
data_cleaned$craving <- rowMeans(data_cleaned[, wisdm_items$CRAVING], na.rm = TRUE)
data_cleaned$taste <- rowMeans(data_cleaned[, wisdm_items$TASTE], na.rm = TRUE)
data_cleaned$cog_e <- rowMeans(data_cleaned[, wisdm_items$COG_E], na.rm = TRUE)
data_cleaned$w_control <- rowMeans(data_cleaned[, wisdm_items$W_CONTROL], na.rm = TRUE)
data_cleaned$cue <- rowMeans(data_cleaned[, wisdm_items$CUE], na.rm = TRUE)
data_cleaned$affect <- rowMeans(data_cleaned[, wisdm_items$AFFECT], na.rm = TRUE)
data_cleaned$attach <- rowMeans(data_cleaned[, wisdm_items$ATTACH], na.rm = TRUE)
data_cleaned$social <- rowMeans(data_cleaned[, wisdm_items$SOCIAL], na.rm = TRUE)

# Select WISDM-37 items and factors
all_items <- data_cleaned[, unlist(wisdm_items)]
all_factors <- data_cleaned[, c("auto","loc","tol","craving","taste",
                               "cog_e","w_control","cue","affect",
                               "attach","social")]

# Calculate item-factor correlations
item_factor_cor <- as.data.frame(cor(all_items, all_factors,
                                     method = "spearman",
                                     use = "pairwise.complete.obs"))

# Write into an excel file
write_xlsx(item_factor_cor, "Item-factor correlation matrix.xlsx")
```

5.4 Factor-factor correlations

```r
# Calculate factor-factor correlations
factor_factor_cor <- as.data.frame(cor(all_factors,
                                       method = "spearman",
                                       use = "pairwise.complete.obs"))

# Write into an excel file
write_xlsx(factor_factor_cor, "Factor-factor correlation matrix.xlsx")
```

5.5 Composite reliability

```r
# Select the most plausible WISDM-37 model
cont_select_fit <- cont_mod3_fit

# Calculate composite reliability for the selected continuous and ordinal WISDM-37 model
cont_rel <- compRelSEM(cont_select_fit)
print(cont_rel, digits = 3)
```

```
## $auto
##
## Composite `auto` is composed of observed variables:
##  auto1, auto2, auto3, auto4
## True-score variance is represented by common factor(s):
##  auto
## Total variance of composite `auto` determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.873
##
## $loc
##
## Composite `loc` is composed of observed variables:
##  loc1, loc2, loc3, loc4
## True-score variance is represented by common factor(s):
##  loc
## Total variance of composite `loc` determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.85
##
## $tol
##
## Composite `tol` is composed of observed variables:
##  tol1, tol2, tol3, tol4
## True-score variance is represented by common factor(s):
##  tol
## Total variance of composite `tol` determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.843
##
## $craving
##
## Composite `craving` is composed of observed variables:
##  craving1, craving2, craving3, craving4
## True-score variance is represented by common factor(s):
##  craving
## Total variance of composite `craving` determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.821
##
```

```
## $taste
##
## Composite ‘taste‘ is composed of observed variables:
##  taste1, taste2, taste3
## True-score variance is represented by common factor(s):
##  taste
## Total variance of composite ‘taste‘ determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.822
##
## $cog_e
##
## Composite ‘cog_e‘ is composed of observed variables:
##  cog_e1, cog_e2, cog_e3
## True-score variance is represented by common factor(s):
##  cog_e
## Total variance of composite ‘cog_e‘ determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.865
##
## $w_control
##
## Composite ‘w_control‘ is composed of observed variables:
##  w_control1, w_control2, w_control3
## True-score variance is represented by common factor(s):
##  w_control
## Total variance of composite ‘w_control‘ determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.684
##
## $cue
##
## Composite ‘cue‘ is composed of observed variables:
##  cue1, cue2, cue3
## True-score variance is represented by common factor(s):
##  cue
## Total variance of composite ‘cue‘ determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.482
##
## $affect
##
## Composite ‘affect‘ is composed of observed variables:
##  affect1, affect2, affect3
## True-score variance is represented by common factor(s):
##  affect
## Total variance of composite ‘affect‘ determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.772
```

```
##
## $attach
##
## Composite ‘attach‘ is composed of observed variables:
##  attach1, attach2, attach3
## True-score variance is represented by common factor(s):
##  attach
## Total variance of composite ‘attach‘ determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.784
##
## $social
##
## Composite ‘social‘ is composed of observed variables:
##  social1, social2, social3
## True-score variance is represented by common factor(s):
##  social
## Total variance of composite ‘social‘ determined from the unrestricted model.
## The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):
##
## [1] 0.839
```

5.6 Modification indices

```r
# Get modification indices
wisdm_mi <- modificationIndices(cont_select_fit)

# Sort by decreasing MI value
wisdm_mi_sorted <- wisdm_mi[order(wisdm_mi$mi, decreasing = TRUE), ]

# Inspect top 10 modifications
head(wisdm_mi_sorted, 10)
```

```
##          lhs op      rhs      mi    epc sepc.lv sepc.all sepc.nox
## 772     tol1 ~~     tol3 267.450  2.105   2.105    0.660    0.660
## 800     tol2 ~~     tol4 210.355  1.210   1.210    0.869    0.869
## 745     loc4 ~~     tol4 203.140  0.952   0.952    0.663    0.663
## 214      tol =~     loc4 154.184  1.243   2.280    1.034    1.034
## 247  craving =~     loc4  82.400 -1.634  -2.466   -1.118   -1.118
## 185      loc =~ craving4  69.196  1.670   2.498    1.140    1.140
## 246  craving =~     loc3  64.388  1.302   1.965    0.989    0.989
## 773     tol1 ~~     tol4  64.357 -0.791  -0.791   -0.414   -0.414
## 218      tol =~ craving4  61.893  0.601   1.104    0.503    0.503
## 1064  cog_e3 ~~  affect2  55.888  0.408   0.408    0.387    0.387
```

6 Smoking Motives and Nicotine Dependence Across Smoking Method

```
# Define vectors to store item names for each smoking method
cigg_items  <- c("cigg_f1","cigg_f2","cigg_f3","cigg_f4","cigg_f5","cigg_f6")
ecigg_items <- c("ecigg_f1","ecigg_f2","ecigg_f3","ecigg_f4","ecigg_f5","ecigg_f6")
h_items     <- c("h_f1","h_f2","h_f3","h_f4","h_f5","h_f6")

# Calculate the Fagerström Test for Nicotine Dependence scores
data_cleaned$cigg_f <- ifelse(rowSums(is.na(data_cleaned[, cigg_items])) == 0,
                              rowSums(data_cleaned[, cigg_items]),NA)
data_cleaned$ecigg_f <- ifelse(rowSums(is.na(data_cleaned[, ecigg_items])) == 0,
                               rowSums(data_cleaned[, ecigg_items]),NA)

data_cleaned$h_f <- ifelse(rowSums(is.na(data_cleaned[, h_items])) == 0,
                           rowSums(data_cleaned[, h_items]),NA)
```

6.1 Distribution of nicotine dependence across smoking methods

6.1.1 Prepare the data

```
# Create color palettes for smoking methods

# 5-category palette (for traditional and E-cigarettes)
cols_5 <- c("No dependence"               = "#22C55E",
            "Low dependence"              = "#FACC15",
            "Low to moderate dependence" = "#F97316",
            "Moderate dependence"         = "#EF4444",
            "High dependence"             = "#000000")

# 3-category palette (for Hookah)
cols_3 <- c("Low dependence"      = "#FACC15",
            "Moderate dependence" = "#F97316",
            "High dependence"     = "#EF4444")
```

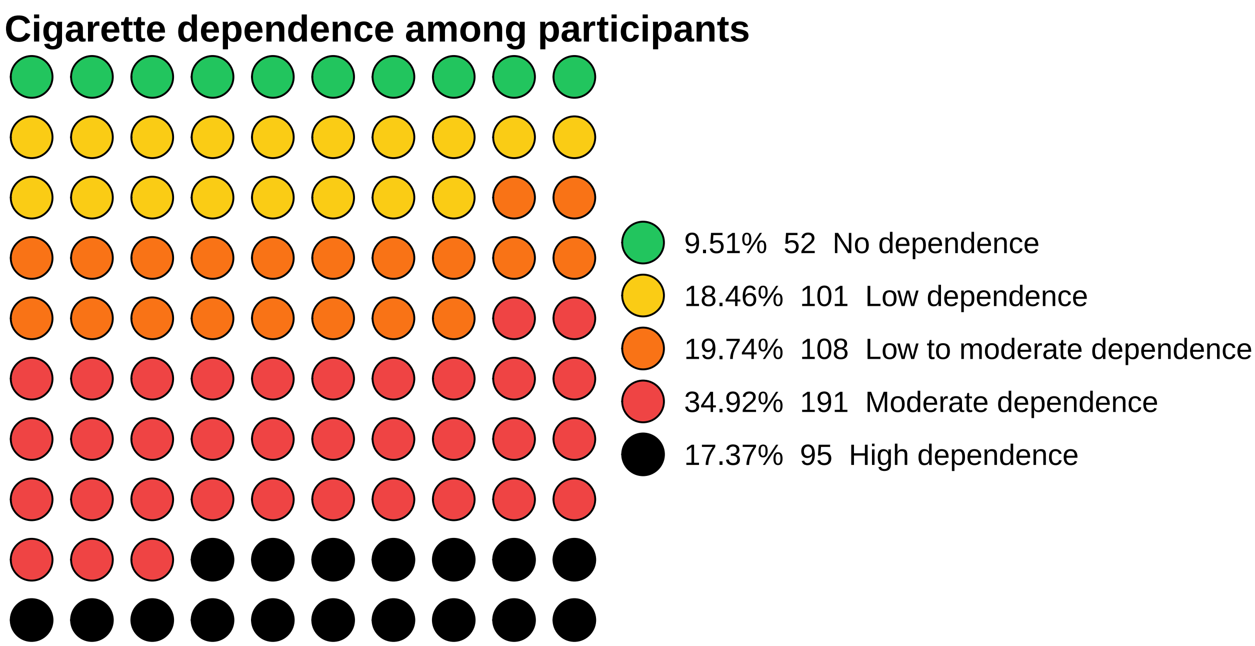
```r
# 5 categories (for traditional and E-cigarettes)
categorize_5 <- function(x) {
  factor(case_when(      x == 0      ~ "No dependence",
                         x %in% 1:2  ~ "Low dependence",
                         x %in% 3:4  ~ "Low to moderate dependence",
                         x %in% 5:7  ~ "Moderate dependence",
                         x >= 8      ~ "High dependence"),
         levels = names(cols_5))}

# 3 categories (for Hookah)
categorize_3 <- function(x) {
  factor(case_when(x %in% 0:1 ~ "Low dependence",
                   x %in% 2:3 ~ "Moderate dependence",
                   x >= 4     ~ "High dependence"),
         levels = names(cols_3))}

# Categorize variables
data_cleaned <- data_cleaned %>%
  mutate(cigg_cat  = categorize_5(cigg_f),
         ecigg_cat = categorize_5(ecigg_f),
         h_cat     = categorize_3(h_f))
```

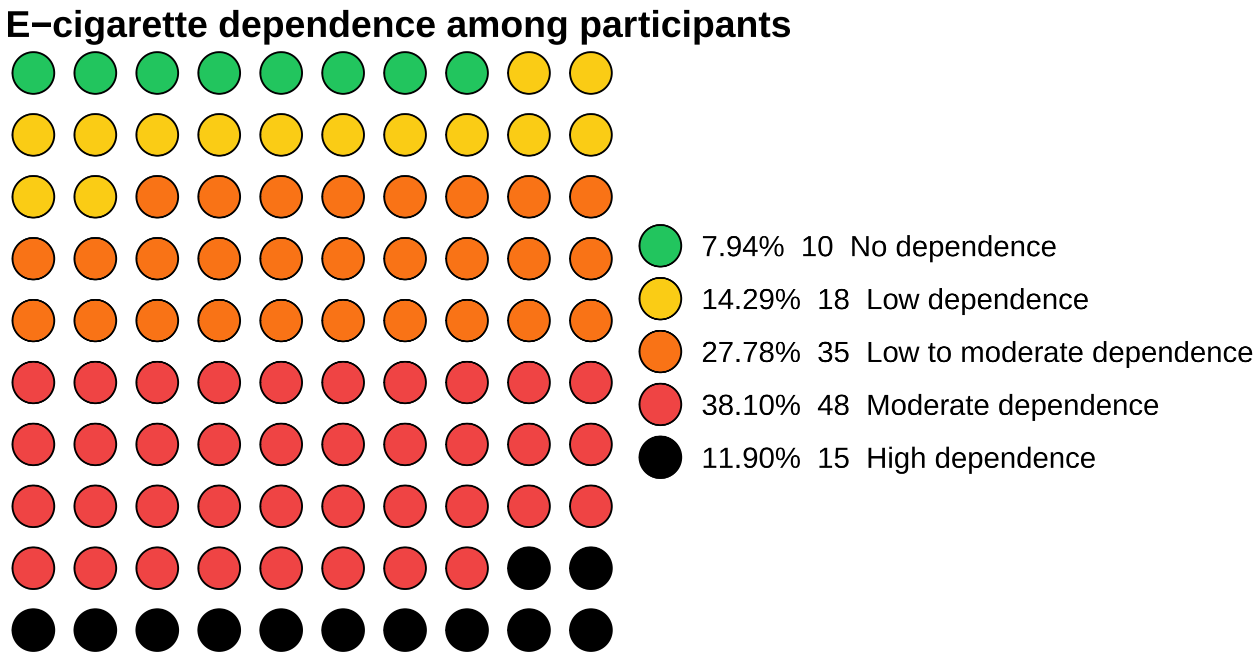
6.1.2 Visualize distributions

```r
# Define a function to create waffle plots
plot_waffle <- function(data, var, colors, title = "") {

  # Summarize counts and percentages
  summary_df <- data %>%
    filter(!is.na(.data[[var]])) %>%
    count(.data[[var]], .drop = FALSE) %>%
    rename(category = 1) %>%
    mutate(pct = n / sum(n) * 100)

  # Convert percentages to whole-number dots using largest-remainder
  floor_pct <- floor(summary_df$pct)
  need      <- 100 - sum(floor_pct)
  top_idx   <- order(summary_df$pct - floor_pct, decreasing = TRUE)
  dots      <- floor_pct
  if (need > 0) dots[top_idx[seq_len(need)]] <- dots[top_idx[seq_len(need)]] + 1
  summary_df$dots <- dots

  # Build 10-column grid coordinates, one row per dot
  waffle_df <- data.frame(category = rep(summary_df$category, summary_df$dots)
  ) %>% mutate(id = row_number(),
               x  = ((id - 1) %%  10) + 1,
               y  = ((id - 1) %/% 10) + 1)

  # Legend labels show real % and real n
  leg_labels <- with(summary_df,
    setNames(sprintf("%.2f%%  %d  %s", pct, n, category), category))

  # Plot
  ggplot(waffle_df, aes(x = x, y = -y, fill = category)) +
    geom_point(shape = 21, size = 7, colour = "black", stroke = 0.5) +
    scale_fill_manual(values = colors, labels = leg_labels, name = NULL) +
    coord_equal() +
    labs(title = title) +
    theme_void() +
    theme(plot.title       = element_text(face = "bold", size = 14, hjust = 0),
          legend.text      = element_text(size = 11),
          legend.key       = element_rect(fill = "white", colour = NA),
          legend.position  = "right",
          plot.background  = element_rect(fill = "white", colour = NA),
          panel.background = element_rect(fill = "white", colour = NA))}
```

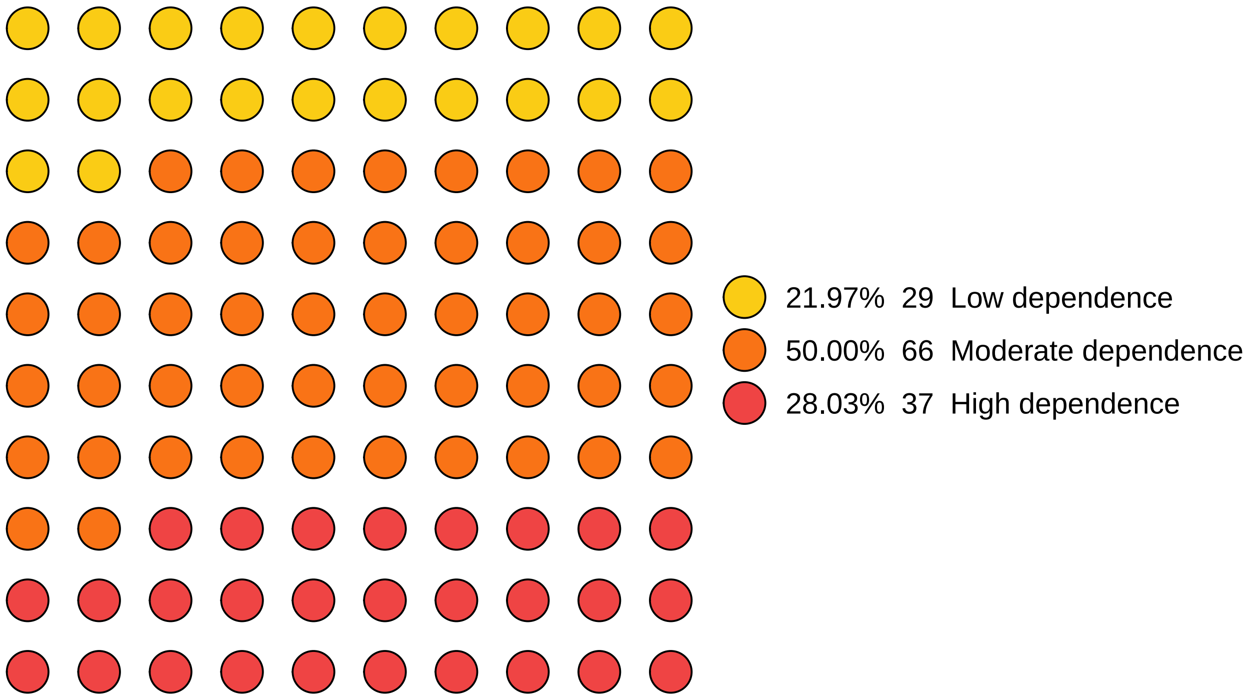
6.1.3 Visualize distributions

```r
# Generate plots
p1 <- plot_waffle(data_cleaned, "cigg_cat",  cols_5,
                  "Cigarette dependence among participants")
p2 <- plot_waffle(data_cleaned, "ecigg_cat", cols_5,
                  "E-cigarette dependence among participants")
p3 <- plot_waffle(data_cleaned, "h_cat",     cols_3,
                  "Hookah dependence among participants")

# Display plots
p1; p2; p3
```




E−cigarette dependence among participants
7.94% 10 No dependence
14.29% 18 Low dependence
27.78% 35 Low to moderate dependence
38.10% 48 Moderate dependence
11.90% 15 High dependence

Hookah dependence among participants



6.2 Hypothesis testing: Data preparation

```r
# Store the names of included motives
wisdm_factors <- c("auto","loc","tol","craving","taste",
                   "cog_e","w_control","affect","attach","social")

# Standardize predictors
data_cleaned[wisdm_factors] <- scale(data_cleaned[wisdm_factors])
```

6.3 Hypothesis testing: Traditional Cigarette Users

```
# Subset data
data_cc <- na.omit(data_cleaned[, c("cigg_f", wisdm_factors)])
```

6.3.1 MM-estimator regression model

```
wisdm37_cigg <- lmrob(cigg_f ~ auto + loc + tol + craving + taste + cog_e +
                        w_control + affect + attach + social,
                      data = data_cc,
                      k.max = 1000)
summary(wisdm37_cigg)
```

```
##
## Call:
## lmrob(formula = cigg_f ~ auto + loc + tol + craving + taste + cog_e + w_control +
##     affect + attach + social, data = data_cc, k.max = 1000)
##  \--> method = "MM"
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.41810 -1.18125  0.01605  1.18283  5.27401
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.10286    0.07678  53.435   <2e-16 ***
## auto         0.07622    0.10442   0.730   0.4657
## loc         -0.04844    0.16239  -0.298   0.7656
## tol          2.07619    0.12670  16.386   <2e-16 ***
## craving      0.30074    0.14370   2.093   0.0368 *
## taste       -0.13515    0.08236  -1.641   0.1014
## cog_e        0.16380    0.09496   1.725   0.0851 .
## w_control    0.06708    0.08642   0.776   0.4379
## affect      -0.13824    0.10310  -1.341   0.1805
## attach       0.13151    0.11884   1.107   0.2689
## social      -0.01261    0.07603  -0.166   0.8684
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Robust residual standard error: 1.731
## Multiple R-squared:  0.6511, Adjusted R-squared:  0.6446
## Convergence in 11 IRWLS iterations
##
## Robustness weights:
##  41 weights are ~= 1. The remaining 506 ones are summarized as
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.3065  0.8758  0.9488  0.9115  0.9865  0.9990
## Algorithmic parameters:
##        tuning.chi                 bb        tuning.psi        refine.tol
##         1.548e+00          5.000e-01         4.685e+00         1.000e-07
##           rel.tol          scale.tol         solve.tol          zero.tol
##         1.000e-07          1.000e-10         1.000e-07         1.000e-10
```

```
##       eps.outlier             eps.x warn.limit.reject warn.limit.meanrw
##         1.828e-04         5.456e-12         5.000e-01         5.000e-01
##      nResample         max.it       best.r.s       k.fast.s          k.max
##            500             50              2              1           1000
##    maxit.scale      trace.lev            mts     compute.rd fast.s.large.n
##            200              0           1000              0           2000
##                   psi           subsampling                   cov
##            "bisquare"         "nonsingular"         ".vcov.avar1"
## compute.outlier.stats
##                  "SM"
## seed : int(0)
```

6.3.2 Squared 0-order Spearman's rank correlation

```r
cor_vals <- cor(data_cc[, wisdm_factors],
                data_cc$cigg_f,
                method = "spearman")
cor.wisdm37.cigg <- data.frame(Factor = wisdm_factors,
                               R2_percent = (cor_vals^2),
                               row.names = NULL)
print(cor.wisdm37.cigg)
```

```
##       Factor  R2_percent
## 1       auto 0.229637508
## 2        loc 0.360738571
## 3        tol 0.620640407
## 4    craving 0.334909284
## 5      taste 0.005230026
## 6      cog_e 0.144783474
## 7  w_control 0.103339480
## 8     affect 0.097664139
## 9     attach 0.244005937
## 10    social 0.016849022
```

6.3.3 Squared partial Spearman's rank correlation

```r
pcor_result <- pcor(data_cc[, c("cigg_f", wisdm_factors)],
                    method = "spearman")
pcor.wisdm37.cigg <- data.frame(Factor = wisdm_factors,
                                Partial_R2_percent =
                                  (pcor_result$estimate["cigg_f",
                                                        wisdm_factors]^2),
                                row.names = NULL)
print(pcor.wisdm37.cigg)
```

```
##       Factor Partial_R2_percent
## 1       auto       1.226713e-03
## 2        loc       2.826588e-04
## 3        tol       3.502580e-01
## 4    craving       1.037289e-02
## 5      taste       9.698552e-03
## 6      cog_e       1.008087e-02
## 7  w_control       4.459605e-03
## 8     affect       2.304409e-03
## 9     attach       1.675156e-03
## 10    social       2.756361e-05
```

6.4 Hypothesis testing: E-Cigarette Users

```r
# Subset data
data_ec <- na.omit(data_cleaned[, c("ecigg_f", wisdm_factors)])
```

6.4.1 MM-estimator regression model

```r
wisdm37_ecigg <- lmrob(ecigg_f ~ auto + loc + tol + craving + taste +
                         cog_e + w_control + affect + attach + social,
                       data = data_ec,
                       k.max = 1000)
summary(wisdm37_ecigg)
```

```
##
## Call:
## lmrob(formula = ecigg_f ~ auto + loc + tol + craving + taste + cog_e + w_control +
##     affect + attach + social, data = data_ec, k.max = 1000)
##  \--> method = "MM"
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.73106 -0.98889  0.04752  1.06810  3.62450
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.484662   0.187395  23.932  < 2e-16 ***
## auto         0.423486   0.181787   2.330 0.021574 *
## loc         -0.484046   0.320622  -1.510 0.133860
## tol          1.413197   0.361250   3.912 0.000155 ***
## craving      0.275467   0.234013   1.177 0.241568
## taste       -0.001509   0.211274  -0.007 0.994315
## cog_e        0.054272   0.169973   0.319 0.750078
## w_control   -0.075949   0.182747  -0.416 0.678480
## affect      -0.016182   0.184736  -0.088 0.930350
## attach       0.483964   0.194156   2.493 0.014103 *
## social       0.145134   0.142407   1.019 0.310272
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Robust residual standard error: 1.662
## Multiple R-squared:  0.5828, Adjusted R-squared:  0.5465
## Convergence in 12 IRWLS iterations
##
## Robustness weights:
##  13 weights are ~= 1. The remaining 113 ones are summarized as
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.5936  0.8687  0.9553  0.9127  0.9832  0.9987
## Algorithmic parameters:
##        tuning.chi                bb        tuning.psi        refine.tol
##         1.548e+00         5.000e-01         4.685e+00         1.000e-07
##           rel.tol         scale.tol         solve.tol          zero.tol
##         1.000e-07         1.000e-10         1.000e-07         1.000e-10
```

```
##       eps.outlier             eps.x warn.limit.reject warn.limit.meanrw
##         7.937e-04         5.456e-12         5.000e-01         5.000e-01
##      nResample         max.it       best.r.s       k.fast.s          k.max
##            500             50              2              1           1000
##    maxit.scale      trace.lev            mts     compute.rd fast.s.large.n
##            200              0           1000              0           2000
##                   psi           subsampling                   cov
##            "bisquare"         "nonsingular"         ".vcov.avar1"
## compute.outlier.stats
##                  "SM"
## seed : int(0)
```

6.4.2 Squared 0-order Spearman's rank correlation

```
cor_vals <- cor(data_ec[, wisdm_factors],
                data_ec$ecigg_f,
                method = "spearman")
cor.wisdm37.ecigg <- data.frame(Factor = wisdm_factors,
                               R2_percent = (cor_vals^2),
                               row.names = NULL)
print(cor.wisdm37.ecigg)
```

```
##       Factor R2_percent
## 1       auto 0.30616298
## 2        loc 0.32556291
## 3        tol 0.52841265
## 4    craving 0.36042440
## 5      taste 0.05737590
## 6      cog_e 0.13541963
## 7  w_control 0.05481778
## 8     affect 0.22688590
## 9     attach 0.33958929
## 10    social 0.03033001
```

6.4.3 Squared partial Spearman's rank correlation

```r
pcor_result <- pcor(data_ec[, c("ecigg_f", wisdm_factors)],
                    method = "spearman")
pcor.wisdm37.ecigg <- data.frame(Factor = wisdm_factors,
                                 Partial_R2_percent =
                                   (pcor_result$estimate["ecigg_f",
                                                         wisdm_factors]^2),
                                 row.names = NULL)
print(pcor.wisdm37.ecigg)
```

```
##       Factor Partial_R2_percent
## 1       auto       0.0391858045
## 2        loc       0.0185427046
## 3        tol       0.1999666334
## 4    craving       0.0078255616
## 5      taste       0.0001559971
## 6      cog_e       0.0002377717
## 7  w_control       0.0044542165
## 8     affect       0.0003346380
## 9     attach       0.0399645922
## 10    social       0.0085471653
```

6.5 Hypothesis testing: Hookah users

```r
# Subset data
data_h <- na.omit(data_cleaned[, c("h_f", wisdm_factors)])
```

6.5.1 MM-estimator regression model

```r
wisdm37_h <- lmrob(h_f ~ auto + loc + tol + craving + taste +
                     cog_e + w_control + affect + attach + social,
                   data = data_h,
                   k.max = 1000)
summary(wisdm37_h)
```

```
##
## Call:
## lmrob(formula = h_f ~ auto + loc + tol + craving + taste + cog_e + w_control +
##     affect + attach + social, data = data_h, k.max = 1000)
##  \--> method = "MM"
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.41875 -0.71643 -0.05955  0.77322  3.79480
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.789320   0.166780  22.720  < 2e-16 ***
## auto         0.082667   0.156411   0.529 0.598106
## loc          0.024270   0.227773   0.107 0.915321
## tol          0.963569   0.284764   3.384 0.000964 ***
## craving      0.259154   0.170329   1.521 0.130746
## taste       -0.009029   0.133447  -0.068 0.946169
## cog_e        0.277982   0.168469   1.650 0.101526
## w_control    0.281237   0.142576   1.973 0.050828 .
## affect      -0.179008   0.113381  -1.579 0.116987
## attach      -0.033697   0.173970  -0.194 0.846741
## social      -0.023472   0.101847  -0.230 0.818122
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Robust residual standard error: 1.107
## Multiple R-squared:  0.5757, Adjusted R-squared:  0.5406
## Convergence in 16 IRWLS iterations
##
## Robustness weights:
##  6 weights are ~= 1. The remaining 126 ones are summarized as
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.2162  0.8926  0.9569  0.9116  0.9884  0.9989
## Algorithmic parameters:
##        tuning.chi                bb        tuning.psi        refine.tol
##         1.548e+00         5.000e-01         4.685e+00         1.000e-07
##           rel.tol         scale.tol         solve.tol          zero.tol
##         1.000e-07         1.000e-10         1.000e-07         1.000e-10
```

```
##       eps.outlier             eps.x warn.limit.reject warn.limit.meanrw
##         7.576e-04         5.456e-12         5.000e-01         5.000e-01
##      nResample         max.it       best.r.s       k.fast.s          k.max
##            500             50              2              1           1000
##    maxit.scale      trace.lev            mts     compute.rd fast.s.large.n
##            200              0           1000              0           2000
##                   psi           subsampling                   cov
##            "bisquare"         "nonsingular"         ".vcov.avar1"
## compute.outlier.stats
##                  "SM"
## seed : int(0)
```

6.5.2 Squared 0-order Spearman's rank correlation

```r
cor_vals <- cor(data_h[, wisdm_factors],
                data_h$h_f,
                method = "spearman")
cor.wisdm37.h <- data.frame(Factor = wisdm_factors,
                            R2_percent = (cor_vals^2),
                            row.names = NULL)
print(cor.wisdm37.h)
```

```
##       Factor R2_percent
## 1       auto 0.16536329
## 2        loc 0.38031152
## 3        tol 0.45620866
## 4    craving 0.37491345
## 5      taste 0.03751169
## 6      cog_e 0.27302335
## 7  w_control 0.15154333
## 8     affect 0.13050529
## 9     attach 0.30822292
## 10    social 0.04739038
```

6.5.3 Squared partial Spearman's rank correlation

```r
pcor_result <- pcor(data_h[, c("h_f", wisdm_factors)],
                    method = "spearman")
pcor.wisdm37.h <- data.frame(Factor = wisdm_factors,
                             Partial_R2_percent =
                               (pcor_result$estimate["h_f",
                                                     wisdm_factors]^2),
                            row.names = NULL)
print(pcor.wisdm37.h)
```

```
##       Factor Partial_R2_percent
## 1       auto       6.857413e-03
## 2        loc       1.595507e-04
## 3        tol       8.261330e-02
## 4    craving       4.818070e-02
## 5      taste       1.471057e-03
## 6      cog_e       4.974435e-02
## 7  w_control       2.807645e-03
## 8     affect       2.772594e-02
## 9     attach       2.168254e-08
## 10    social       3.695372e-04
```

7 RFQ psychometric analysis

7.1 Confirmatory factor analysis

```
df_exsmoker <- data_cleaned %>%
  filter(smoke_stat == "ex_smoker")
```

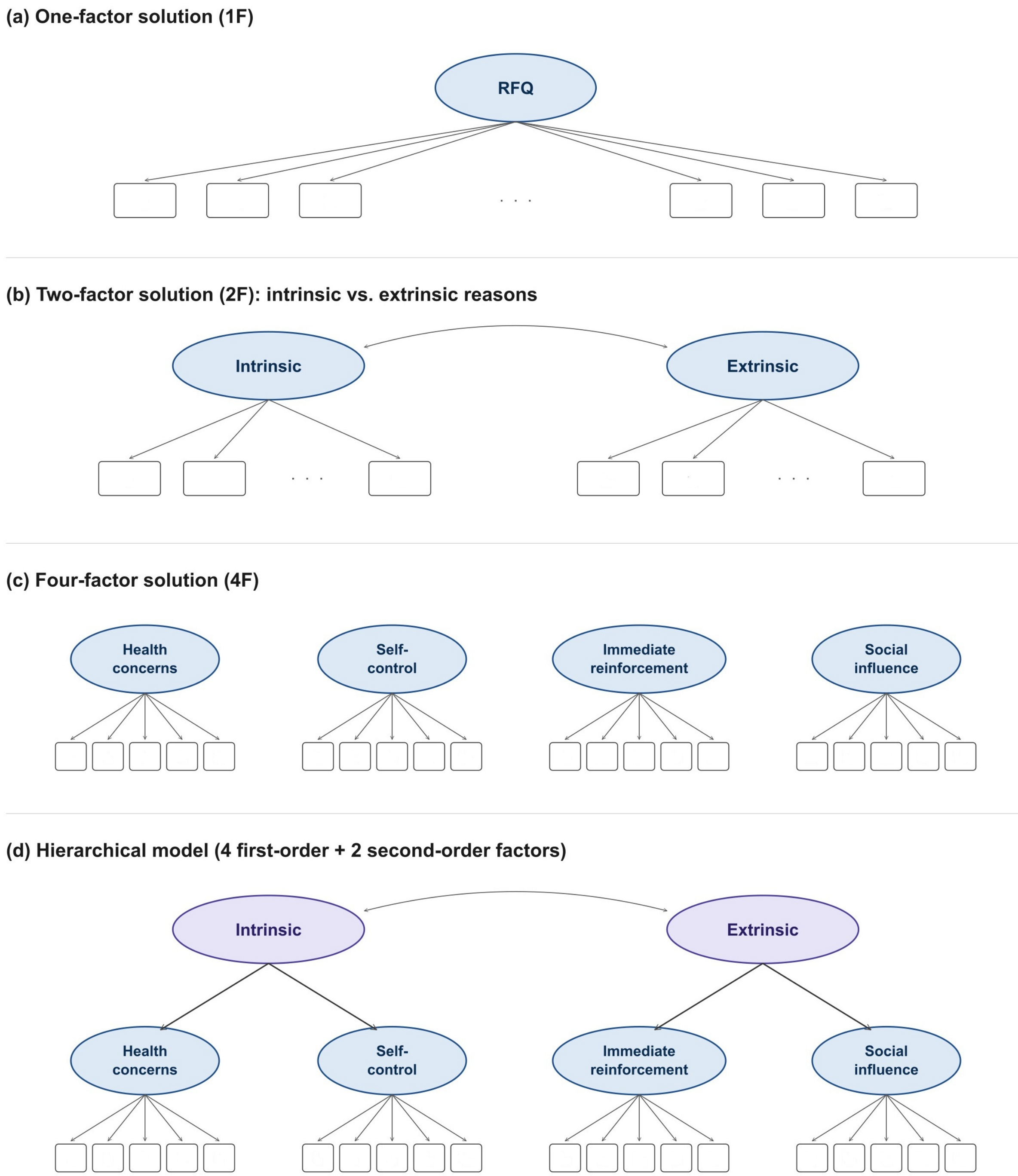


Figure 2: RFQ models

7.1.1 MLR model 1: 1-factor solution

```r
rfq_cmod1 <-'f =~ hc1 + hc2 + hc3 + hc4 + hc5 + sc1 + sc2 + sc3 + sc4 + sc5 +
                  ir1 + ir2 + ir3 + ir4 + ir5 + sp1 + sp2 + sp3 + sp4 + sp5
            '
rfq_cmod1_fit <- cfa(rfq_cmod1, data = df_exsmoker, estimator = "MLR")
summary(rfq_cmod1_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 42 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        40
##
##   Number of observations                           186
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               824.463     750.441
##   Degrees of freedom                               170         170
##   P-value (Chi-square)                           0.000       0.000
##   Scaling correction factor                                  1.099
##     Yuan-Bentler correction (Mplus variant)
##
## Model Test Baseline Model:
##
##   Test statistic                              1598.299    1428.834
##   Degrees of freedom                               190         190
##   P-value                                        0.000       0.000
##   Scaling correction factor                                  1.119
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.535       0.531
##   Tucker-Lewis Index (TLI)                       0.481       0.476
##
##   Robust Comparative Fit Index (CFI)                         0.540
##   Robust Tucker-Lewis Index (TLI)                            0.486
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -6124.959   -6124.959
##   Scaling correction factor                                  1.221
##       for the MLR correction
##   Loglikelihood unrestricted model (H1)      -5712.727   -5712.727
##   Scaling correction factor                                  1.122
##       for the MLR correction
##
##   Akaike (AIC)                               12329.918   12329.918
##   Bayesian (BIC)                             12458.948   12458.948
##   Sample-size adjusted Bayesian (SABIC)      12332.254   12332.254
##
## Root Mean Square Error of Approximation:
```

```
##
##   RMSEA                                        0.144       0.135
##   90 Percent confidence interval - lower       0.134       0.126
##   90 Percent confidence interval - upper       0.154       0.145
##   P-value H_0: RMSEA <= 0.050                  0.000       0.000
##   P-value H_0: RMSEA >= 0.080                  1.000       1.000
##
##   Robust RMSEA                                             0.142
##   90 Percent confidence interval - lower                   0.132
##   90 Percent confidence interval - upper                   0.152
##   P-value H_0: Robust RMSEA <= 0.050                       0.000
##   P-value H_0: Robust RMSEA >= 0.080                       1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                         0.110       0.110
##
## Parameter Estimates:
##
##   Standard errors                           Sandwich
##   Information bread                         Observed
##   Observed information based on              Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   f =~
##     hc1               1.000                               0.530    0.379
##     hc2               1.058    0.196    5.388    0.000    0.560    0.354
##     hc3               1.303    0.211    6.186    0.000    0.690    0.493
##     hc4               1.533    0.320    4.783    0.000    0.812    0.517
##     hc5               1.414    0.272    5.208    0.000    0.749    0.495
##     sc1               0.856    0.291    2.942    0.003    0.454    0.298
##     sc2               1.665    0.336    4.960    0.000    0.882    0.546
##     sc3               1.596    0.386    4.135    0.000    0.845    0.564
##     sc4               1.560    0.381    4.092    0.000    0.826    0.553
##     sc5               1.198    0.316    3.793    0.000    0.635    0.438
##     ir1               1.551    0.357    4.348    0.000    0.822    0.499
##     ir2               2.117    0.462    4.582    0.000    1.121    0.703
##     ir3               1.454    0.356    4.089    0.000    0.770    0.631
##     ir4               1.992    0.464    4.295    0.000    1.055    0.668
##     ir5               1.233    0.304    4.053    0.000    0.653    0.594
##     sp1               1.584    0.387    4.093    0.000    0.839    0.521
##     sp2               0.836    0.295    2.831    0.005    0.443    0.317
##     sp3               0.859    0.285    3.015    0.003    0.455    0.519
##     sp4               1.430    0.376    3.802    0.000    0.758    0.500
##     sp5               0.674    0.224    3.009    0.003    0.357    0.462
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hc1               1.675    0.141   11.856    0.000    1.675    0.856
##    .hc2               2.191    0.177   12.350    0.000    2.191    0.875
##    .hc3               1.486    0.143   10.415    0.000    1.486    0.757
##    .hc4               1.809    0.191    9.491    0.000    1.809    0.733
##    .hc5               1.727    0.166   10.401    0.000    1.727    0.755
```

```
##    .sc1              2.119    0.167   12.703    0.000    2.119    0.911
##    .sc2              1.835    0.163   11.262    0.000    1.835    0.702
##    .sc3              1.535    0.151   10.171    0.000    1.535    0.682
##    .sc4              1.551    0.170    9.112    0.000    1.551    0.694
##    .sc5              1.694    0.155   10.940    0.000    1.694    0.808
##    .ir1              2.039    0.175   11.621    0.000    2.039    0.751
##    .ir2              1.286    0.185    6.938    0.000    1.286    0.506
##    .ir3              0.896    0.154    5.803    0.000    0.896    0.602
##    .ir4              1.380    0.212    6.493    0.000    1.380    0.554
##    .ir5              0.783    0.136    5.757    0.000    0.783    0.647
##    .sp1              1.886    0.203    9.270    0.000    1.886    0.728
##    .sp2              1.755    0.205    8.560    0.000    1.755    0.900
##    .sp3              0.563    0.115    4.890    0.000    0.563    0.731
##    .sp4              1.724    0.191    9.035    0.000    1.724    0.750
##    .sp5              0.470    0.121    3.880    0.000    0.470    0.787
##     f                0.281    0.111    2.519    0.012    1.000    1.000
```

7.1.2 MLR model 2: 2-factor solution

```
rfq_cmod2 <-'intrinsic =~ hc1 + hc2 + hc3 + hc4 + hc5 + sc1 + sc2 + sc3 + sc4 + sc5
             extrinsic =~ ir1 + ir2 + ir3 + ir4 + ir5 + sp1 + sp2 + sp3 + sp4 + sp5
             '
rfq_cmod2_fit <- cfa(rfq_cmod2, data = df_exsmoker, estimator = "MLR")
summary(rfq_cmod2_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 52 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        41
##
##   Number of observations                           186
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               758.841     717.185
##   Degrees of freedom                               169         169
##   P-value (Chi-square)                           0.000       0.000
##   Scaling correction factor                                  1.058
##     Yuan-Bentler correction (Mplus variant)
##
## Model Test Baseline Model:
##
##   Test statistic                              1598.299    1428.834
##   Degrees of freedom                               190         190
##   P-value                                        0.000       0.000
##   Scaling correction factor                                  1.119
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.581       0.557
##   Tucker-Lewis Index (TLI)                       0.529       0.503
##
##   Robust Comparative Fit Index (CFI)                         0.581
##   Robust Tucker-Lewis Index (TLI)                            0.529
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -6092.148   -6092.148
##   Scaling correction factor                                  1.385
##       for the MLR correction
##   Loglikelihood unrestricted model (H1)      -5712.727   -5712.727
##   Scaling correction factor                                  1.122
##       for the MLR correction
##
##   Akaike (AIC)                               12266.296   12266.296
##   Bayesian (BIC)                             12398.552   12398.552
##   Sample-size adjusted Bayesian (SABIC)      12268.690   12268.690
##
## Root Mean Square Error of Approximation:
```

```
##
##   RMSEA                                         0.137       0.132
##   90 Percent confidence interval - lower        0.127       0.122
##   90 Percent confidence interval - upper        0.147       0.142
##   P-value H_0: RMSEA <= 0.050                   0.000       0.000
##   P-value H_0: RMSEA >= 0.080                   1.000       1.000
##
##   Robust RMSEA                                              0.136
##   90 Percent confidence interval - lower                    0.126
##   90 Percent confidence interval - upper                    0.146
##   P-value H_0: Robust RMSEA <= 0.050                        0.000
##   P-value H_0: Robust RMSEA >= 0.080                        1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                          0.105       0.105
##
## Parameter Estimates:
##
##   Standard errors                            Sandwich
##   Information bread                          Observed
##   Observed information based on               Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   intrinsic =~
##     hc1               1.000                               0.624    0.446
##     hc2               1.071    0.192    5.590    0.000    0.669    0.422
##     hc3               1.330    0.232    5.732    0.000    0.830    0.593
##     hc4               1.316    0.248    5.296    0.000    0.821    0.523
##     hc5               1.260    0.214    5.895    0.000    0.786    0.520
##     sc1               0.789    0.495    1.593    0.111    0.493    0.323
##     sc2               1.481    0.511    2.899    0.004    0.924    0.572
##     sc3               1.561    0.759    2.056    0.040    0.974    0.650
##     sc4               1.465    0.716    2.047    0.041    0.914    0.612
##     sc5               1.205    0.606    1.988    0.047    0.752    0.519
##   extrinsic =~
##     ir1               1.000                               0.793    0.481
##     ir2               1.468    0.262    5.604    0.000    1.164    0.730
##     ir3               1.027    0.134    7.661    0.000    0.814    0.667
##     ir4               1.387    0.290    4.776    0.000    1.100    0.697
##     ir5               0.850    0.150    5.665    0.000    0.673    0.612
##     sp1               1.101    0.202    5.460    0.000    0.873    0.542
##     sp2               0.601    0.182    3.302    0.001    0.477    0.341
##     sp3               0.613    0.178    3.436    0.001    0.486    0.554
##     sp4               1.043    0.214    4.869    0.000    0.827    0.545
##     sp5               0.486    0.143    3.399    0.001    0.385    0.498
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   intrinsic ~~
##     extrinsic         0.366    0.123    2.984    0.003    0.740    0.740
##
## Variances:
```

```
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hc1              1.566    0.230    6.814    0.000    1.566    0.801
##    .hc2              2.058    0.247    8.336    0.000    2.058    0.822
##    .hc3              1.274    0.233    5.475    0.000    1.274    0.649
##    .hc4              1.794    0.342    5.252    0.000    1.794    0.727
##    .hc5              1.671    0.329    5.073    0.000    1.671    0.730
##    .sc1              2.082    0.219    9.487    0.000    2.082    0.896
##    .sc2              1.759    0.189    9.288    0.000    1.759    0.673
##    .sc3              1.300    0.362    3.589    0.000    1.300    0.578
##    .sc4              1.398    0.348    4.012    0.000    1.398    0.626
##    .sc5              1.532    0.262    5.836    0.000    1.532    0.730
##    .ir1              2.086    0.180   11.564    0.000    2.086    0.769
##    .ir2              1.190    0.267    4.457    0.000    1.190    0.468
##    .ir3              0.827    0.158    5.232    0.000    0.827    0.555
##    .ir4              1.283    0.318    4.039    0.000    1.283    0.515
##    .ir5              0.756    0.143    5.300    0.000    0.756    0.625
##    .sp1              1.828    0.213    8.589    0.000    1.828    0.706
##    .sp2              1.724    0.207    8.345    0.000    1.724    0.884
##    .sp3              0.534    0.109    4.884    0.000    0.534    0.693
##    .sp4              1.615    0.195    8.263    0.000    1.615    0.703
##    .sp5              0.449    0.115    3.908    0.000    0.449    0.752
##     intrinsic        0.389    0.232    1.679    0.093    1.000    1.000
##     extrinsic        0.628    0.173    3.635    0.000    1.000    1.000
```

7.1.3 MLR model 3: 4-factor solution

```
rfq_cmod3 <-'hc =~ hc1 + hc2 + hc3 + hc4 + hc5
             sc =~ sc1 + sc2 + sc3 + sc4 + sc5
             ir =~ ir1 + ir2 + ir3 + ir4 + ir5
             sp =~ sp1 + sp2 + sp3 + sp4 + sp5
             '
rfq_cmod3_fit <- cfa(rfq_cmod3, data = df_exsmoker, estimator = "MLR")
summary(rfq_cmod3_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 47 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        46
##
##   Number of observations                           186
##
## Model Test User Model:
##                                              Standard      Scaled
##   Test Statistic                              533.228     520.774
##   Degrees of freedom                              164         164
##   P-value (Chi-square)                          0.000       0.000
##   Scaling correction factor                                 1.024
##     Yuan-Bentler correction (Mplus variant)
##
## Model Test Baseline Model:
##
##   Test statistic                             1598.299    1428.834
##   Degrees of freedom                              190         190
##   P-value                                       0.000       0.000
##   Scaling correction factor                                 1.119
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                   0.738       0.712
##   Tucker-Lewis Index (TLI)                      0.696       0.666
##
##   Robust Comparative Fit Index (CFI)                        0.736
##   Robust Tucker-Lewis Index (TLI)                           0.695
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)             -5979.341   -5979.341
##   Scaling correction factor                                 1.471
##       for the MLR correction
##   Loglikelihood unrestricted model (H1)     -5712.727   -5712.727
##   Scaling correction factor                                 1.122
##       for the MLR correction
##
##   Akaike (AIC)                              12050.683   12050.683
##   Bayesian (BIC)                            12199.067   12199.067
##   Sample-size adjusted Bayesian (SABIC)     12053.369   12053.369
```

```
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.110       0.108
##   90 Percent confidence interval - lower         0.100       0.098
##   90 Percent confidence interval - upper         0.121       0.119
##   P-value H_0: RMSEA <= 0.050                    0.000       0.000
##   P-value H_0: RMSEA >= 0.080                    1.000       1.000
##
##   Robust RMSEA                                               0.109
##   90 Percent confidence interval - lower                     0.099
##   90 Percent confidence interval - upper                     0.120
##   P-value H_0: Robust RMSEA <= 0.050                         0.000
##   P-value H_0: Robust RMSEA >= 0.080                         1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.090       0.090
##
## Parameter Estimates:
##
##   Standard errors                             Sandwich
##   Information bread                           Observed
##   Observed information based on                Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   hc =~
##     hc1               1.000                               0.878    0.628
##     hc2               0.929    0.140    6.636    0.000    0.815    0.515
##     hc3               1.109    0.141    7.885    0.000    0.974    0.695
##     hc4               1.216    0.198    6.147    0.000    1.067    0.679
##     hc5               1.242    0.182    6.808    0.000    1.090    0.721
##   sc =~
##     sc1               1.000                               0.696    0.457
##     sc2               1.202    0.290    4.139    0.000    0.837    0.518
##     sc3               1.758    0.311    5.651    0.000    1.224    0.816
##     sc4               1.638    0.325    5.047    0.000    1.141    0.763
##     sc5               1.341    0.195    6.880    0.000    0.934    0.645
##   ir =~
##     ir1               1.000                               0.738    0.448
##     ir2               1.845    0.513    3.595    0.000    1.362    0.854
##     ir3               0.987    0.148    6.685    0.000    0.729    0.597
##     ir4               1.742    0.514    3.393    0.001    1.286    0.815
##     ir5               0.752    0.153    4.923    0.000    0.555    0.504
##   sp =~
##     sp1               1.000                               1.010    0.628
##     sp2               0.694    0.137    5.084    0.000    0.701    0.502
##     sp3               0.504    0.297    1.694    0.090    0.509    0.580
##     sp4               1.103    0.105   10.460    0.000    1.113    0.735
##     sp5               0.409    0.256    1.594    0.111    0.413    0.534
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
```

```
##   hc ~~
##     sc                0.221    0.064    3.468    0.001    0.361    0.361
##     ir                0.346    0.114    3.034    0.002    0.534    0.534
##     sp                0.367    0.090    4.083    0.000    0.414    0.414
##   sc ~~
##     ir                0.298    0.107    2.795    0.005    0.581    0.581
##     sp                0.318    0.092    3.476    0.001    0.453    0.453
##   ir ~~
##     sp                0.480    0.147    3.268    0.001    0.644    0.644
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hc1               1.185    0.151    7.832    0.000    1.185    0.606
##    .hc2               1.841    0.226    8.130    0.000    1.841    0.735
##    .hc3               1.015    0.174    5.831    0.000    1.015    0.517
##    .hc4               1.329    0.226    5.873    0.000    1.329    0.538
##    .hc5               1.100    0.177    6.216    0.000    1.100    0.480
##    .sc1               1.840    0.197    9.347    0.000    1.840    0.791
##    .sc2               1.912    0.214    8.952    0.000    1.912    0.732
##    .sc3               0.751    0.175    4.287    0.000    0.751    0.334
##    .sc4               0.933    0.167    5.596    0.000    0.933    0.418
##    .sc5               1.225    0.166    7.369    0.000    1.225    0.584
##    .ir1               2.170    0.238    9.124    0.000    2.170    0.799
##    .ir2               0.689    0.314    2.196    0.028    0.689    0.271
##    .ir3               0.959    0.198    4.836    0.000    0.959    0.644
##    .ir4               0.839    0.330    2.540    0.011    0.839    0.337
##    .ir5               0.902    0.168    5.368    0.000    0.902    0.746
##    .sp1               1.570    0.347    4.519    0.000    1.570    0.606
##    .sp2               1.459    0.282    5.169    0.000    1.459    0.748
##    .sp3               0.511    0.169    3.024    0.002    0.511    0.664
##    .sp4               1.058    0.475    2.229    0.026    1.058    0.461
##    .sp5               0.427    0.149    2.863    0.004    0.427    0.715
##     hc                0.770    0.166    4.629    0.000    1.000    1.000
##     sc                0.485    0.169    2.875    0.004    1.000    1.000
##     ir                0.545    0.232    2.345    0.019    1.000    1.000
##     sp                1.020    0.351    2.902    0.004    1.000    1.000
```

7.1.4 MLR model 4: Hierarchial factor solution

```
rfq_cmod4 <-'#1st order
             hc =~ hc1 + hc2 + hc3 + hc4 + hc5
             sc =~ sc1 + sc2 + sc3 + sc4 + sc5
             ir =~ ir1 + ir2 + ir3 + ir4 + ir5
             sp =~ sp1 + sp2 + sp3 + sp4 + sp5

             #2nd order
             intrinsic =~ hc + sc
             extrinsic =~ ir + sp
             '
rfq_cmod4_fit <- cfa(rfq_cmod4, data = df_exsmoker, estimator = "MLR")
```

```
## Warning: lavaan->lav_object_post_check():
##    covariance matrix of latent variables is not positive definite ; use
##    lavInspect(fit, "cov.lv") to investigate.
```

```
summary(rfq_cmod4_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 51 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        45
##
##   Number of observations                           186
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               533.228     521.543
##   Degrees of freedom                               165         165
##   P-value (Chi-square)                           0.000       0.000
##   Scaling correction factor                                  1.022
##     Yuan-Bentler correction (Mplus variant)
##
## Model Test Baseline Model:
##
##   Test statistic                              1598.299    1428.834
##   Degrees of freedom                               190         190
##   P-value                                        0.000       0.000
##   Scaling correction factor                                  1.119
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.739       0.712
##   Tucker-Lewis Index (TLI)                       0.699       0.669
##
##   Robust Comparative Fit Index (CFI)                         0.737
##   Robust Tucker-Lewis Index (TLI)                            0.697
##
## Loglikelihood and Information Criteria:
```

```
##
##   Loglikelihood user model (H0)             -5979.342   -5979.342
##   Scaling correction factor                                 1.487
##       for the MLR correction
##   Loglikelihood unrestricted model (H1)     -5712.727   -5712.727
##   Scaling correction factor                                 1.122
##       for the MLR correction
##
##   Akaike (AIC)                              12048.683   12048.683
##   Bayesian (BIC)                            12193.842   12193.842
##   Sample-size adjusted Bayesian (SABIC)     12051.311   12051.311
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                         0.110       0.108
##   90 Percent confidence interval - lower        0.099       0.098
##   90 Percent confidence interval - upper        0.120       0.118
##   P-value H_0: RMSEA <= 0.050                   0.000       0.000
##   P-value H_0: RMSEA >= 0.080                   1.000       1.000
##
##   Robust RMSEA                                              0.109
##   90 Percent confidence interval - lower                    0.099
##   90 Percent confidence interval - upper                    0.120
##   P-value H_0: Robust RMSEA <= 0.050                        0.000
##   P-value H_0: Robust RMSEA >= 0.080                        1.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                          0.090       0.090
##
## Parameter Estimates:
##
##   Standard errors                            Sandwich
##   Information bread                          Observed
##   Observed information based on               Hessian
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   hc =~
##     hc1               1.000                               0.878    0.628
##     hc2               0.929    0.140    6.637    0.000    0.815    0.515
##     hc3               1.109    0.140    7.941    0.000    0.973    0.695
##     hc4               1.217    0.194    6.255    0.000    1.068    0.680
##     hc5               1.243    0.181    6.859    0.000    1.091    0.721
##   sc =~
##     sc1               1.000                               0.696    0.457
##     sc2               1.202    0.290    4.150    0.000    0.837    0.518
##     sc3               1.758    0.311    5.662    0.000    1.224    0.816
##     sc4               1.638    0.324    5.055    0.000    1.141    0.763
##     sc5               1.341    0.195    6.879    0.000    0.934    0.645
##   ir =~
##     ir1               1.000                               0.738    0.448
##     ir2               1.845    0.513    3.600    0.000    1.362    0.854
##     ir3               0.987    0.148    6.686    0.000    0.729    0.597
```

```
##     ir4              1.742    0.513    3.394    0.001    1.286    0.815
##     ir5              0.751    0.153    4.923    0.000    0.555    0.504
##   sp =~
##     sp1              1.000                               1.010    0.627
##     sp2              0.694    0.137    5.077    0.000    0.701    0.502
##     sp3              0.504    0.294    1.712    0.087    0.509    0.580
##     sp4              1.102    0.105   10.528    0.000    1.113    0.734
##     sp5              0.409    0.254    1.610    0.108    0.413    0.534
##   intrinsic =~
##     hc               1.000                               0.576    0.576
##     sc               0.863    0.217    3.984    0.000    0.627    0.627
##   extrinsic =~
##     ir               1.000                               0.910    0.910
##     sp               1.064    0.349    3.046    0.002    0.708    0.708
##
## Covariances:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   intrinsic ~~
##     extrinsic        0.346    0.115    3.001    0.003    1.019    1.019
##
## Variances:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hc1              1.185    0.150    7.901    0.000    1.185    0.606
##    .hc2              1.841    0.226    8.131    0.000    1.841    0.735
##    .hc3              1.016    0.169    6.027    0.000    1.016    0.517
##    .hc4              1.329    0.220    6.033    0.000    1.329    0.538
##    .hc5              1.099    0.175    6.287    0.000    1.099    0.480
##    .sc1              1.840    0.197    9.355    0.000    1.840    0.791
##    .sc2              1.912    0.213    8.961    0.000    1.912    0.732
##    .sc3              0.751    0.175    4.287    0.000    0.751    0.334
##    .sc4              0.933    0.167    5.599    0.000    0.933    0.418
##    .sc5              1.225    0.167    7.355    0.000    1.225    0.584
##    .ir1              2.169    0.238    9.130    0.000    2.169    0.799
##    .ir2              0.689    0.313    2.200    0.028    0.689    0.271
##    .ir3              0.959    0.198    4.838    0.000    0.959    0.643
##    .ir4              0.839    0.330    2.539    0.011    0.839    0.337
##    .ir5              0.902    0.168    5.369    0.000    0.902    0.746
##    .sp1              1.570    0.345    4.554    0.000    1.570    0.606
##    .sp2              1.459    0.282    5.181    0.000    1.459    0.748
##    .sp3              0.511    0.167    3.051    0.002    0.511    0.663
##    .sp4              1.059    0.467    2.267    0.023    1.059    0.461
##    .sp5              0.427    0.148    2.884    0.004    0.427    0.715
##    .hc               0.515    0.154    3.335    0.001    0.668    0.668
##    .sc               0.294    0.125    2.351    0.019    0.607    0.607
##    .ir               0.094    0.077    1.221    0.222    0.173    0.173
##    .sp               0.509    0.264    1.932    0.053    0.499    0.499
##     intrinsic        0.255    0.091    2.792    0.005    1.000    1.000
##     extrinsic        0.451    0.247    1.826    0.068    1.000    1.000
```

7.1.5 Relative fitness between MLR models

```r
# Model 1 vs Model 2
lavTestLRT(rfq_cmod1_fit, rfq_cmod2_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##                Df   AIC   BIC  Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
## rfq_cmod2_fit 169 12266 12399 758.84
## rfq_cmod1_fit 170 12330 12459 824.46     8.2517 0.55682       1   0.004071 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Model 1 vs Model 3
lavTestLRT(rfq_cmod1_fit, rfq_cmod3_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##                Df   AIC   BIC  Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
## rfq_cmod3_fit 164 12051 12199 533.23
## rfq_cmod1_fit 170 12330 12459 824.46     92.717 0.49404       6  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Model 1 vs Model 4
lavTestLRT(rfq_cmod1_fit, rfq_cmod4_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##                Df   AIC   BIC  Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
## rfq_cmod4_fit 165 12049 12194 533.23
## rfq_cmod1_fit 170 12330 12459 824.46     80.578 0.54196       5  6.352e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Model 3 vs Model 4
lavTestLRT(rfq_cmod3_fit, rfq_cmod4_fit)
```

```
##
## Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")
##
## lavaan->lavTestLRT():
##    lavaan NOTE: The "Chisq" column contains standard test statistics, not the
##    robust test that should be reported per model. A robust difference test is
##    a function of two standard (not robust) statistics.
##
##                Df   AIC   BIC  Chisq Chisq diff RMSEA Df diff Pr(>Chisq)
## rfq_cmod3_fit 164 12051 12199 533.23
## rfq_cmod4_fit 165 12049 12194 533.23 0.00070427     0       1     0.9788
```

7.1.6 WLSMV model 1: 1-factor solution

```r
#Model 1: 1-factor solution
rfq_omod1 <- 'f =~ hc1 + hc2 + hc3 + hc4 + hc5 + sc1 + sc2 + sc3 + sc4 + sc5 +
                   ir1 + ir2 + ir3 + ir4 + ir5 + sp1 + sp2 + sp3 + sp4 + sp5
              '
rfq_omod1_fit <- cfa(rfq_omod1, data = df_exsmoker, estimator = "WLSMV",
                     ordered = rfq_items)
summary(rfq_omod1_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 35 iterations
##
##   Estimator                                       DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                       100
##
##   Number of observations                           186
##
```

```
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               792.531     708.383
##   Degrees of freedom                               170         170
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  1.258
##   Shift parameter                                           78.612
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              6816.274    2957.357
##   Degrees of freedom                               190         190
##   P-value                                           NA       0.000
##   Scaling correction factor                                  2.394
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.906       0.805
##   Tucker-Lewis Index (TLI)                       0.895       0.783
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.141       0.131
##   90 Percent confidence interval - lower         0.131       0.121
##   90 Percent confidence interval - upper         0.151       0.141
##   P-value H_0: RMSEA <= 0.050                    0.000       0.000
##   P-value H_0: RMSEA >= 0.080                    1.000       1.000
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.143       0.143
##
## Parameter Estimates:
##
##   Parameterization                               Delta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   f =~
##     hc1               1.000                               0.472    0.472
##     hc2               0.967    0.130    7.414    0.000    0.457    0.457
```

```
##     hc3              1.254    0.141    8.894    0.000    0.592    0.592
##     hc4              1.298    0.179    7.232    0.000    0.613    0.613
##     hc5              1.196    0.159    7.506    0.000    0.564    0.564
##     sc1              0.777    0.159    4.893    0.000    0.367    0.367
##     sc2              1.275    0.155    8.219    0.000    0.602    0.602
##     sc3              1.479    0.192    7.707    0.000    0.698    0.698
##     sc4              1.351    0.185    7.290    0.000    0.638    0.638
##     sc5              1.143    0.168    6.796    0.000    0.540    0.540
##     ir1              1.138    0.172    6.613    0.000    0.537    0.537
##     ir2              1.828    0.230    7.935    0.000    0.863    0.863
##     ir3              1.538    0.210    7.332    0.000    0.726    0.726
##     ir4              1.752    0.225    7.805    0.000    0.827    0.827
##     ir5              1.531    0.210    7.292    0.000    0.723    0.723
##     sp1              1.250    0.188    6.659    0.000    0.590    0.590
##     sp2              0.991    0.205    4.841    0.000    0.468    0.468
##     sp3              1.848    0.244    7.560    0.000    0.872    0.872
##     sp4              1.259    0.185    6.801    0.000    0.594    0.594
##     sp5              1.721    0.242    7.108    0.000    0.813    0.813
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     hc1|t1           -1.300    0.127  -10.248    0.000   -1.300   -1.300
##     hc1|t2           -0.683    0.100   -6.803    0.000   -0.683   -0.683
##     hc1|t3           -0.204    0.093   -2.192    0.028   -0.204   -0.204
##     hc1|t4            0.190    0.093    2.046    0.041    0.190    0.190
##     hc2|t1           -0.845    0.105   -8.039    0.000   -0.845   -0.845
##     hc2|t2           -0.568    0.098   -5.812    0.000   -0.568   -0.568
##     hc2|t3           -0.231    0.093   -2.484    0.013   -0.231   -0.231
##     hc2|t4            0.135    0.092    1.462    0.144    0.135    0.135
##     hc3|t1           -1.211    0.121   -9.973    0.000   -1.211   -1.211
##     hc3|t2           -0.735    0.102   -7.221    0.000   -0.735   -0.735
##     hc3|t3           -0.315    0.094   -3.358    0.001   -0.315   -0.315
##     hc3|t4            0.204    0.093    2.192    0.028    0.204    0.204
##     hc4|t1           -0.416    0.095   -4.374    0.000   -0.416   -0.416
##     hc4|t2           -0.027    0.092   -0.293    0.770   -0.027   -0.027
##     hc4|t3            0.387    0.095    4.084    0.000    0.387    0.387
##     hc4|t4            0.753    0.102    7.359    0.000    0.753    0.753
##     hc5|t1           -0.666    0.100   -6.662    0.000   -0.666   -0.666
##     hc5|t2           -0.162    0.093   -1.754    0.079   -0.162   -0.162
##     hc5|t3            0.204    0.093    2.192    0.028    0.204    0.204
##     hc5|t4            0.753    0.102    7.359    0.000    0.753    0.753
##     sc1|t1           -1.057    0.114   -9.313    0.000   -1.057   -1.057
##     sc1|t2           -0.521    0.097   -5.383    0.000   -0.521   -0.521
##     sc1|t3           -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sc1|t4            0.135    0.092    1.462    0.144    0.135    0.135
##     sc2|t1           -0.771    0.103   -7.496    0.000   -0.771   -0.771
##     sc2|t2           -0.372    0.095   -3.939    0.000   -0.372   -0.372
##     sc2|t3           -0.054    0.092   -0.585    0.559   -0.054   -0.054
##     sc2|t4            0.259    0.093    2.776    0.006    0.259    0.259
##     sc3|t1           -0.946    0.109   -8.694    0.000   -0.946   -0.946
##     sc3|t2           -0.491    0.096   -5.096    0.000   -0.491   -0.491
##     sc3|t3           -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sc3|t4            0.387    0.095    4.084    0.000    0.387    0.387
##     sc4|t1           -0.905    0.107   -8.436    0.000   -0.905   -0.905
```

```
##     sc4|t2           -0.401    0.095   -4.229    0.000   -0.401   -0.401
##     sc4|t3            0.000    0.092    0.000    1.000    0.000    0.000
##     sc4|t4            0.506    0.097    5.239    0.000    0.506    0.506
##     sc5|t1           -1.131    0.117   -9.657    0.000   -1.131   -1.131
##     sc5|t2           -0.568    0.098   -5.812    0.000   -0.568   -0.568
##     sc5|t3           -0.204    0.093   -2.192    0.028   -0.204   -0.204
##     sc5|t4            0.301    0.094    3.213    0.001    0.301    0.301
##     ir1|t1           -0.552    0.097   -5.669    0.000   -0.552   -0.552
##     ir1|t2           -0.054    0.092   -0.585    0.559   -0.054   -0.054
##     ir1|t3            0.176    0.093    1.900    0.057    0.176    0.176
##     ir1|t4            0.491    0.096    5.096    0.000    0.491    0.491
##     ir2|t1           -0.273    0.093   -2.922    0.003   -0.273   -0.273
##     ir2|t2            0.162    0.093    1.754    0.079    0.162    0.162
##     ir2|t3            0.446    0.096    4.663    0.000    0.446    0.446
##     ir2|t4            0.789    0.103    7.633    0.000    0.789    0.789
##     ir3|t1            0.358    0.094    3.794    0.000    0.358    0.358
##     ir3|t2            0.865    0.106    8.172    0.000    0.865    0.865
##     ir3|t3            1.157    0.118    9.766    0.000    1.157    1.157
##     ir3|t4            1.477    0.140   10.564    0.000    1.477    1.477
##     ir4|t1           -0.149    0.093   -1.608    0.108   -0.149   -0.149
##     ir4|t2            0.315    0.094    3.358    0.001    0.315    0.315
##     ir4|t3            0.506    0.097    5.239    0.000    0.506    0.506
##     ir4|t4            0.885    0.107    8.305    0.000    0.885    0.885
##     ir5|t1            0.568    0.098    5.812    0.000    0.568    0.568
##     ir5|t2            0.925    0.108    8.566    0.000    0.925    0.925
##     ir5|t3            1.401    0.134   10.466    0.000    1.401    1.401
##     ir5|t4            1.661    0.157   10.576    0.000    1.661    1.661
##     sp1|t1           -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sp1|t2            0.231    0.093    2.484    0.013    0.231    0.231
##     sp1|t3            0.521    0.097    5.383    0.000    0.521    0.521
##     sp1|t4            0.845    0.105    8.039    0.000    0.845    0.845
##     sp2|t1            0.600    0.098    6.097    0.000    0.600    0.600
##     sp2|t2            0.735    0.102    7.221    0.000    0.735    0.735
##     sp2|t3            0.865    0.106    8.172    0.000    0.865    0.865
##     sp2|t4            1.332    0.129   10.328    0.000    1.332    1.332
##     sp3|t1            1.081    0.115    9.430    0.000    1.081    1.081
##     sp3|t2            1.438    0.137   10.520    0.000    1.438    1.438
##     sp3|t3            1.609    0.152   10.604    0.000    1.609    1.609
##     sp3|t4            1.849    0.180   10.282    0.000    1.849    1.849
##     sp4|t1           -0.190    0.093   -2.046    0.041   -0.190   -0.190
##     sp4|t2            0.217    0.093    2.338    0.019    0.217    0.217
##     sp4|t3            0.616    0.099    6.239    0.000    0.616    0.616
##     sp4|t4            0.967    0.110    8.822    0.000    0.967    0.967
##     sp5|t1            1.131    0.117    9.657    0.000    1.131    1.131
##     sp5|t2            1.518    0.143   10.594    0.000    1.518    1.518
##     sp5|t3            1.779    0.171   10.425    0.000    1.779    1.779
##     sp5|t4            2.024    0.207    9.769    0.000    2.024    2.024
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hc1               0.777                               0.777    0.777
##    .hc2               0.792                               0.792    0.792
##    .hc3               0.650                               0.650    0.650
##    .hc4               0.625                               0.625    0.625
```

```
##    .hc5              0.681                               0.681    0.681
##    .sc1              0.865                               0.865    0.865
##    .sc2              0.638                               0.638    0.638
##    .sc3              0.512                               0.512    0.512
##    .sc4              0.593                               0.593    0.593
##    .sc5              0.709                               0.709    0.709
##    .ir1              0.711                               0.711    0.711
##    .ir2              0.255                               0.255    0.255
##    .ir3              0.473                               0.473    0.473
##    .ir4              0.316                               0.316    0.316
##    .ir5              0.478                               0.478    0.478
##    .sp1              0.652                               0.652    0.652
##    .sp2              0.781                               0.781    0.781
##    .sp3              0.239                               0.239    0.239
##    .sp4              0.647                               0.647    0.647
##    .sp5              0.340                               0.340    0.340
##     f                0.223    0.053    4.175    0.000    1.000    1.000
```

7.1.7 WLSMV model 2: 2-factor solution

```
rfq_omod2 <- 'intrinsic =~ hc1 + hc2 + hc3 + hc4 + hc5 + sc1 + sc2 + sc3 + sc4 + sc5
              extrinsic =~ ir1 + ir2 + ir3 + ir4 + ir5 + sp1 + sp2 + sp3 + sp4 + sp5
              '
rfq_omod2_fit <- cfa(rfq_omod2, data = df_exsmoker, estimator = "WLSMV",
                     ordered = rfq_items)
summary(rfq_omod2_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 36 iterations
##
##   Estimator                                         DWLS
##   Optimization method                             NLMINB
##   Number of model parameters                         101
##
##   Number of observations                             186
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               638.168     573.827
##   Degrees of freedom                               169         169
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  1.291
##   Shift parameter                                           79.413
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              6816.274    2957.357
##   Degrees of freedom                               190         190
##   P-value                                           NA       0.000
##   Scaling correction factor                                  2.394
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.929       0.854
##   Tucker-Lewis Index (TLI)                       0.920       0.836
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.122       0.114
##   90 Percent confidence interval - lower         0.112       0.104
##   90 Percent confidence interval - upper         0.133       0.124
##   P-value H_0: RMSEA <= 0.050                    0.000       0.000
##   P-value H_0: RMSEA >= 0.080                    1.000       1.000
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
```

```
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                          0.135       0.135
##
## Parameter Estimates:
##
##   Parameterization                               Delta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   intrinsic =~
##     hc1               1.000                               0.533    0.533
##     hc2               0.959    0.125    7.644    0.000    0.511    0.511
##     hc3               1.239    0.131    9.480    0.000    0.660    0.660
##     hc4               1.294    0.170    7.606    0.000    0.689    0.689
##     hc5               1.169    0.144    8.107    0.000    0.623    0.623
##     sc1               0.766    0.146    5.241    0.000    0.408    0.408
##     sc2               1.239    0.143    8.661    0.000    0.660    0.660
##     sc3               1.443    0.173    8.359    0.000    0.768    0.768
##     sc4               1.314    0.165    7.958    0.000    0.700    0.700
##     sc5               1.116    0.150    7.456    0.000    0.595    0.595
##   extrinsic =~
##     ir1               1.000                               0.568    0.568
##     ir2               1.568    0.169    9.284    0.000    0.891    0.891
##     ir3               1.344    0.133   10.145    0.000    0.764    0.764
##     ir4               1.492    0.160    9.322    0.000    0.847    0.847
##     ir5               1.331    0.153    8.686    0.000    0.756    0.756
##     sp1               1.110    0.135    8.212    0.000    0.630    0.630
##     sp2               0.889    0.165    5.383    0.000    0.505    0.505
##     sp3               1.588    0.176    9.048    0.000    0.902    0.902
##     sp4               1.132    0.146    7.733    0.000    0.643    0.643
##     sp5               1.471    0.170    8.645    0.000    0.836    0.836
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   intrinsic ~~
##     extrinsic         0.211    0.037    5.784    0.000    0.698    0.698
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     hc1|t1           -1.300    0.127  -10.248    0.000   -1.300   -1.300
##     hc1|t2           -0.683    0.100   -6.803    0.000   -0.683   -0.683
##     hc1|t3           -0.204    0.093   -2.192    0.028   -0.204   -0.204
##     hc1|t4            0.190    0.093    2.046    0.041    0.190    0.190
##     hc2|t1           -0.845    0.105   -8.039    0.000   -0.845   -0.845
##     hc2|t2           -0.568    0.098   -5.812    0.000   -0.568   -0.568
##     hc2|t3           -0.231    0.093   -2.484    0.013   -0.231   -0.231
##     hc2|t4            0.135    0.092    1.462    0.144    0.135    0.135
##     hc3|t1           -1.211    0.121   -9.973    0.000   -1.211   -1.211
```

```
##     hc3|t2            -0.735    0.102   -7.221    0.000   -0.735   -0.735
##     hc3|t3            -0.315    0.094   -3.358    0.001   -0.315   -0.315
##     hc3|t4             0.204    0.093    2.192    0.028    0.204    0.204
##     hc4|t1            -0.416    0.095   -4.374    0.000   -0.416   -0.416
##     hc4|t2            -0.027    0.092   -0.293    0.770   -0.027   -0.027
##     hc4|t3             0.387    0.095    4.084    0.000    0.387    0.387
##     hc4|t4             0.753    0.102    7.359    0.000    0.753    0.753
##     hc5|t1            -0.666    0.100   -6.662    0.000   -0.666   -0.666
##     hc5|t2            -0.162    0.093   -1.754    0.079   -0.162   -0.162
##     hc5|t3             0.204    0.093    2.192    0.028    0.204    0.204
##     hc5|t4             0.753    0.102    7.359    0.000    0.753    0.753
##     sc1|t1            -1.057    0.114   -9.313    0.000   -1.057   -1.057
##     sc1|t2            -0.521    0.097   -5.383    0.000   -0.521   -0.521
##     sc1|t3            -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sc1|t4             0.135    0.092    1.462    0.144    0.135    0.135
##     sc2|t1            -0.771    0.103   -7.496    0.000   -0.771   -0.771
##     sc2|t2            -0.372    0.095   -3.939    0.000   -0.372   -0.372
##     sc2|t3            -0.054    0.092   -0.585    0.559   -0.054   -0.054
##     sc2|t4             0.259    0.093    2.776    0.006    0.259    0.259
##     sc3|t1            -0.946    0.109   -8.694    0.000   -0.946   -0.946
##     sc3|t2            -0.491    0.096   -5.096    0.000   -0.491   -0.491
##     sc3|t3            -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sc3|t4             0.387    0.095    4.084    0.000    0.387    0.387
##     sc4|t1            -0.905    0.107   -8.436    0.000   -0.905   -0.905
##     sc4|t2            -0.401    0.095   -4.229    0.000   -0.401   -0.401
##     sc4|t3             0.000    0.092    0.000    1.000    0.000    0.000
##     sc4|t4             0.506    0.097    5.239    0.000    0.506    0.506
##     sc5|t1            -1.131    0.117   -9.657    0.000   -1.131   -1.131
##     sc5|t2            -0.568    0.098   -5.812    0.000   -0.568   -0.568
##     sc5|t3            -0.204    0.093   -2.192    0.028   -0.204   -0.204
##     sc5|t4             0.301    0.094    3.213    0.001    0.301    0.301
##     ir1|t1            -0.552    0.097   -5.669    0.000   -0.552   -0.552
##     ir1|t2            -0.054    0.092   -0.585    0.559   -0.054   -0.054
##     ir1|t3             0.176    0.093    1.900    0.057    0.176    0.176
##     ir1|t4             0.491    0.096    5.096    0.000    0.491    0.491
##     ir2|t1            -0.273    0.093   -2.922    0.003   -0.273   -0.273
##     ir2|t2             0.162    0.093    1.754    0.079    0.162    0.162
##     ir2|t3             0.446    0.096    4.663    0.000    0.446    0.446
##     ir2|t4             0.789    0.103    7.633    0.000    0.789    0.789
##     ir3|t1             0.358    0.094    3.794    0.000    0.358    0.358
##     ir3|t2             0.865    0.106    8.172    0.000    0.865    0.865
##     ir3|t3             1.157    0.118    9.766    0.000    1.157    1.157
##     ir3|t4             1.477    0.140   10.564    0.000    1.477    1.477
##     ir4|t1            -0.149    0.093   -1.608    0.108   -0.149   -0.149
##     ir4|t2             0.315    0.094    3.358    0.001    0.315    0.315
##     ir4|t3             0.506    0.097    5.239    0.000    0.506    0.506
##     ir4|t4             0.885    0.107    8.305    0.000    0.885    0.885
##     ir5|t1             0.568    0.098    5.812    0.000    0.568    0.568
##     ir5|t2             0.925    0.108    8.566    0.000    0.925    0.925
##     ir5|t3             1.401    0.134   10.466    0.000    1.401    1.401
##     ir5|t4             1.661    0.157   10.576    0.000    1.661    1.661
##     sp1|t1            -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sp1|t2             0.231    0.093    2.484    0.013    0.231    0.231
##     sp1|t3             0.521    0.097    5.383    0.000    0.521    0.521
```

```
##     sp1|t4           0.845    0.105    8.039    0.000    0.845    0.845
##     sp2|t1           0.600    0.098    6.097    0.000    0.600    0.600
##     sp2|t2           0.735    0.102    7.221    0.000    0.735    0.735
##     sp2|t3           0.865    0.106    8.172    0.000    0.865    0.865
##     sp2|t4           1.332    0.129   10.328    0.000    1.332    1.332
##     sp3|t1           1.081    0.115    9.430    0.000    1.081    1.081
##     sp3|t2           1.438    0.137   10.520    0.000    1.438    1.438
##     sp3|t3           1.609    0.152   10.604    0.000    1.609    1.609
##     sp3|t4           1.849    0.180   10.282    0.000    1.849    1.849
##     sp4|t1          -0.190    0.093   -2.046    0.041   -0.190   -0.190
##     sp4|t2           0.217    0.093    2.338    0.019    0.217    0.217
##     sp4|t3           0.616    0.099    6.239    0.000    0.616    0.616
##     sp4|t4           0.967    0.110    8.822    0.000    0.967    0.967
##     sp5|t1           1.131    0.117    9.657    0.000    1.131    1.131
##     sp5|t2           1.518    0.143   10.594    0.000    1.518    1.518
##     sp5|t3           1.779    0.171   10.425    0.000    1.779    1.779
##     sp5|t4           2.024    0.207    9.769    0.000    2.024    2.024
##
## Variances:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hc1              0.716                               0.716    0.716
##    .hc2              0.739                               0.739    0.739
##    .hc3              0.564                               0.564    0.564
##    .hc4              0.525                               0.525    0.525
##    .hc5              0.612                               0.612    0.612
##    .sc1              0.833                               0.833    0.833
##    .sc2              0.565                               0.565    0.565
##    .sc3              0.410                               0.410    0.410
##    .sc4              0.510                               0.510    0.510
##    .sc5              0.646                               0.646    0.646
##    .ir1              0.677                               0.677    0.677
##    .ir2              0.207                               0.207    0.207
##    .ir3              0.417                               0.417    0.417
##    .ir4              0.282                               0.282    0.282
##    .ir5              0.428                               0.428    0.428
##    .sp1              0.603                               0.603    0.603
##    .sp2              0.745                               0.745    0.745
##    .sp3              0.186                               0.186    0.186
##    .sp4              0.587                               0.587    0.587
##    .sp5              0.302                               0.302    0.302
##     intrinsic        0.284    0.060    4.738    0.000    1.000    1.000
##     extrinsic        0.323    0.064    5.004    0.000    1.000    1.000
```

7.1.8 WLSMV model 3: 4-factor solution

```
rfq_omod3 <- 'hc =~ hc1 + hc2 + hc3 + hc4 + hc5
              sc =~ sc1 + sc2 + sc3 + sc4 + sc5
              ir =~ ir1 + ir2 + ir3 + ir4 + ir5
              sp =~ sp1 + sp2 + sp3 + sp4 + sp5
              '
rfq_omod3_fit <- cfa(rfq_omod3, data = df_exsmoker, estimator = "WLSMV",
                     ordered = rfq_items)
summary(rfq_omod3_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 52 iterations
##
##   Estimator                                         DWLS
##   Optimization method                             NLMINB
##   Number of model parameters                         106
##
##   Number of observations                             186
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               347.642     371.631
##   Degrees of freedom                               164         164
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  1.167
##   Shift parameter                                           73.769
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              6816.274    2957.357
##   Degrees of freedom                               190         190
##   P-value                                           NA       0.000
##   Scaling correction factor                                  2.394
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.972       0.925
##   Tucker-Lewis Index (TLI)                       0.968       0.913
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.078       0.083
##   90 Percent confidence interval - lower         0.066       0.072
##   90 Percent confidence interval - upper         0.089       0.094
##   P-value H_0: RMSEA <= 0.050                    0.000       0.000
##   P-value H_0: RMSEA >= 0.080                    0.385       0.665
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
```

```
##   90 Percent confidence interval - upper                          NA
##   P-value H_0: Robust RMSEA <= 0.050                              NA
##   P-value H_0: Robust RMSEA >= 0.080                              NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                         0.107       0.107
##
## Parameter Estimates:
##
##   Parameterization                               Delta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   hc =~
##     hc1               1.000                               0.641    0.641
##     hc2               0.949    0.115    8.249    0.000    0.609    0.609
##     hc3               1.199    0.118   10.161    0.000    0.769    0.769
##     hc4               1.272    0.149    8.514    0.000    0.816    0.816
##     hc5               1.156    0.124    9.307    0.000    0.741    0.741
##   sc =~
##     sc1               1.000                               0.493    0.493
##     sc2               1.484    0.234    6.337    0.000    0.731    0.731
##     sc3               1.720    0.256    6.711    0.000    0.847    0.847
##     sc4               1.590    0.244    6.525    0.000    0.783    0.783
##     sc5               1.379    0.204    6.752    0.000    0.679    0.679
##   ir =~
##     ir1               1.000                               0.594    0.594
##     ir2               1.524    0.161    9.458    0.000    0.906    0.906
##     ir3               1.336    0.130   10.246    0.000    0.794    0.794
##     ir4               1.455    0.153    9.494    0.000    0.865    0.865
##     ir5               1.352    0.154    8.787    0.000    0.804    0.804
##   sp =~
##     sp1               1.000                               0.729    0.729
##     sp2               0.805    0.122    6.578    0.000    0.587    0.587
##     sp3               1.333    0.133   10.018    0.000    0.971    0.971
##     sp4               1.031    0.099   10.464    0.000    0.752    0.752
##     sp5               1.207    0.135    8.963    0.000    0.879    0.879
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   hc ~~
##     sc                0.129    0.034    3.821    0.000    0.409    0.409
##     ir                0.215    0.038    5.703    0.000    0.565    0.565
##     sp                0.237    0.042    5.586    0.000    0.508    0.508
##   sc ~~
##     ir                0.187    0.041    4.602    0.000    0.639    0.639
##     sp                0.188    0.043    4.427    0.000    0.525    0.525
##   ir ~~
##     sp                0.303    0.045    6.745    0.000    0.699    0.699
##
```

```
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     hc1|t1           -1.300    0.127  -10.248    0.000   -1.300   -1.300
##     hc1|t2           -0.683    0.100   -6.803    0.000   -0.683   -0.683
##     hc1|t3           -0.204    0.093   -2.192    0.028   -0.204   -0.204
##     hc1|t4            0.190    0.093    2.046    0.041    0.190    0.190
##     hc2|t1           -0.845    0.105   -8.039    0.000   -0.845   -0.845
##     hc2|t2           -0.568    0.098   -5.812    0.000   -0.568   -0.568
##     hc2|t3           -0.231    0.093   -2.484    0.013   -0.231   -0.231
##     hc2|t4            0.135    0.092    1.462    0.144    0.135    0.135
##     hc3|t1           -1.211    0.121   -9.973    0.000   -1.211   -1.211
##     hc3|t2           -0.735    0.102   -7.221    0.000   -0.735   -0.735
##     hc3|t3           -0.315    0.094   -3.358    0.001   -0.315   -0.315
##     hc3|t4            0.204    0.093    2.192    0.028    0.204    0.204
##     hc4|t1           -0.416    0.095   -4.374    0.000   -0.416   -0.416
##     hc4|t2           -0.027    0.092   -0.293    0.770   -0.027   -0.027
##     hc4|t3            0.387    0.095    4.084    0.000    0.387    0.387
##     hc4|t4            0.753    0.102    7.359    0.000    0.753    0.753
##     hc5|t1           -0.666    0.100   -6.662    0.000   -0.666   -0.666
##     hc5|t2           -0.162    0.093   -1.754    0.079   -0.162   -0.162
##     hc5|t3            0.204    0.093    2.192    0.028    0.204    0.204
##     hc5|t4            0.753    0.102    7.359    0.000    0.753    0.753
##     sc1|t1           -1.057    0.114   -9.313    0.000   -1.057   -1.057
##     sc1|t2           -0.521    0.097   -5.383    0.000   -0.521   -0.521
##     sc1|t3           -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sc1|t4            0.135    0.092    1.462    0.144    0.135    0.135
##     sc2|t1           -0.771    0.103   -7.496    0.000   -0.771   -0.771
##     sc2|t2           -0.372    0.095   -3.939    0.000   -0.372   -0.372
##     sc2|t3           -0.054    0.092   -0.585    0.559   -0.054   -0.054
##     sc2|t4            0.259    0.093    2.776    0.006    0.259    0.259
##     sc3|t1           -0.946    0.109   -8.694    0.000   -0.946   -0.946
##     sc3|t2           -0.491    0.096   -5.096    0.000   -0.491   -0.491
##     sc3|t3           -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sc3|t4            0.387    0.095    4.084    0.000    0.387    0.387
##     sc4|t1           -0.905    0.107   -8.436    0.000   -0.905   -0.905
##     sc4|t2           -0.401    0.095   -4.229    0.000   -0.401   -0.401
##     sc4|t3            0.000    0.092    0.000    1.000    0.000    0.000
##     sc4|t4            0.506    0.097    5.239    0.000    0.506    0.506
##     sc5|t1           -1.131    0.117   -9.657    0.000   -1.131   -1.131
##     sc5|t2           -0.568    0.098   -5.812    0.000   -0.568   -0.568
##     sc5|t3           -0.204    0.093   -2.192    0.028   -0.204   -0.204
##     sc5|t4            0.301    0.094    3.213    0.001    0.301    0.301
##     ir1|t1           -0.552    0.097   -5.669    0.000   -0.552   -0.552
##     ir1|t2           -0.054    0.092   -0.585    0.559   -0.054   -0.054
##     ir1|t3            0.176    0.093    1.900    0.057    0.176    0.176
##     ir1|t4            0.491    0.096    5.096    0.000    0.491    0.491
##     ir2|t1           -0.273    0.093   -2.922    0.003   -0.273   -0.273
##     ir2|t2            0.162    0.093    1.754    0.079    0.162    0.162
##     ir2|t3            0.446    0.096    4.663    0.000    0.446    0.446
##     ir2|t4            0.789    0.103    7.633    0.000    0.789    0.789
##     ir3|t1            0.358    0.094    3.794    0.000    0.358    0.358
##     ir3|t2            0.865    0.106    8.172    0.000    0.865    0.865
##     ir3|t3            1.157    0.118    9.766    0.000    1.157    1.157
##     ir3|t4            1.477    0.140   10.564    0.000    1.477    1.477
```

```
##     ir4|t1           -0.149    0.093   -1.608    0.108   -0.149   -0.149
##     ir4|t2            0.315    0.094    3.358    0.001    0.315    0.315
##     ir4|t3            0.506    0.097    5.239    0.000    0.506    0.506
##     ir4|t4            0.885    0.107    8.305    0.000    0.885    0.885
##     ir5|t1            0.568    0.098    5.812    0.000    0.568    0.568
##     ir5|t2            0.925    0.108    8.566    0.000    0.925    0.925
##     ir5|t3            1.401    0.134   10.466    0.000    1.401    1.401
##     ir5|t4            1.661    0.157   10.576    0.000    1.661    1.661
##     sp1|t1           -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sp1|t2            0.231    0.093    2.484    0.013    0.231    0.231
##     sp1|t3            0.521    0.097    5.383    0.000    0.521    0.521
##     sp1|t4            0.845    0.105    8.039    0.000    0.845    0.845
##     sp2|t1            0.600    0.098    6.097    0.000    0.600    0.600
##     sp2|t2            0.735    0.102    7.221    0.000    0.735    0.735
##     sp2|t3            0.865    0.106    8.172    0.000    0.865    0.865
##     sp2|t4            1.332    0.129   10.328    0.000    1.332    1.332
##     sp3|t1            1.081    0.115    9.430    0.000    1.081    1.081
##     sp3|t2            1.438    0.137   10.520    0.000    1.438    1.438
##     sp3|t3            1.609    0.152   10.604    0.000    1.609    1.609
##     sp3|t4            1.849    0.180   10.282    0.000    1.849    1.849
##     sp4|t1           -0.190    0.093   -2.046    0.041   -0.190   -0.190
##     sp4|t2            0.217    0.093    2.338    0.019    0.217    0.217
##     sp4|t3            0.616    0.099    6.239    0.000    0.616    0.616
##     sp4|t4            0.967    0.110    8.822    0.000    0.967    0.967
##     sp5|t1            1.131    0.117    9.657    0.000    1.131    1.131
##     sp5|t2            1.518    0.143   10.594    0.000    1.518    1.518
##     sp5|t3            1.779    0.171   10.425    0.000    1.779    1.779
##     sp5|t4            2.024    0.207    9.769    0.000    2.024    2.024
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hc1               0.589                               0.589    0.589
##    .hc2               0.629                               0.629    0.629
##    .hc3               0.409                               0.409    0.409
##    .hc4               0.334                               0.334    0.334
##    .hc5               0.450                               0.450    0.450
##    .sc1               0.757                               0.757    0.757
##    .sc2               0.466                               0.466    0.466
##    .sc3               0.282                               0.282    0.282
##    .sc4               0.387                               0.387    0.387
##    .sc5               0.539                               0.539    0.539
##    .ir1               0.647                               0.647    0.647
##    .ir2               0.180                               0.180    0.180
##    .ir3               0.369                               0.369    0.369
##    .ir4               0.252                               0.252    0.252
##    .ir5               0.354                               0.354    0.354
##    .sp1               0.469                               0.469    0.469
##    .sp2               0.656                               0.656    0.656
##    .sp3               0.057                               0.057    0.057
##    .sp4               0.435                               0.435    0.435
##    .sp5               0.227                               0.227    0.227
##     hc                0.411    0.072    5.714    0.000    1.000    1.000
##     sc                0.243    0.070    3.475    0.001    1.000    1.000
##     ir                0.353    0.069    5.129    0.000    1.000    1.000
```

```
##     sp                   0.531    0.088    6.036    0.000    1.000    1.000
```

7.1.9 WLSMV model 4: hierarchial factor solution

```
rfq_omod4 <- '#1st order
             hc =~ hc1 + hc2 + hc3 + hc4 + hc5
             sc =~ sc1 + sc2 + sc3 + sc4 + sc5
             ir =~ ir1 + ir2 + ir3 + ir4 + ir5
             sp =~ sp1 + sp2 + sp3 + sp4 + sp5

             #2nd order
             intrinsic =~ hc + sc
             extrinsic =~ ir + sp
             '
rfq_omod4_fit <- cfa(rfq_omod4, data = df_exsmoker, estimator = "WLSMV",
                     ordered = rfq_items)
```

```
## Warning: lavaan->lav_object_post_check():
##    covariance matrix of latent variables is not positive definite ; use
##    lavInspect(fit, "cov.lv") to investigate.
```

```
summary(rfq_omod4_fit, standardized = TRUE, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 44 iterations
##
##   Estimator                                         DWLS
##   Optimization method                             NLMINB
##   Number of model parameters                         105
##
##   Number of observations                             186
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               348.317     370.429
##   Degrees of freedom                               165         165
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  1.176
##   Shift parameter                                           74.200
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              6816.274    2957.357
##   Degrees of freedom                               190         190
##   P-value                                           NA       0.000
##   Scaling correction factor                                  2.394
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.972       0.926
##   Tucker-Lewis Index (TLI)                       0.968       0.915
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
```

```
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.077       0.082
##   90 Percent confidence interval - lower         0.066       0.071
##   90 Percent confidence interval - upper         0.089       0.093
##   P-value H_0: RMSEA <= 0.050                    0.000       0.000
##   P-value H_0: RMSEA >= 0.080                    0.368       0.628
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.107       0.107
##
## Parameter Estimates:
##
##   Parameterization                               Delta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   hc =~
##     hc1               1.000                               0.642    0.642
##     hc2               0.950    0.115    8.260    0.000    0.609    0.609
##     hc3               1.199    0.118   10.151    0.000    0.770    0.770
##     hc4               1.271    0.149    8.521    0.000    0.815    0.815
##     hc5               1.155    0.124    9.304    0.000    0.741    0.741
##   sc =~
##     sc1               1.000                               0.493    0.493
##     sc2               1.482    0.234    6.330    0.000    0.731    0.731
##     sc3               1.718    0.256    6.709    0.000    0.847    0.847
##     sc4               1.588    0.243    6.526    0.000    0.783    0.783
##     sc5               1.379    0.204    6.746    0.000    0.680    0.680
##   ir =~
##     ir1               1.000                               0.594    0.594
##     ir2               1.524    0.161    9.447    0.000    0.906    0.906
##     ir3               1.336    0.131   10.234    0.000    0.794    0.794
##     ir4               1.455    0.153    9.490    0.000    0.865    0.865
##     ir5               1.353    0.154    8.781    0.000    0.804    0.804
##   sp =~
##     sp1               1.000                               0.729    0.729
##     sp2               0.804    0.122    6.587    0.000    0.586    0.586
##     sp3               1.332    0.133   10.045    0.000    0.971    0.971
##     sp4               1.032    0.098   10.490    0.000    0.752    0.752
##     sp5               1.206    0.134    8.978    0.000    0.879    0.879
##   intrinsic =~
##     hc                1.000                               0.610    0.610
```

```
##     sc                 0.842    0.174    4.850    0.000    0.669    0.669
##   extrinsic =~
##     ir                 1.000                               0.904    0.904
##     sp                 1.050    0.149    7.032    0.000    0.774    0.774
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   intrinsic ~~
##     extrinsic          0.219    0.038    5.757    0.000    1.042    1.042
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     hc1|t1            -1.300    0.127  -10.248    0.000   -1.300   -1.300
##     hc1|t2            -0.683    0.100   -6.803    0.000   -0.683   -0.683
##     hc1|t3            -0.204    0.093   -2.192    0.028   -0.204   -0.204
##     hc1|t4             0.190    0.093    2.046    0.041    0.190    0.190
##     hc2|t1            -0.845    0.105   -8.039    0.000   -0.845   -0.845
##     hc2|t2            -0.568    0.098   -5.812    0.000   -0.568   -0.568
##     hc2|t3            -0.231    0.093   -2.484    0.013   -0.231   -0.231
##     hc2|t4             0.135    0.092    1.462    0.144    0.135    0.135
##     hc3|t1            -1.211    0.121   -9.973    0.000   -1.211   -1.211
##     hc3|t2            -0.735    0.102   -7.221    0.000   -0.735   -0.735
##     hc3|t3            -0.315    0.094   -3.358    0.001   -0.315   -0.315
##     hc3|t4             0.204    0.093    2.192    0.028    0.204    0.204
##     hc4|t1            -0.416    0.095   -4.374    0.000   -0.416   -0.416
##     hc4|t2            -0.027    0.092   -0.293    0.770   -0.027   -0.027
##     hc4|t3             0.387    0.095    4.084    0.000    0.387    0.387
##     hc4|t4             0.753    0.102    7.359    0.000    0.753    0.753
##     hc5|t1            -0.666    0.100   -6.662    0.000   -0.666   -0.666
##     hc5|t2            -0.162    0.093   -1.754    0.079   -0.162   -0.162
##     hc5|t3             0.204    0.093    2.192    0.028    0.204    0.204
##     hc5|t4             0.753    0.102    7.359    0.000    0.753    0.753
##     sc1|t1            -1.057    0.114   -9.313    0.000   -1.057   -1.057
##     sc1|t2            -0.521    0.097   -5.383    0.000   -0.521   -0.521
##     sc1|t3            -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sc1|t4             0.135    0.092    1.462    0.144    0.135    0.135
##     sc2|t1            -0.771    0.103   -7.496    0.000   -0.771   -0.771
##     sc2|t2            -0.372    0.095   -3.939    0.000   -0.372   -0.372
##     sc2|t3            -0.054    0.092   -0.585    0.559   -0.054   -0.054
##     sc2|t4             0.259    0.093    2.776    0.006    0.259    0.259
##     sc3|t1            -0.946    0.109   -8.694    0.000   -0.946   -0.946
##     sc3|t2            -0.491    0.096   -5.096    0.000   -0.491   -0.491
##     sc3|t3            -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sc3|t4             0.387    0.095    4.084    0.000    0.387    0.387
##     sc4|t1            -0.905    0.107   -8.436    0.000   -0.905   -0.905
##     sc4|t2            -0.401    0.095   -4.229    0.000   -0.401   -0.401
##     sc4|t3             0.000    0.092    0.000    1.000    0.000    0.000
##     sc4|t4             0.506    0.097    5.239    0.000    0.506    0.506
##     sc5|t1            -1.131    0.117   -9.657    0.000   -1.131   -1.131
##     sc5|t2            -0.568    0.098   -5.812    0.000   -0.568   -0.568
##     sc5|t3            -0.204    0.093   -2.192    0.028   -0.204   -0.204
##     sc5|t4             0.301    0.094    3.213    0.001    0.301    0.301
##     ir1|t1            -0.552    0.097   -5.669    0.000   -0.552   -0.552
##     ir1|t2            -0.054    0.092   -0.585    0.559   -0.054   -0.054
```

```
##     ir1|t3           0.176    0.093    1.900    0.057    0.176    0.176
##     ir1|t4           0.491    0.096    5.096    0.000    0.491    0.491
##     ir2|t1          -0.273    0.093   -2.922    0.003   -0.273   -0.273
##     ir2|t2           0.162    0.093    1.754    0.079    0.162    0.162
##     ir2|t3           0.446    0.096    4.663    0.000    0.446    0.446
##     ir2|t4           0.789    0.103    7.633    0.000    0.789    0.789
##     ir3|t1           0.358    0.094    3.794    0.000    0.358    0.358
##     ir3|t2           0.865    0.106    8.172    0.000    0.865    0.865
##     ir3|t3           1.157    0.118    9.766    0.000    1.157    1.157
##     ir3|t4           1.477    0.140   10.564    0.000    1.477    1.477
##     ir4|t1          -0.149    0.093   -1.608    0.108   -0.149   -0.149
##     ir4|t2           0.315    0.094    3.358    0.001    0.315    0.315
##     ir4|t3           0.506    0.097    5.239    0.000    0.506    0.506
##     ir4|t4           0.885    0.107    8.305    0.000    0.885    0.885
##     ir5|t1           0.568    0.098    5.812    0.000    0.568    0.568
##     ir5|t2           0.925    0.108    8.566    0.000    0.925    0.925
##     ir5|t3           1.401    0.134   10.466    0.000    1.401    1.401
##     ir5|t4           1.661    0.157   10.576    0.000    1.661    1.661
##     sp1|t1          -0.067    0.092   -0.731    0.465   -0.067   -0.067
##     sp1|t2           0.231    0.093    2.484    0.013    0.231    0.231
##     sp1|t3           0.521    0.097    5.383    0.000    0.521    0.521
##     sp1|t4           0.845    0.105    8.039    0.000    0.845    0.845
##     sp2|t1           0.600    0.098    6.097    0.000    0.600    0.600
##     sp2|t2           0.735    0.102    7.221    0.000    0.735    0.735
##     sp2|t3           0.865    0.106    8.172    0.000    0.865    0.865
##     sp2|t4           1.332    0.129   10.328    0.000    1.332    1.332
##     sp3|t1           1.081    0.115    9.430    0.000    1.081    1.081
##     sp3|t2           1.438    0.137   10.520    0.000    1.438    1.438
##     sp3|t3           1.609    0.152   10.604    0.000    1.609    1.609
##     sp3|t4           1.849    0.180   10.282    0.000    1.849    1.849
##     sp4|t1          -0.190    0.093   -2.046    0.041   -0.190   -0.190
##     sp4|t2           0.217    0.093    2.338    0.019    0.217    0.217
##     sp4|t3           0.616    0.099    6.239    0.000    0.616    0.616
##     sp4|t4           0.967    0.110    8.822    0.000    0.967    0.967
##     sp5|t1           1.131    0.117    9.657    0.000    1.131    1.131
##     sp5|t2           1.518    0.143   10.594    0.000    1.518    1.518
##     sp5|t3           1.779    0.171   10.425    0.000    1.779    1.779
##     sp5|t4           2.024    0.207    9.769    0.000    2.024    2.024
##
## Variances:
##                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .hc1              0.588                               0.588    0.588
##    .hc2              0.629                               0.629    0.629
##    .hc3              0.408                               0.408    0.408
##    .hc4              0.335                               0.335    0.335
##    .hc5              0.451                               0.451    0.451
##    .sc1              0.757                               0.757    0.757
##    .sc2              0.466                               0.466    0.466
##    .sc3              0.283                               0.283    0.283
##    .sc4              0.387                               0.387    0.387
##    .sc5              0.538                               0.538    0.538
##    .ir1              0.647                               0.647    0.647
##    .ir2              0.180                               0.180    0.180
##    .ir3              0.369                               0.369    0.369
```

```
##    .ir4              0.252                               0.252    0.252
##    .ir5              0.354                               0.354    0.354
##    .sp1              0.469                               0.469    0.469
##    .sp2              0.657                               0.657    0.657
##    .sp3              0.057                               0.057    0.057
##    .sp4              0.435                               0.435    0.435
##    .sp5              0.227                               0.227    0.227
##    .hc               0.258    0.059    4.403    0.000    0.627    0.627
##    .sc               0.134    0.044    3.083    0.002    0.552    0.552
##    .ir               0.065    0.032    2.003    0.045    0.183    0.183
##    .sp               0.213    0.061    3.482    0.000    0.401    0.401
##     intrinsic        0.153    0.047    3.235    0.001    1.000    1.000
##     extrinsic        0.288    0.061    4.720    0.000    1.000    1.000
```

8 Relations between smoking cessation motives and time since cessation

```r
# Define a function to perform repeated Kruskall-Wallis tests
rfq_kruskal_table <- function(data, items, group_var, digits = 2) {

  # Epsilon-squared effect size for Kruskal-Wallis.
  epsilon_squared <- function(H, n, k) (H - k + 1) / (n - k)

  # Format summary statistics
  fmt_mq <- function(x, digits) {
    sprintf(
      "%s (%s, %s)",
      formatC(median(x),                           format = "f", digits = digits),
      formatC(quantile(x, 0.25, names = FALSE),  format = "f", digits = digits),
      formatC(quantile(x, 0.75, names = FALSE),  format = "f", digits = digits)
    )
  }

  # Loop over numeric variables
  results <- lapply(items, function(var) {

    # Subset data
    df <- na.omit(data[, c(var, group_var)])
    names(df) <- c("value", "group")

    # Perform Kruskal-Wallis test
    kw   <- kruskal.test(value ~ group, data = df)
    H    <- as.numeric(kw$statistic)
    n    <- nrow(df)
    k    <- length(unique(df$group))
    eps2 <- epsilon_squared(H, n, k)

    # Create "Median (Q1, Q3)" format for reporting
    groups <- sort(unique(df$group))
    group_cells <- setNames(lapply(groups,
                                   function(g) fmt_mq(df$value[df$group == g],
                                                      digits)),
                            as.character(groups))

    # Build data frame rows
    data.frame(Variable    = var,
               group_cells,
               Statistic   = round(H, 3),
               p_value     = kw$p.value,
               Effect_size = round(eps2, 3),
               check.names      = FALSE,
               stringsAsFactors = FALSE)
  })

  # Assemble rows into a single table
  results_df <- do.call(rbind, results)
```

```r
  # Adjust p-values for multiple comparisons (Benjamin-Hochberg)
  results_df$adj_p_value <- p.adjust(results_df$p_value, method = "BH")

  # Format p-values for reporting
  fmt_p <- function(p) ifelse(p < 0.001, "<0.001",
                              formatC(round(p, 3), format = "f", digits = 3))
  results_df$p_value     <- fmt_p(results_df$p_value)
  results_df$adj_p_value <- fmt_p(results_df$adj_p_value)

  # Return results
  results_df
}
```

```r
# Perform repeated Kruskal-Wallis tests
results_df <- rfq_kruskal_table(data      = data_cleaned,
                                items     = rfq_items,
                                group_var = "time_since_quitting")
results_df
```

```
##    Variable         <6_months         >= 1 year >6_months_to_<1_year Statistic
## 1       hc1 3.00 (1.00, 4.00) 3.00 (2.00, 4.00)    3.00 (2.00, 4.00)     1.219
## 2       hc2 3.00 (1.00, 4.00) 3.00 (1.00, 4.00)    3.00 (2.00, 4.00)     1.036
## 3       hc3 3.00 (2.00, 4.00) 3.00 (2.00, 4.00)    3.00 (2.00, 4.00)     0.341
## 4       hc4 1.00 (0.00, 3.00) 2.00 (0.00, 3.50)    1.00 (0.00, 3.75)     1.589
## 5       hc5 2.00 (1.00, 3.00) 2.00 (0.50, 3.00)    1.00 (0.00, 3.00)     1.152
## 6       sc1 2.00 (1.00, 4.00) 4.00 (1.00, 4.00)    2.00 (1.00, 3.75)     4.745
## 7       sc2 3.00 (1.00, 4.00) 3.00 (1.00, 4.00)    1.50 (0.00, 4.00)     2.374
## 8       sc3 2.00 (1.00, 4.00) 3.00 (1.00, 4.00)    2.00 (1.00, 3.75)     4.778
## 9       sc4 3.00 (1.00, 4.00) 3.00 (1.00, 4.00)    2.00 (0.25, 3.00)     3.665
## 10      sc5 3.00 (1.00, 4.00) 4.00 (1.50, 4.00)    2.00 (1.00, 3.00)     7.156
## 11      ir1 2.00 (0.00, 3.00) 3.00 (0.00, 4.00)    1.00 (1.00, 4.00)     1.288
## 12      ir2 1.00 (0.00, 3.00) 1.00 (0.00, 3.00)    1.00 (0.00, 2.00)     1.293
## 13      ir3 0.00 (0.00, 1.00) 0.00 (0.00, 1.00)    0.00 (0.00, 0.75)     2.116
## 14      ir4 1.00 (0.00, 3.00) 1.00 (0.00, 3.00)    0.00 (0.00, 2.00)     1.973
## 15      ir5 0.00 (0.00, 0.00) 0.00 (0.00, 1.00)    0.00 (0.00, 1.00)     3.653
## 16      sp1 0.00 (0.00, 3.00) 1.00 (0.00, 3.00)    0.50 (0.00, 2.00)     3.265
## 17      sp2 0.00 (0.00, 1.00) 0.00 (0.00, 2.00)    0.00 (0.00, 0.00)     2.417
## 18      sp3 0.00 (0.00, 0.00) 0.00 (0.00, 0.00)    0.00 (0.00, 0.00)     4.158
## 19      sp4 1.00 (0.00, 3.00) 1.00 (0.00, 3.00)    1.00 (0.00, 2.00)     0.764
## 20      sp5 0.00 (0.00, 0.00) 0.00 (0.00, 0.00)    0.00 (0.00, 0.00)     4.531
##    p_value Effect_size adj_p_value
## 1    0.544      -0.004       0.661
## 2    0.596      -0.005       0.662
## 3    0.843      -0.009       0.843
## 4    0.452      -0.002       0.661
## 5    0.562      -0.005       0.661
## 6    0.093       0.015       0.460
## 7    0.305       0.002       0.610
## 8    0.092       0.015       0.460
## 9    0.160       0.009       0.460
## 10   0.028       0.028       0.460
## 11   0.525      -0.004       0.661
## 12   0.524      -0.004       0.661
## 13   0.347       0.001       0.621
## 14   0.373       0.000       0.621
## 15   0.161       0.009       0.460
## 16   0.195       0.007       0.489
## 17   0.299       0.002       0.610
## 18   0.125       0.012       0.460
## 19   0.682      -0.007       0.718
## 20   0.104       0.014       0.460
```

Characteristic	**ex_smoker** N = 186[1]	**smoker** N = 805[1]	**p-value**[2]
age	22 (20, 25)	22 (20, 24)	0.978
sex			<0.001
female	38 (20%)	86 (11%)	
male	148 (80%)	719 (89%)	
ethnicity			0.214
arab	180 (97%)	787 (98%)	
kurd	2 (1.1%)	11 (1.4%)	
others	0 (0%)	2 (0.2%)	
turkmen	4 (2.2%)	5 (0.6%)	
marital_stat			0.959
divorced	1 (0.5%)	5 (0.6%)	
married	21 (11%)	89 (11%)	
single	164 (88%)	711 (88%)	
residence			0.541
rural	26 (14%)	99 (12%)	
urban	160 (86%)	706 (88%)	
speciality			
college_of_administrative_technology	3 (1.6%)	16 (2.0%)	
college_of_education	25 (13%)	73 (9.1%)	
college_of_science	12 (6.5%)	66 (8.2%)	
dentistry	4 (2.2%)	32 (4.0%)	
engineering	20 (11%)	118 (15%)	
engineering_technology	5 (2.7%)	40 (5.0%)	
humanitarian_college	8 (4.3%)	49 (6.1%)	
institute_of_medical_technology	5 (2.7%)	29 (3.6%)	
institute_of_technology	0 (0%)	3 (0.4%)	
law_school	9 (4.8%)	30 (3.7%)	
medical_technology	27 (15%)	116 (14%)	
medicine	24 (13%)	74 (9.2%)	
nursing	5 (2.7%)	14 (1.7%)	
other_colleges	19 (10%)	77 (9.6%)	
other_institutes	2 (1.1%)	24 (3.0%)	
pharmacology	16 (8.6%)	43 (5.3%)	
veterinary	2 (1.1%)	1 (0.1%)	
cigarettes	146 (78%)	609 (76%)	0.446
e_cigarettes	62 (33%)	228 (28%)	0.181
hookah	74 (40%)	251 (31%)	0.030
smoke_most			0.677
cigarettes	127 (68%)	547 (68%)	
e_cigarettes	25 (13%)	126 (16%)	
hookah	34 (18%)	132 (16%)	

[1] Median (Q1, Q3); n (%)

[2] Wilcoxon rank sum test; Fisher's exact test; NA